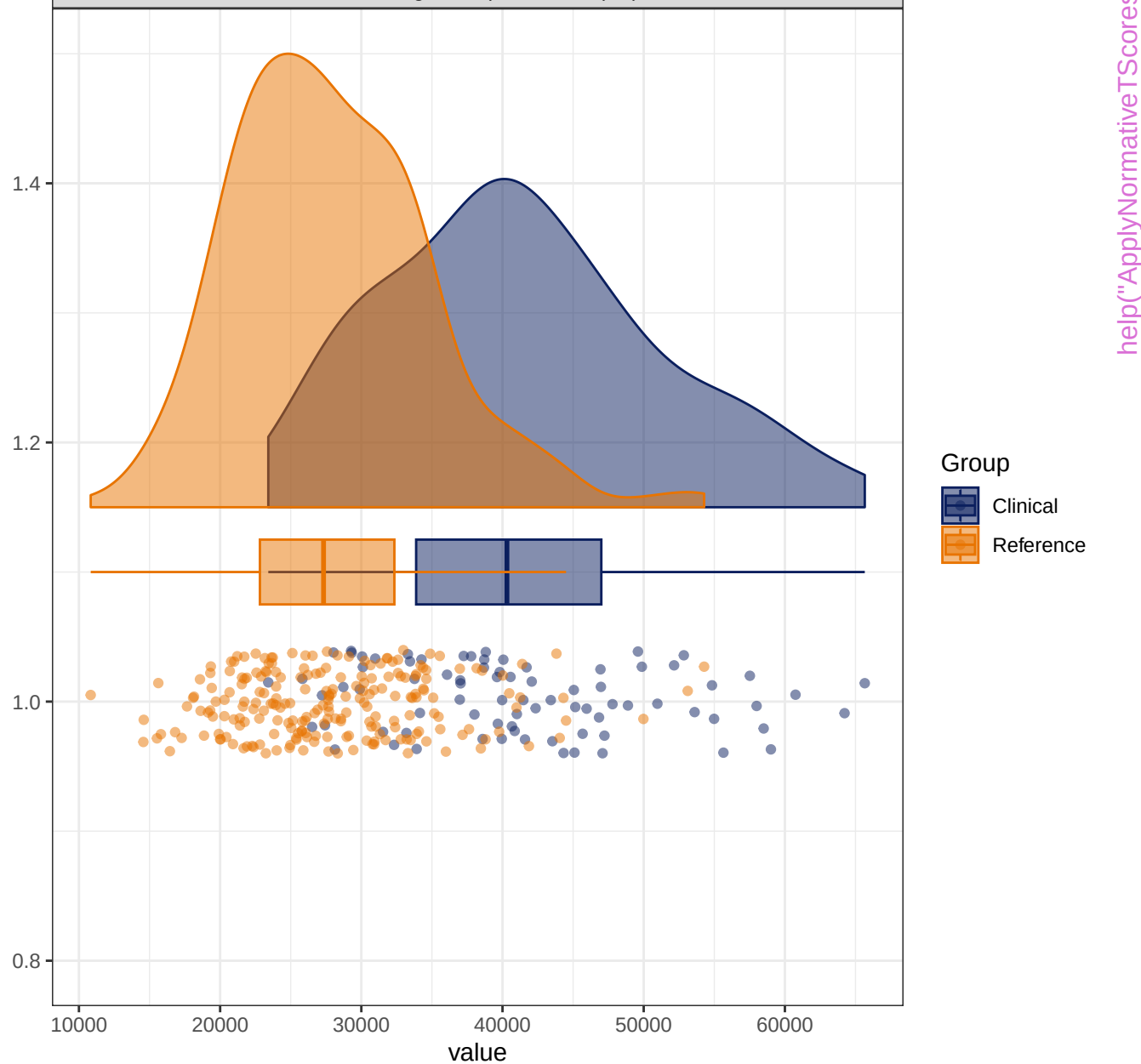
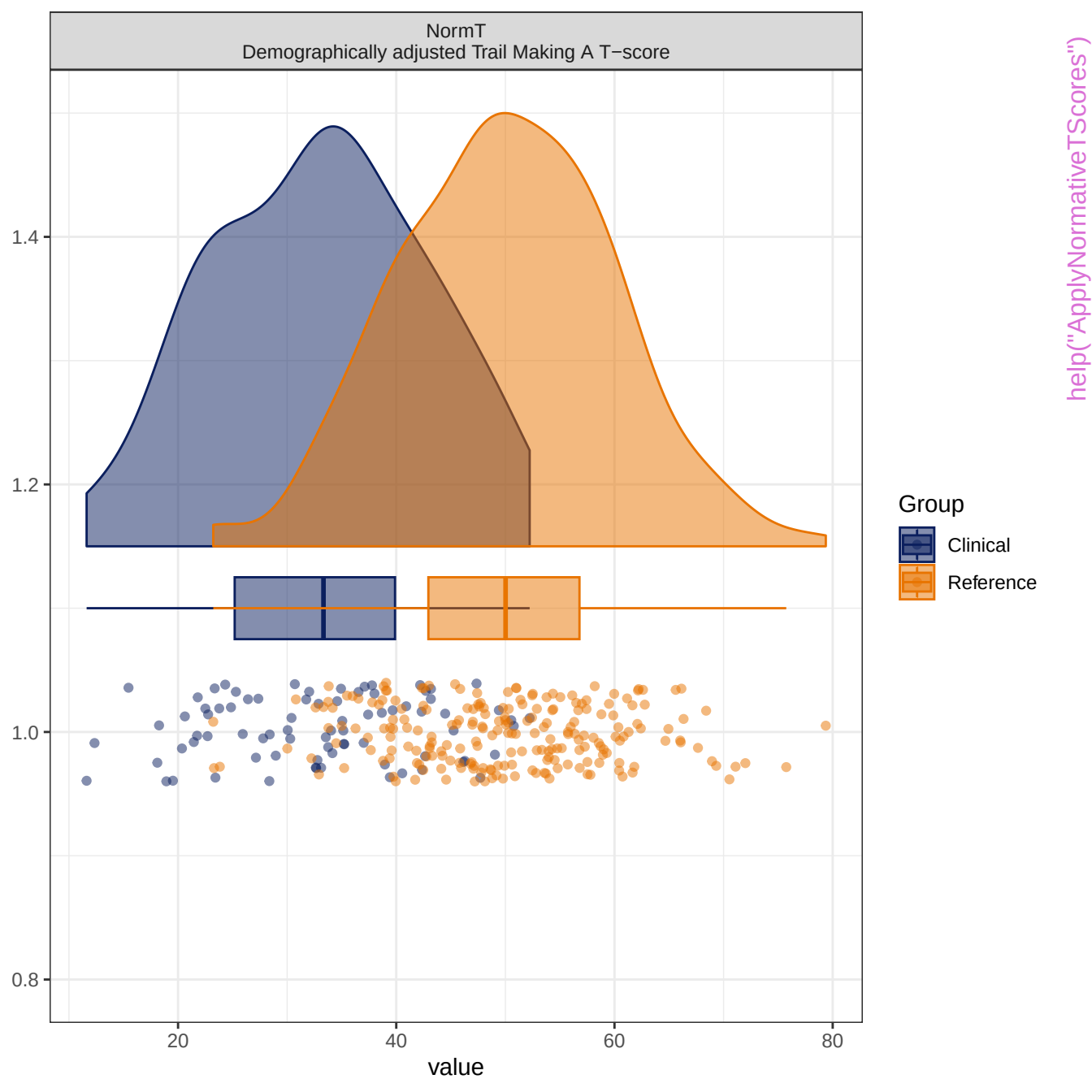


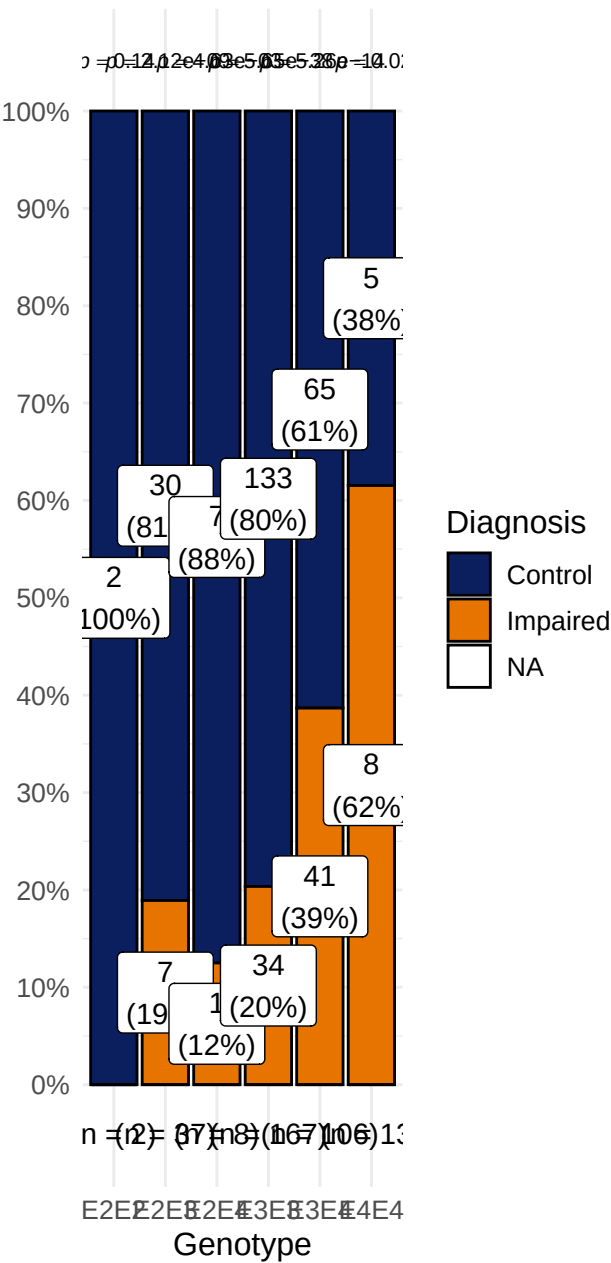
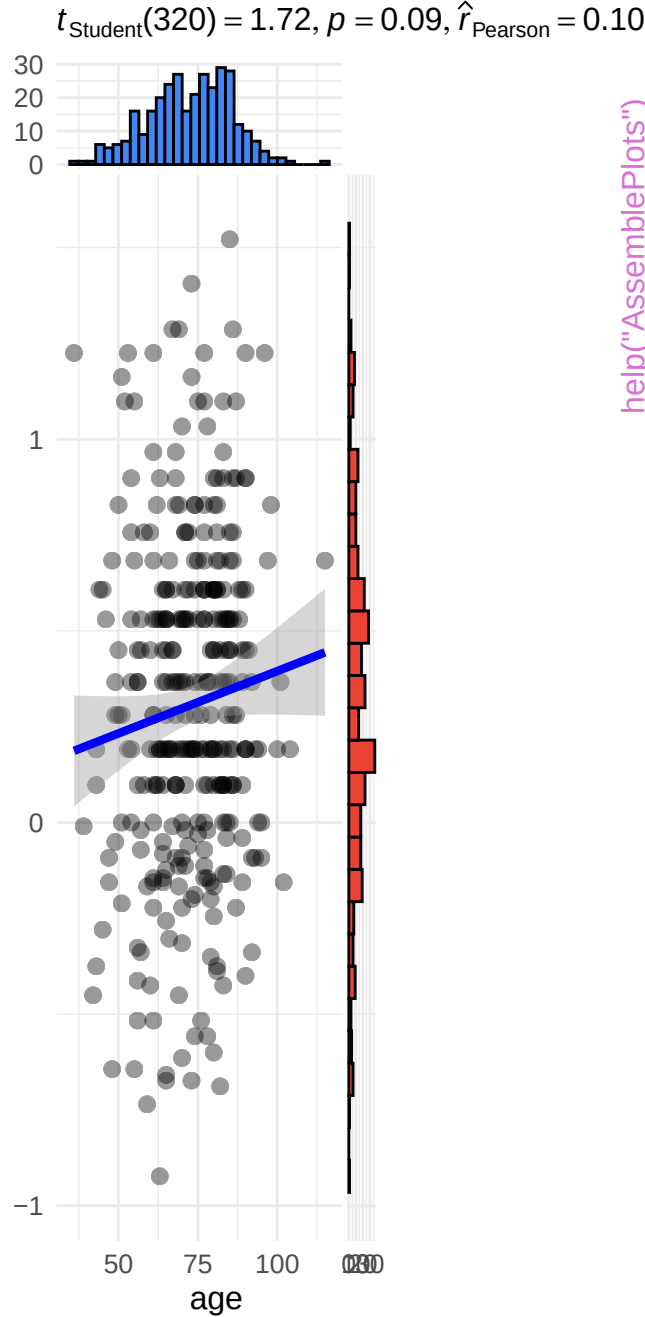
NormRaw
Trail Making A completion time (ms)

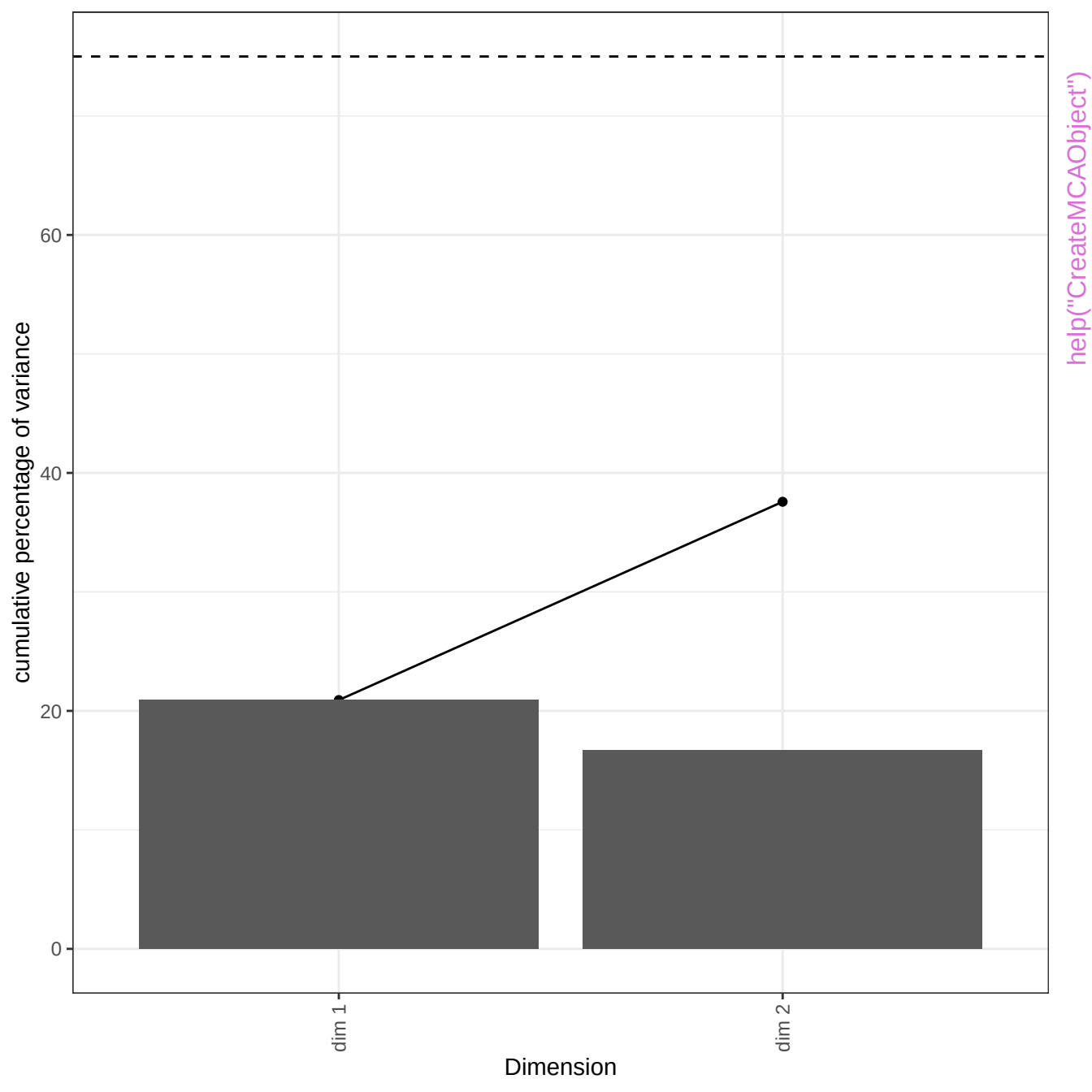


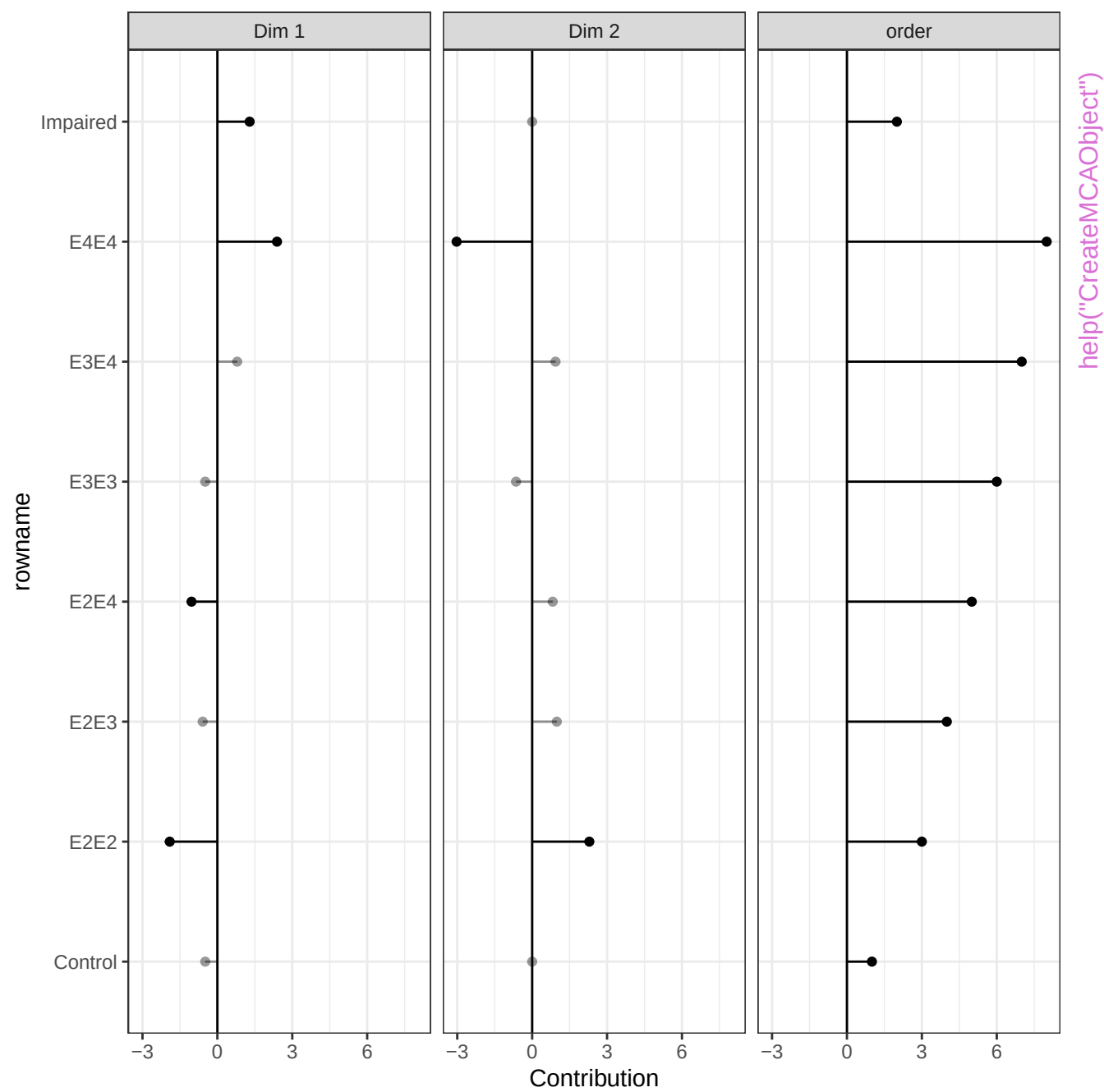
NormT
Demographically adjusted Trail Making A T-score



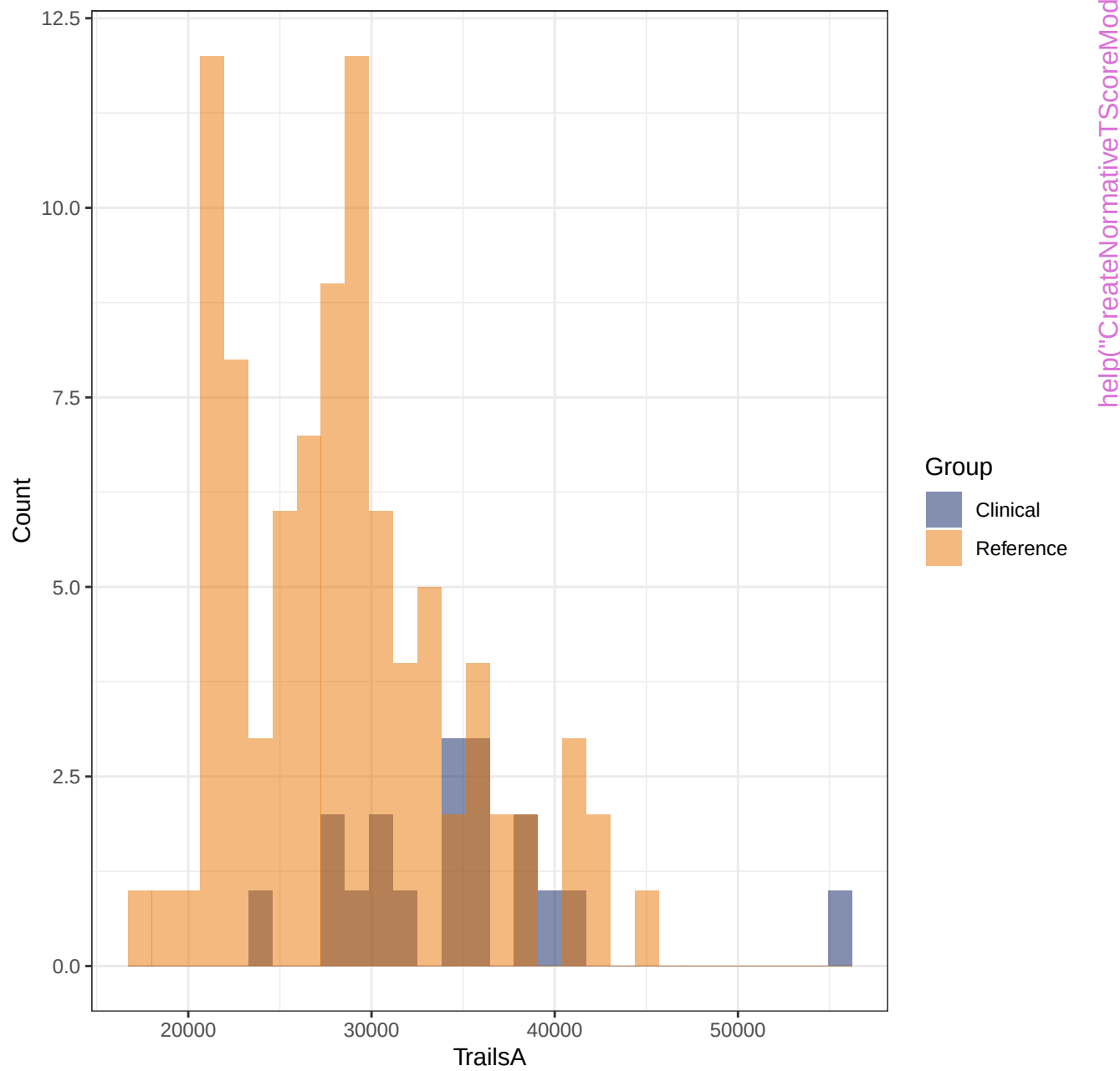
help("ApplyNormativeTScores")

A**B**

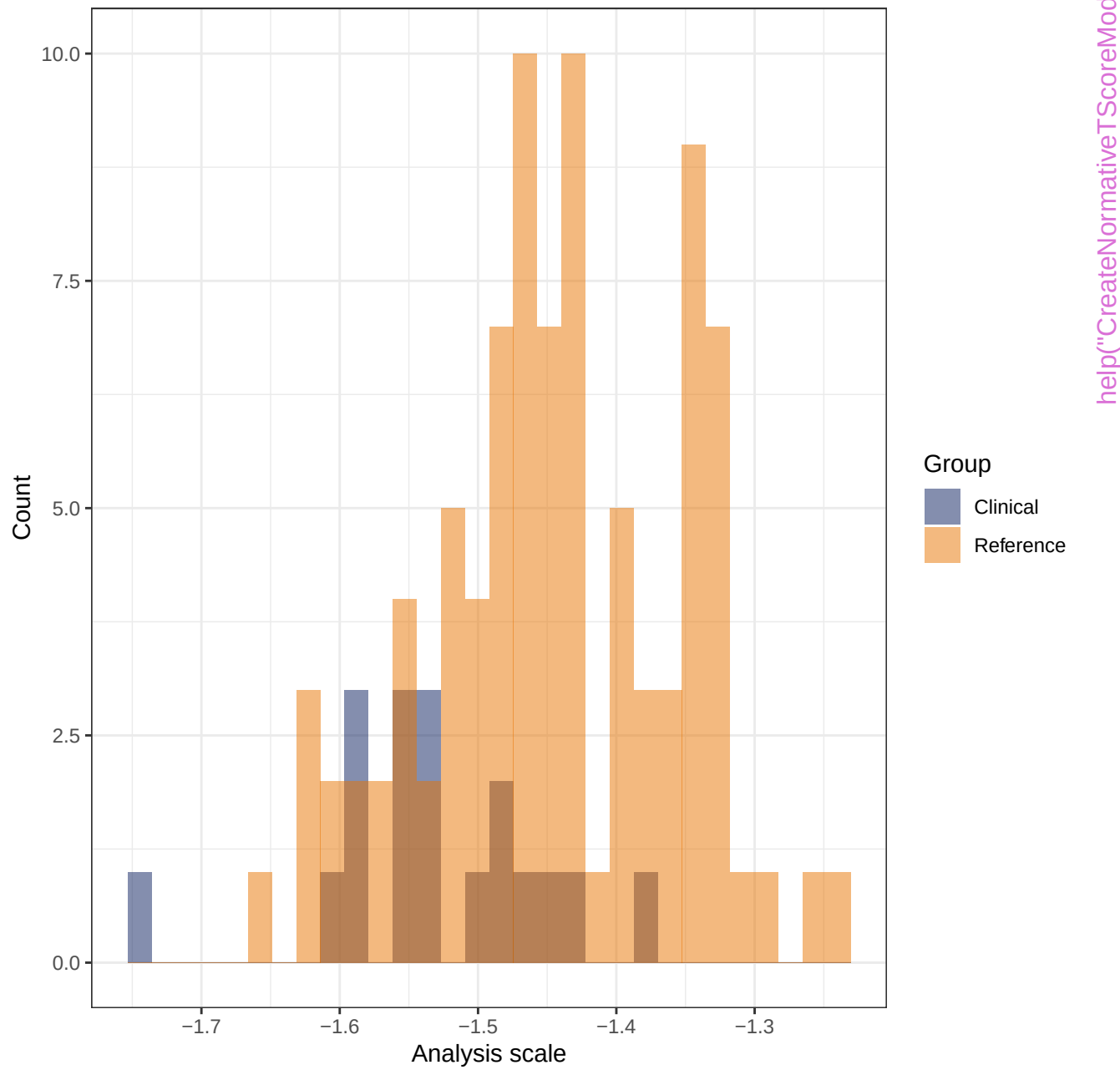




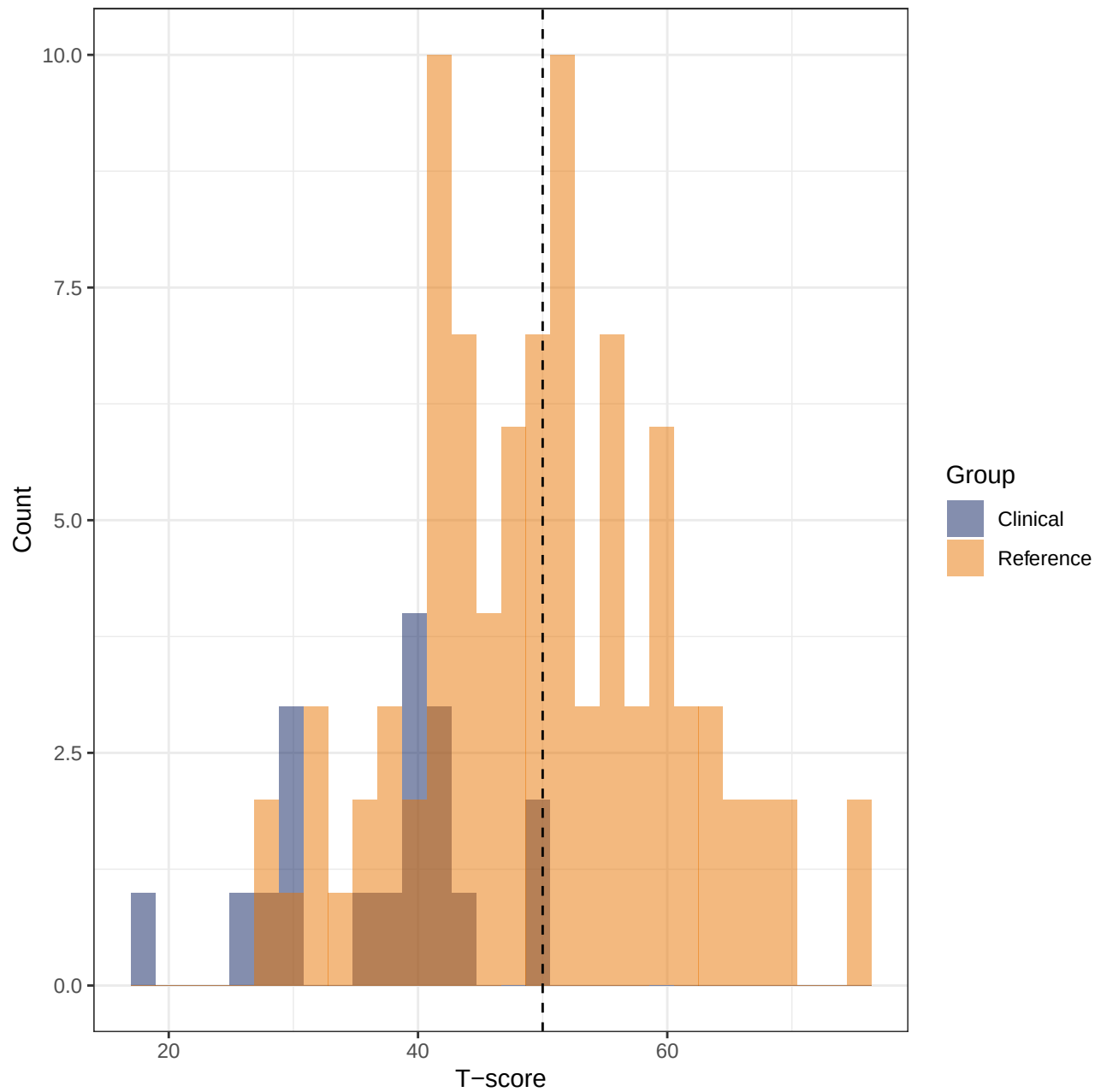
Raw score distribution



Scaled score distribution

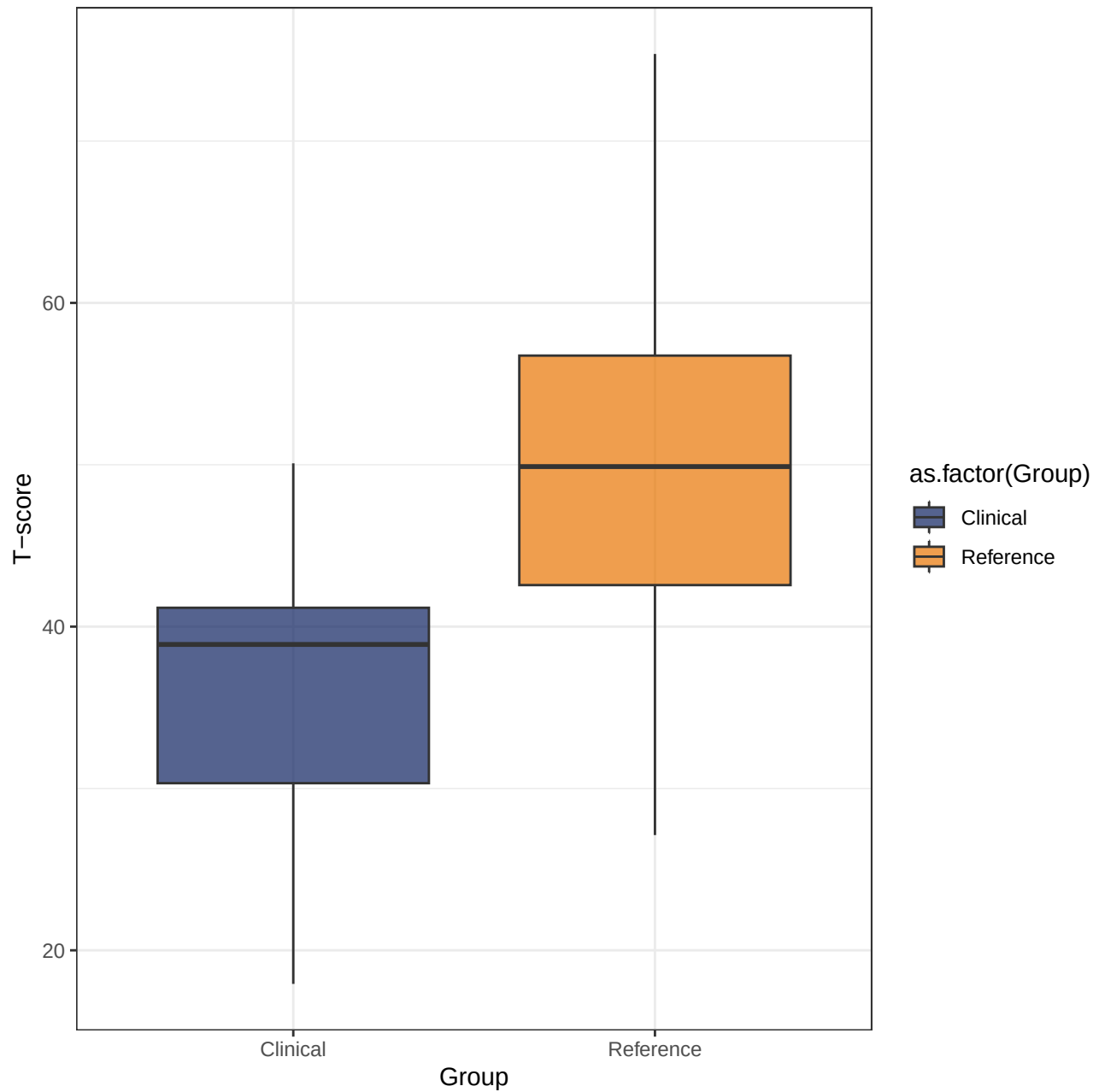


T-score distribution

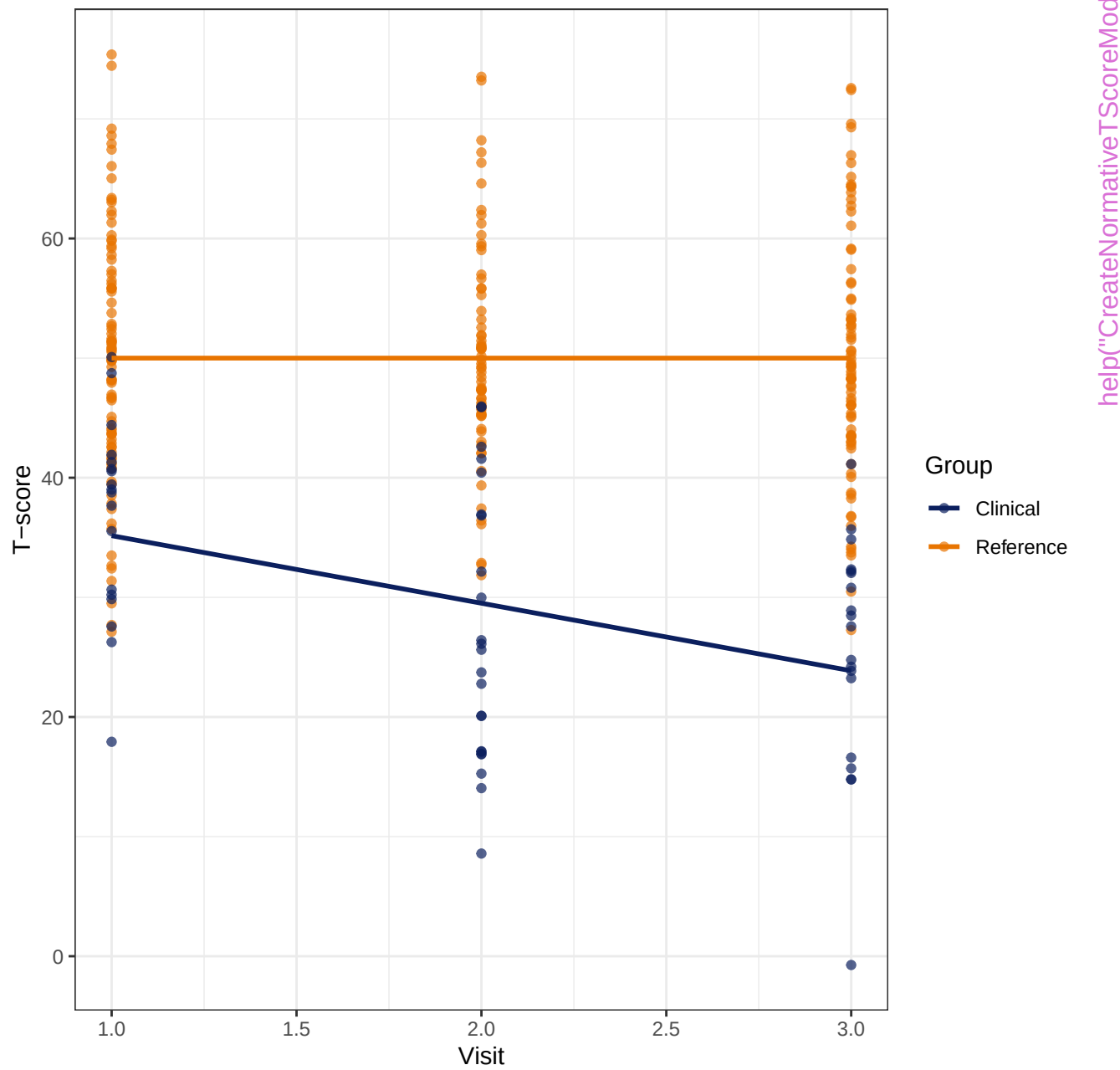


help("CreateNormativeTScoreModel")

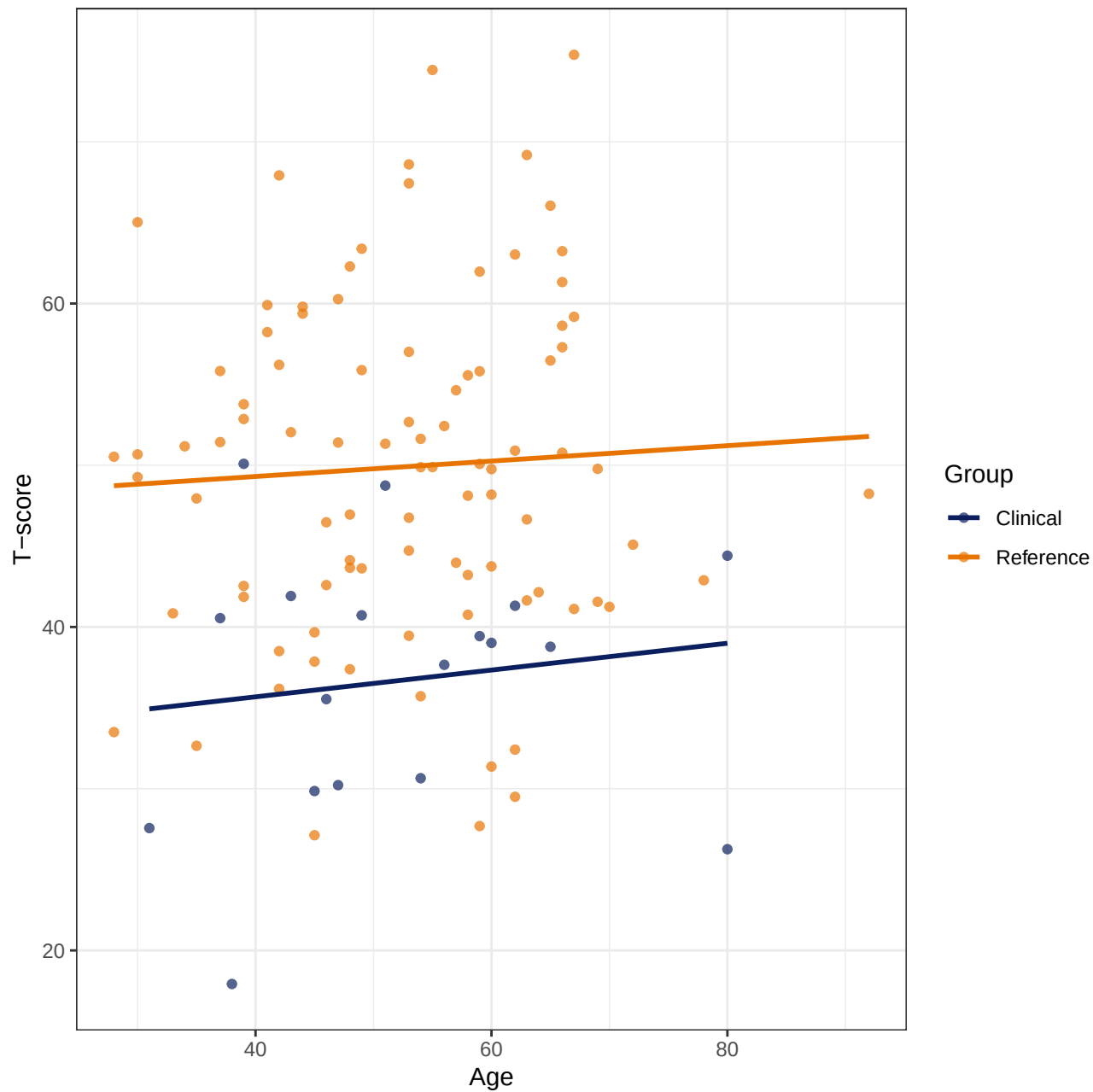
T-scores by reference group



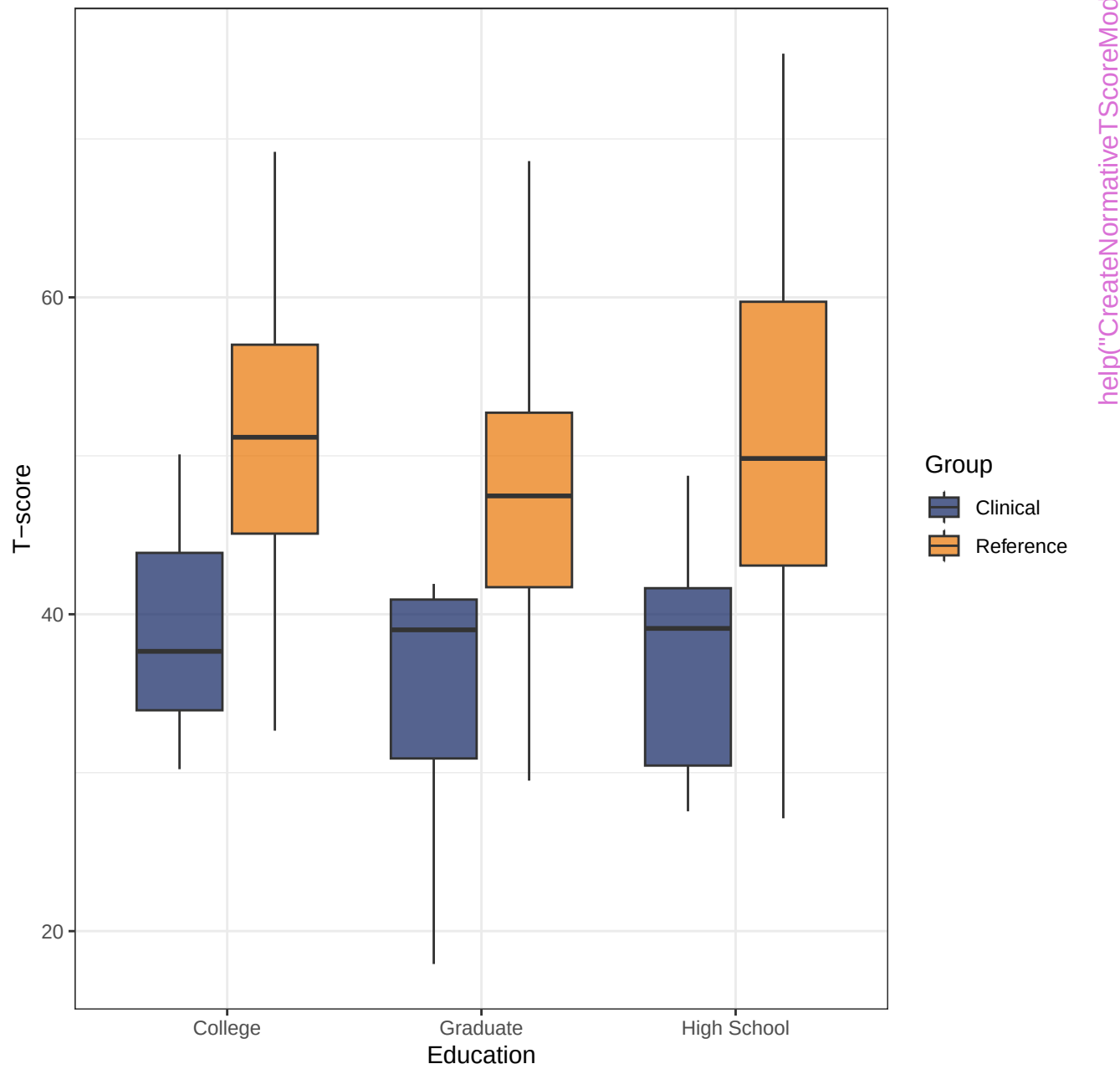
T-scores by visit count



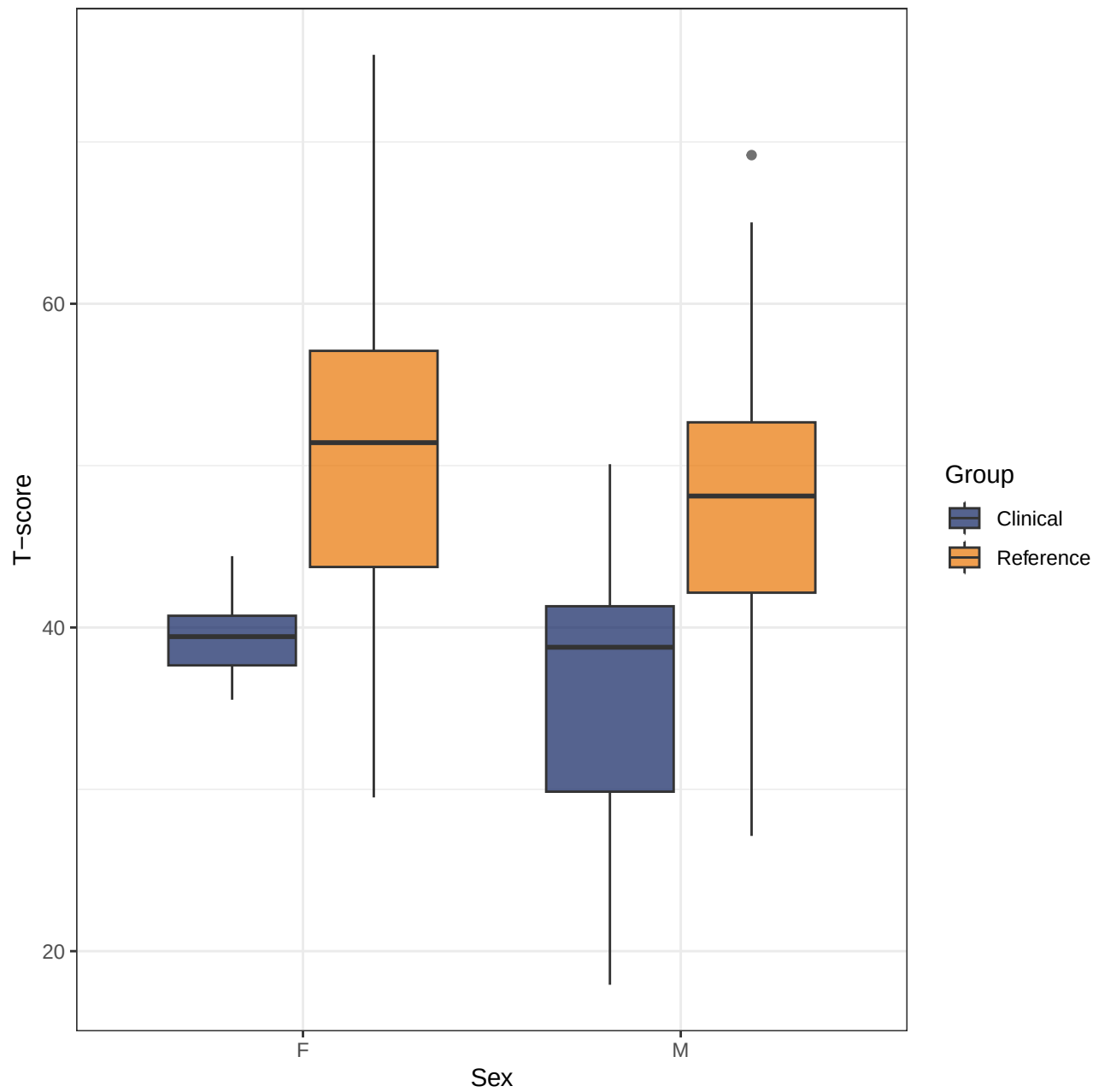
T-scores by Age



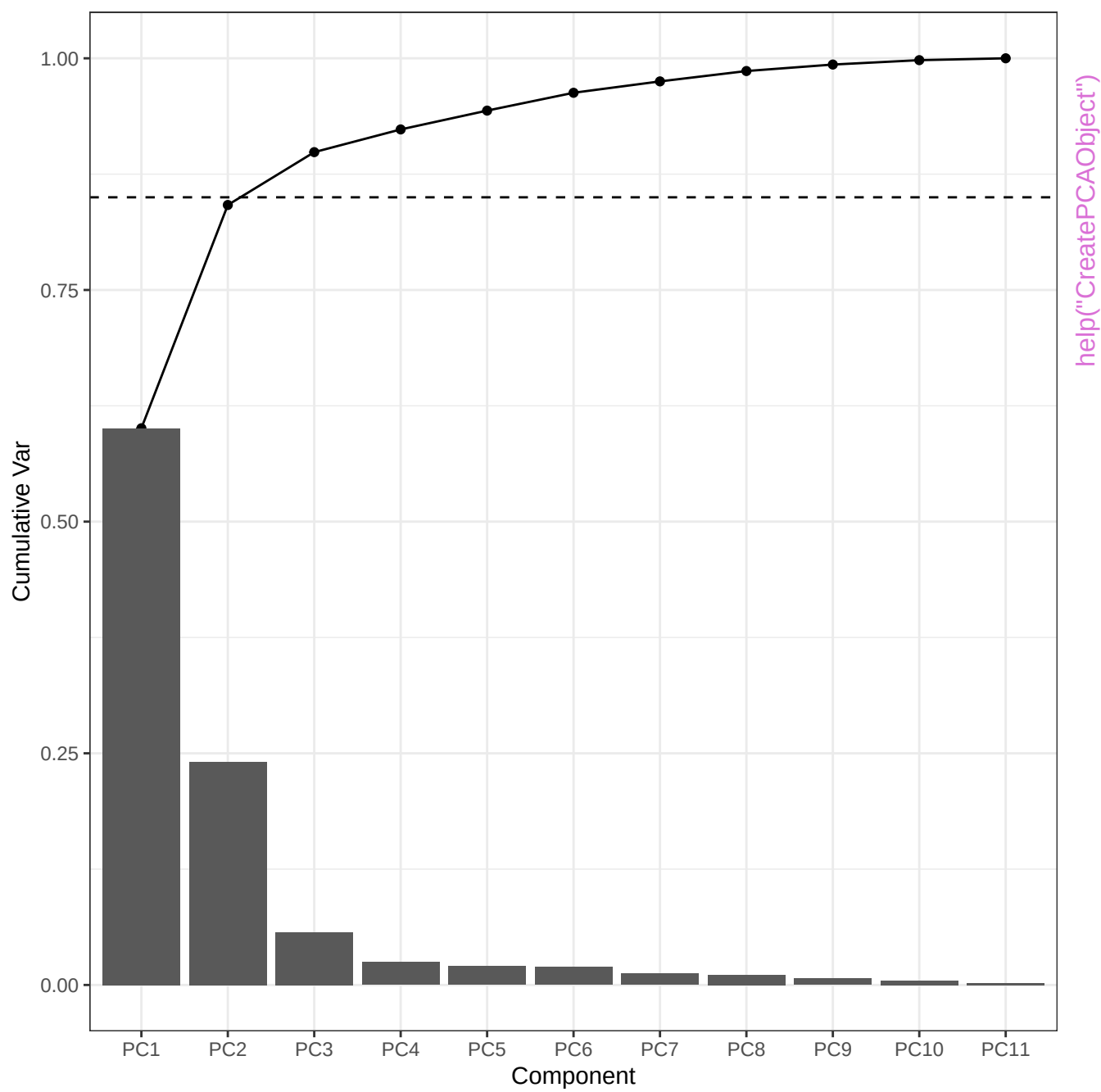
T-scores by Education

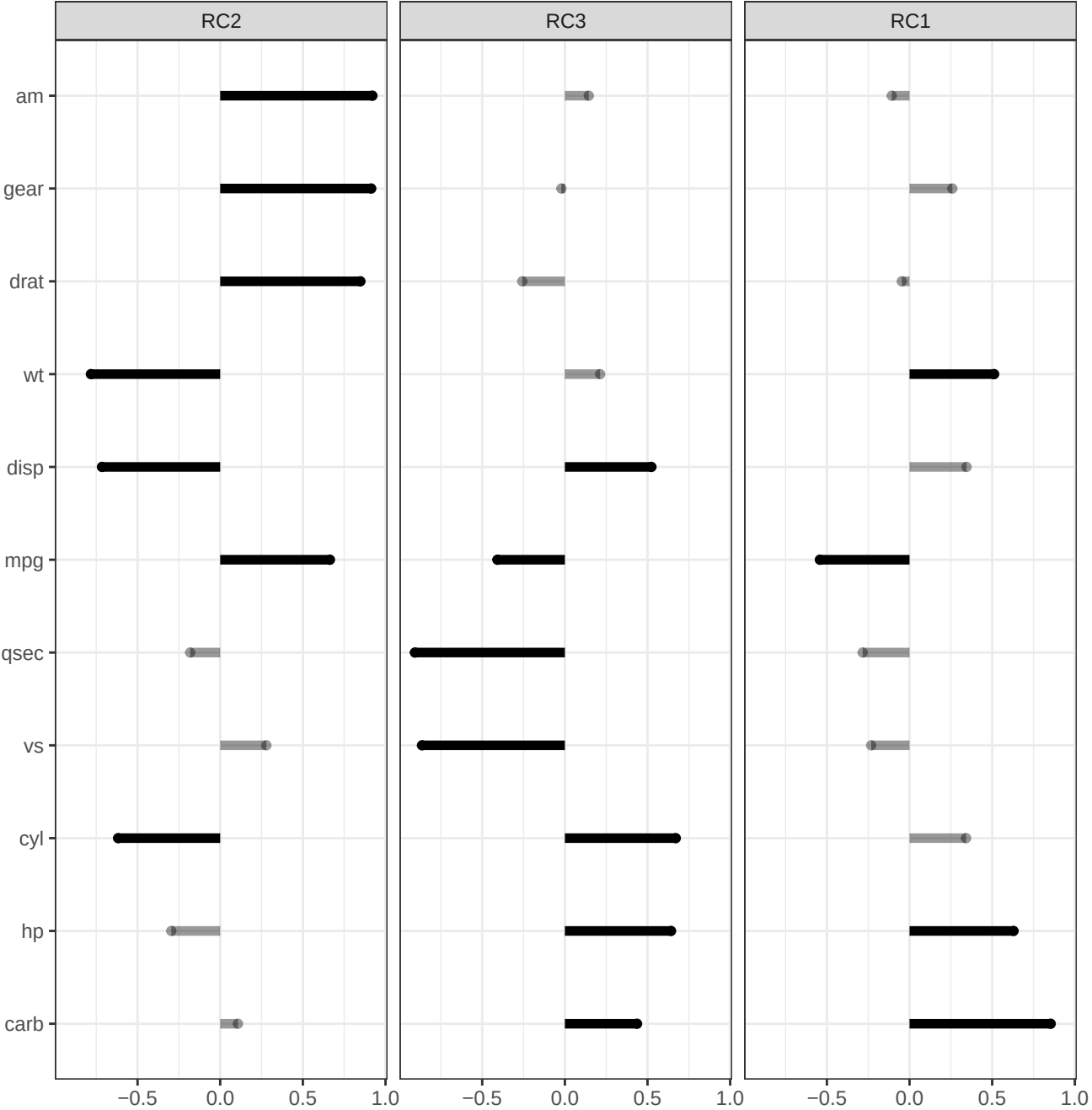


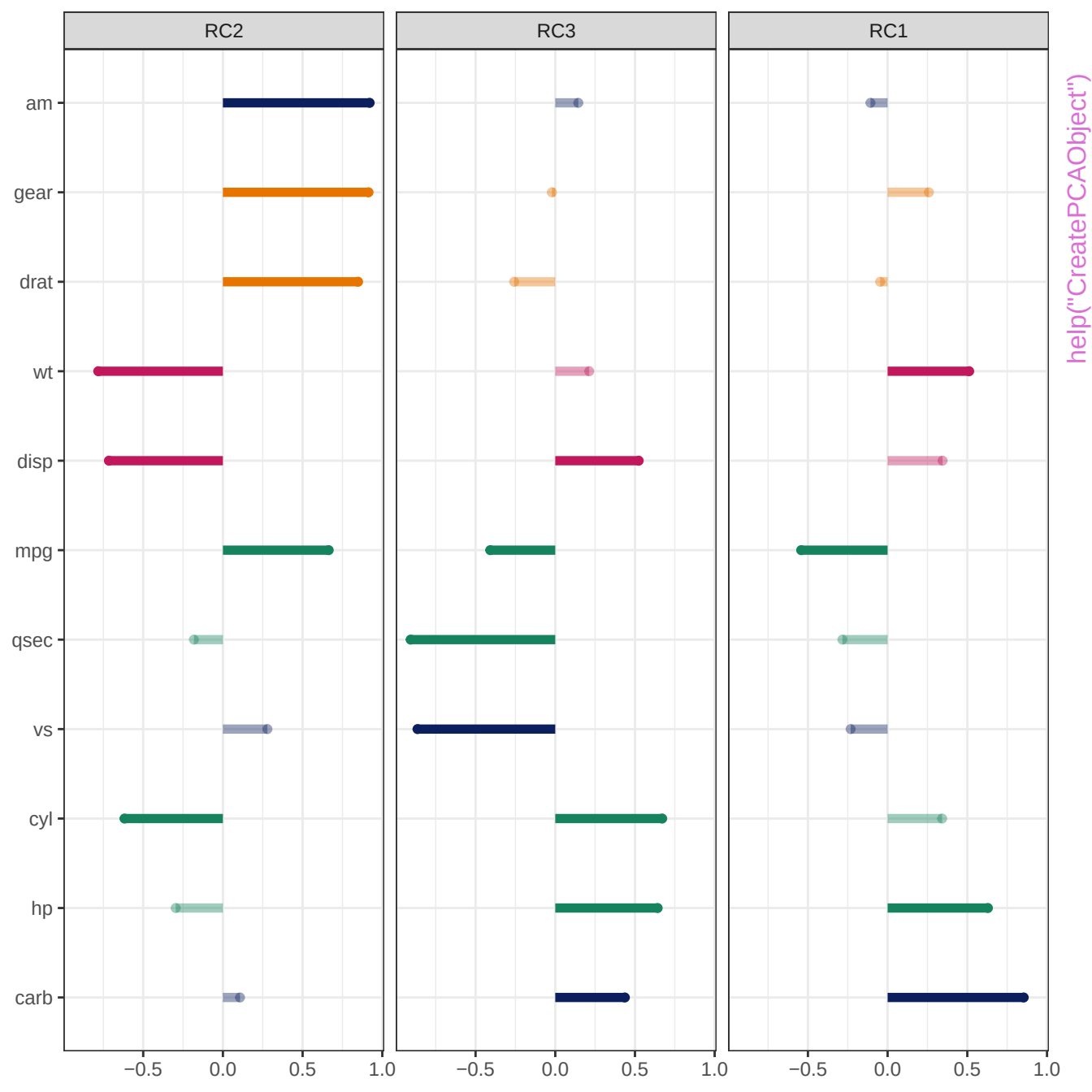
T-scores by Sex



help("CreateNormativeTScoreModel")

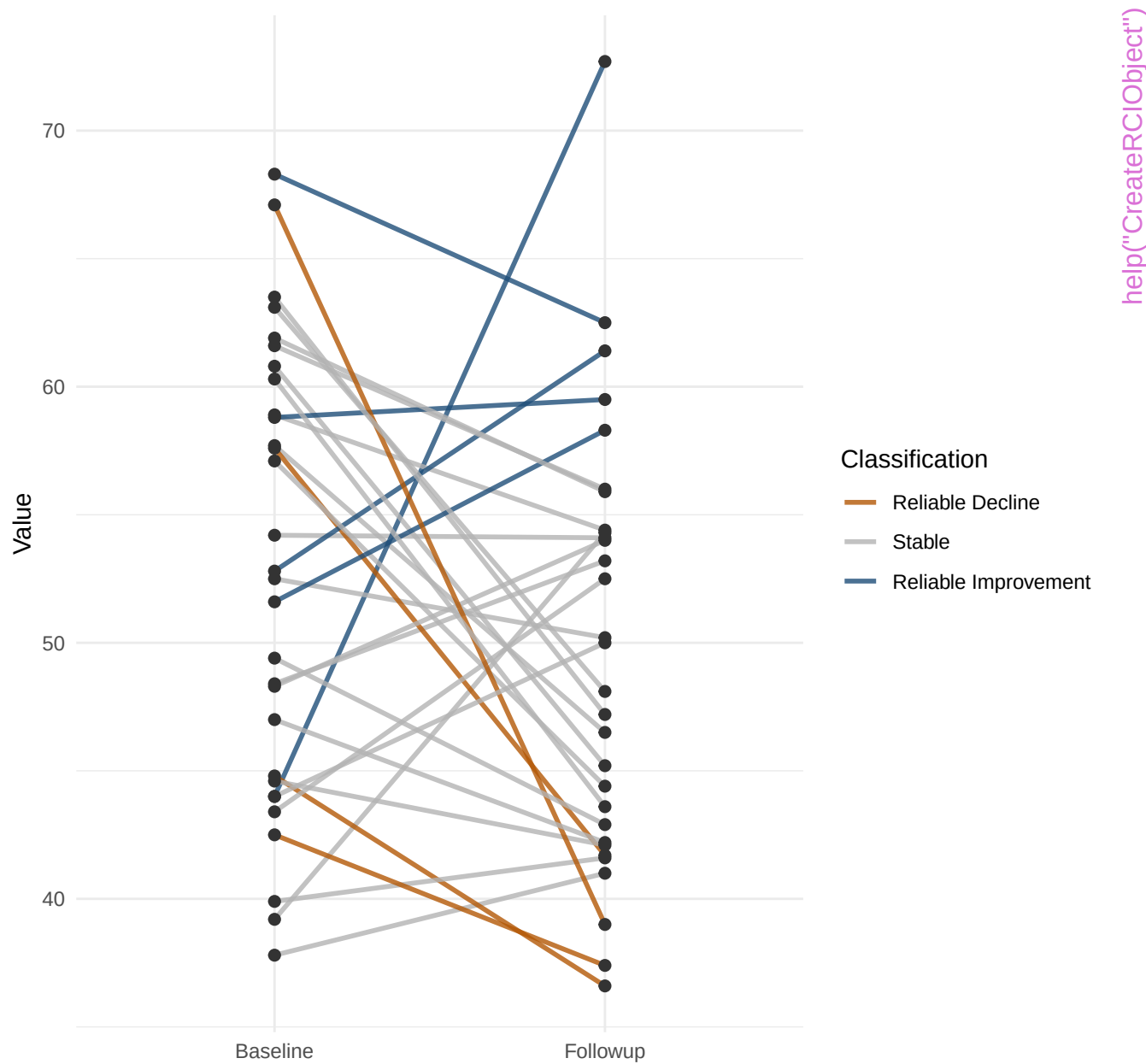






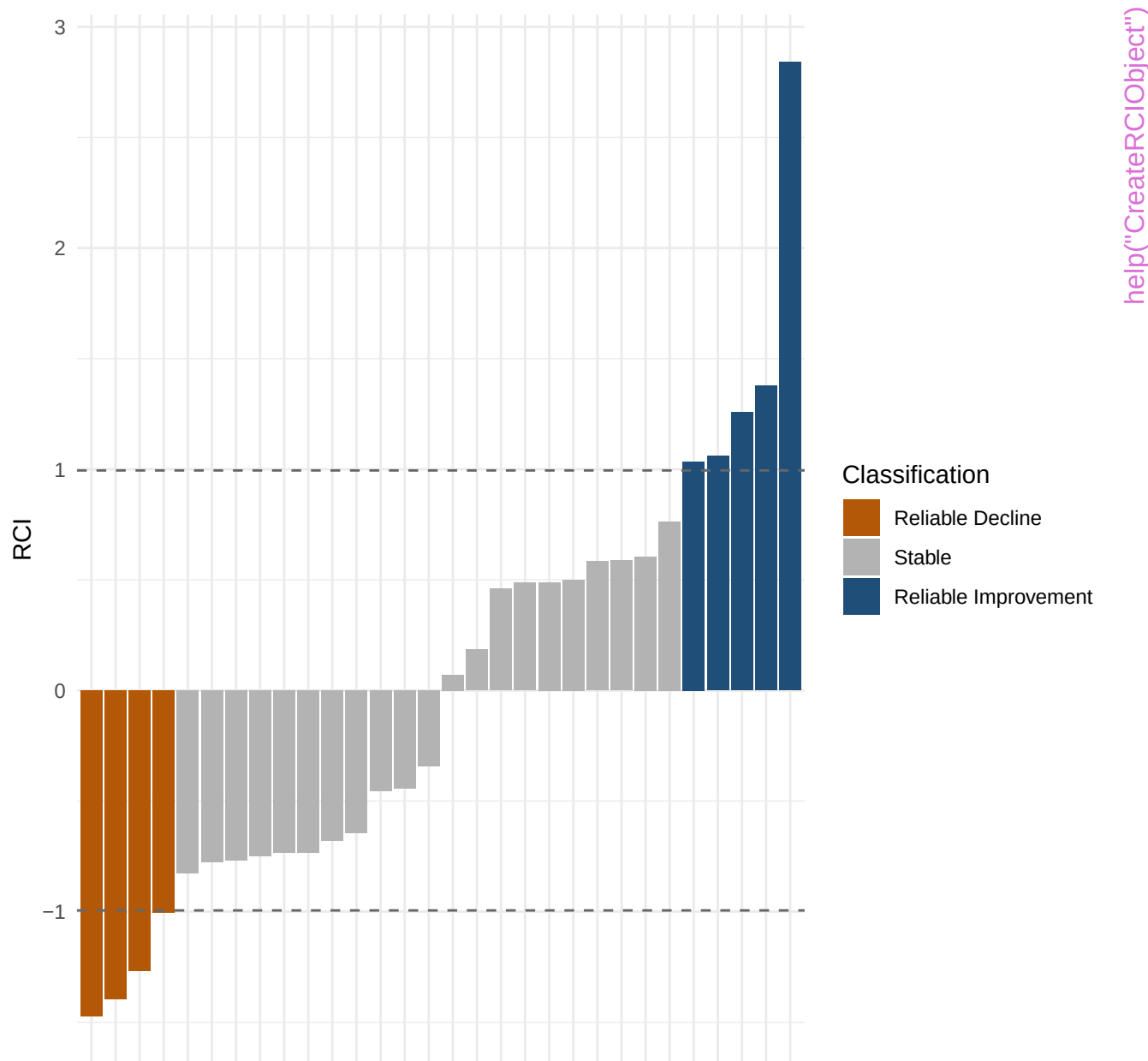
Score

Reference visit: Baseline



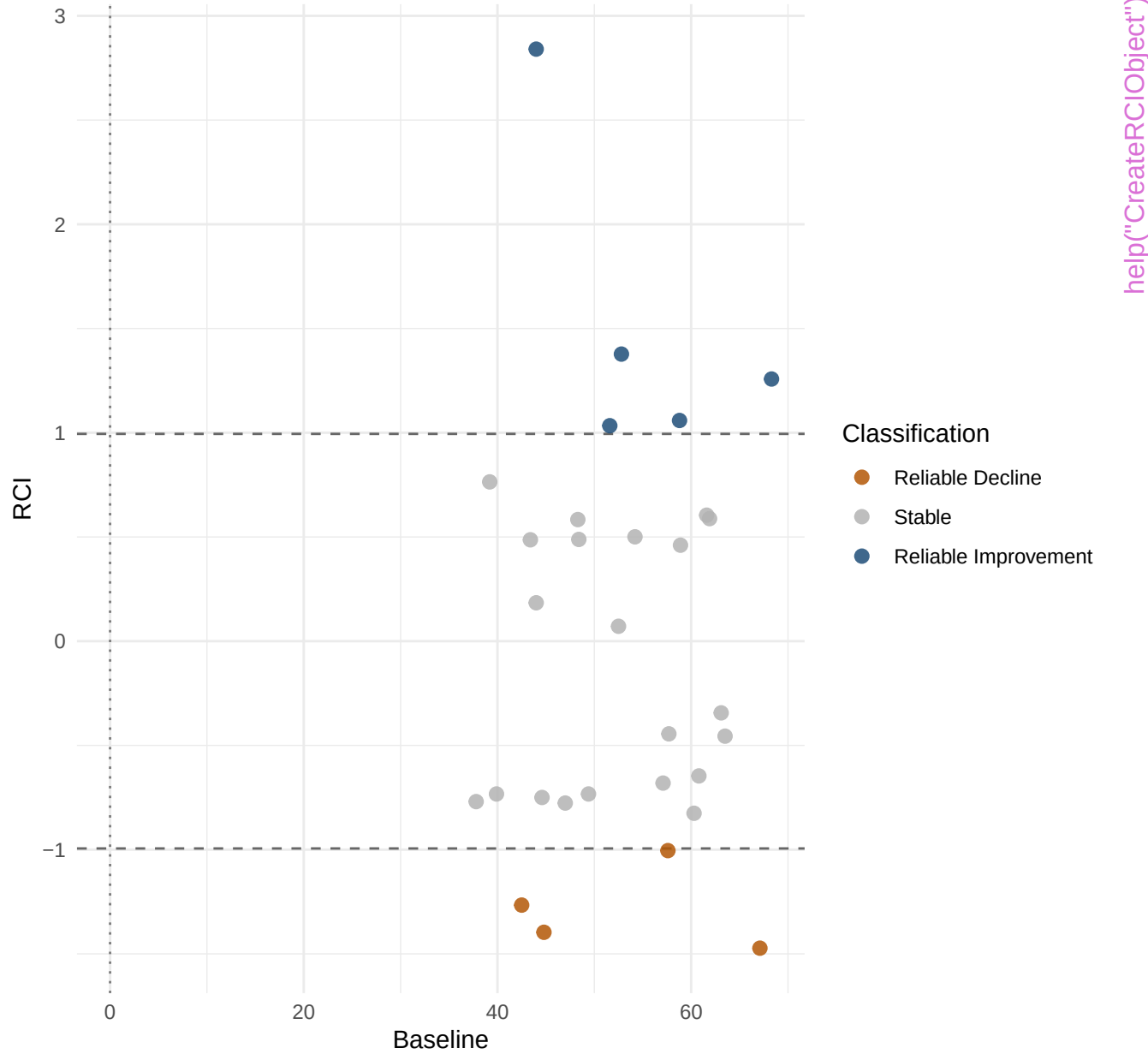
Score

Reference visit: Baseline

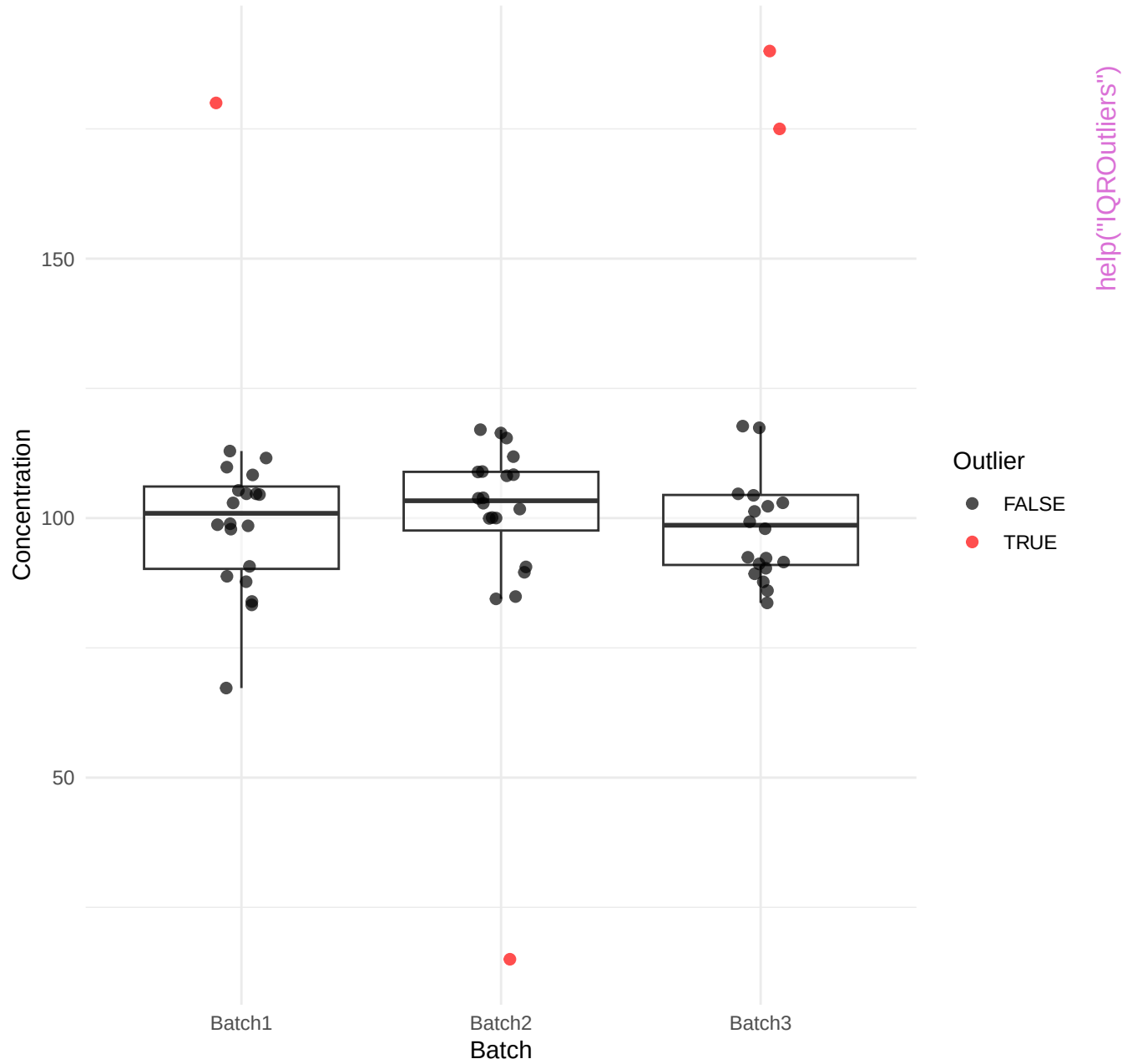


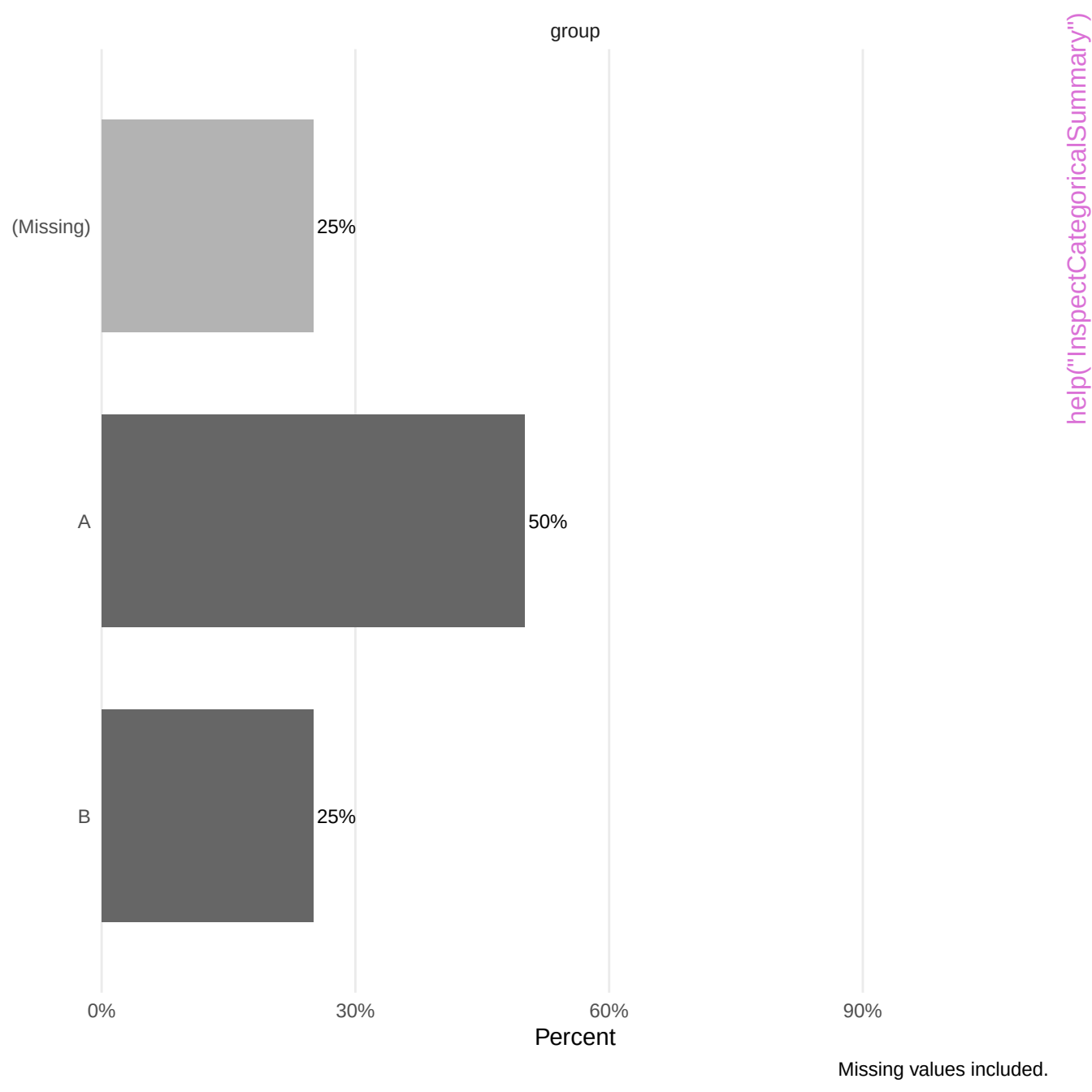
Score

Reference visit: Baseline

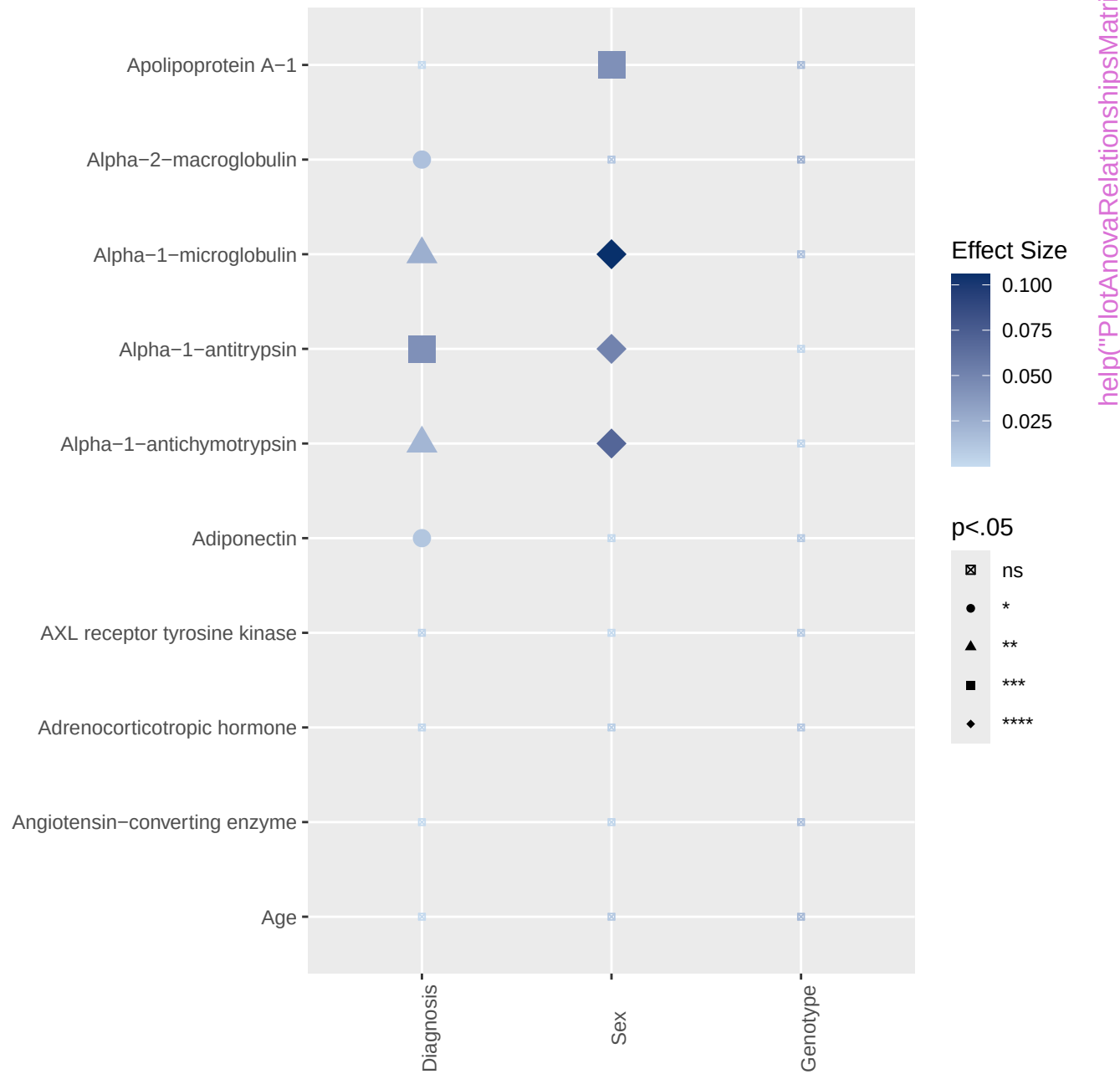


IQR Outlier Detection for Concentration

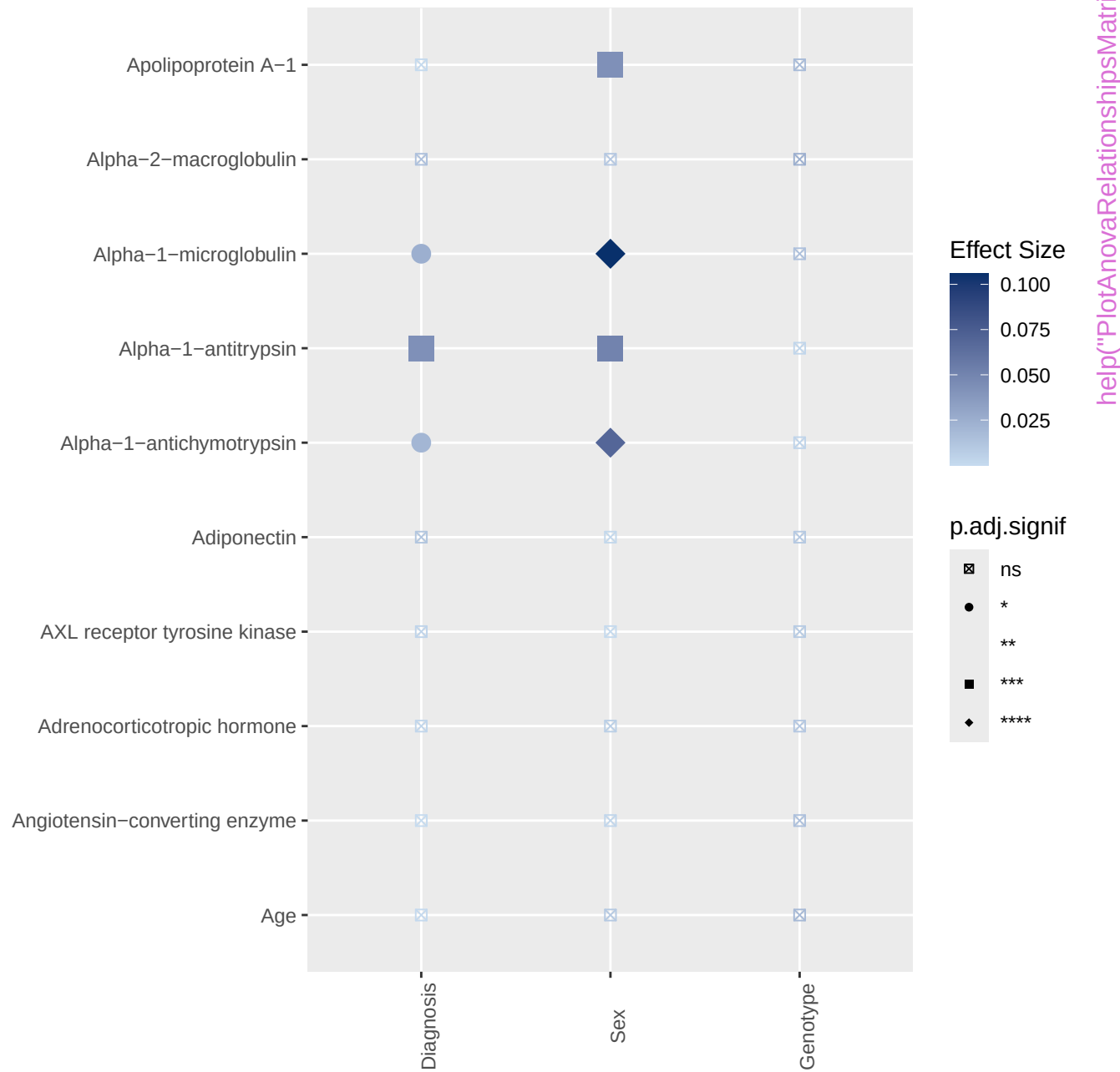




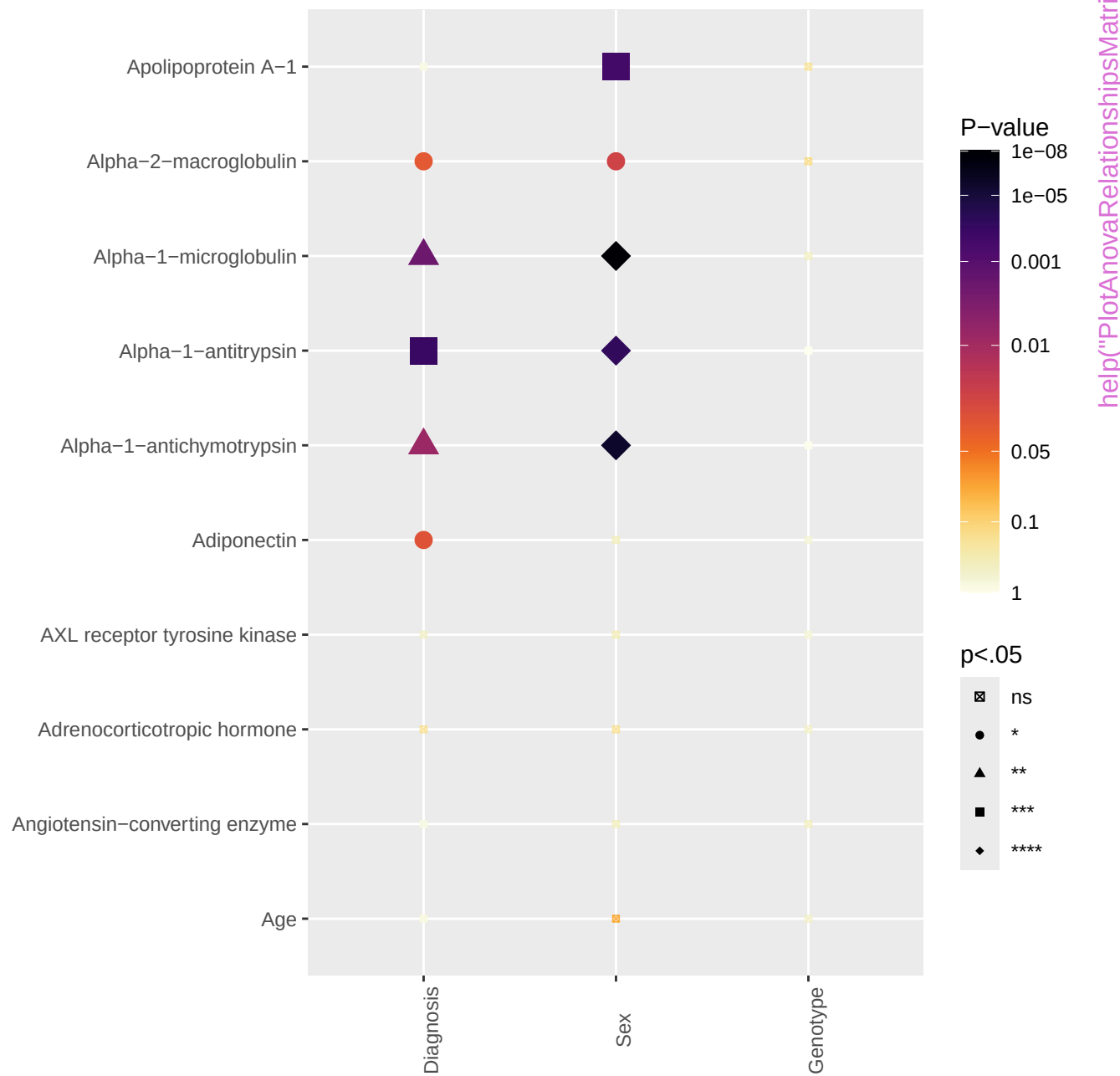
No Multiple Comparison Correction



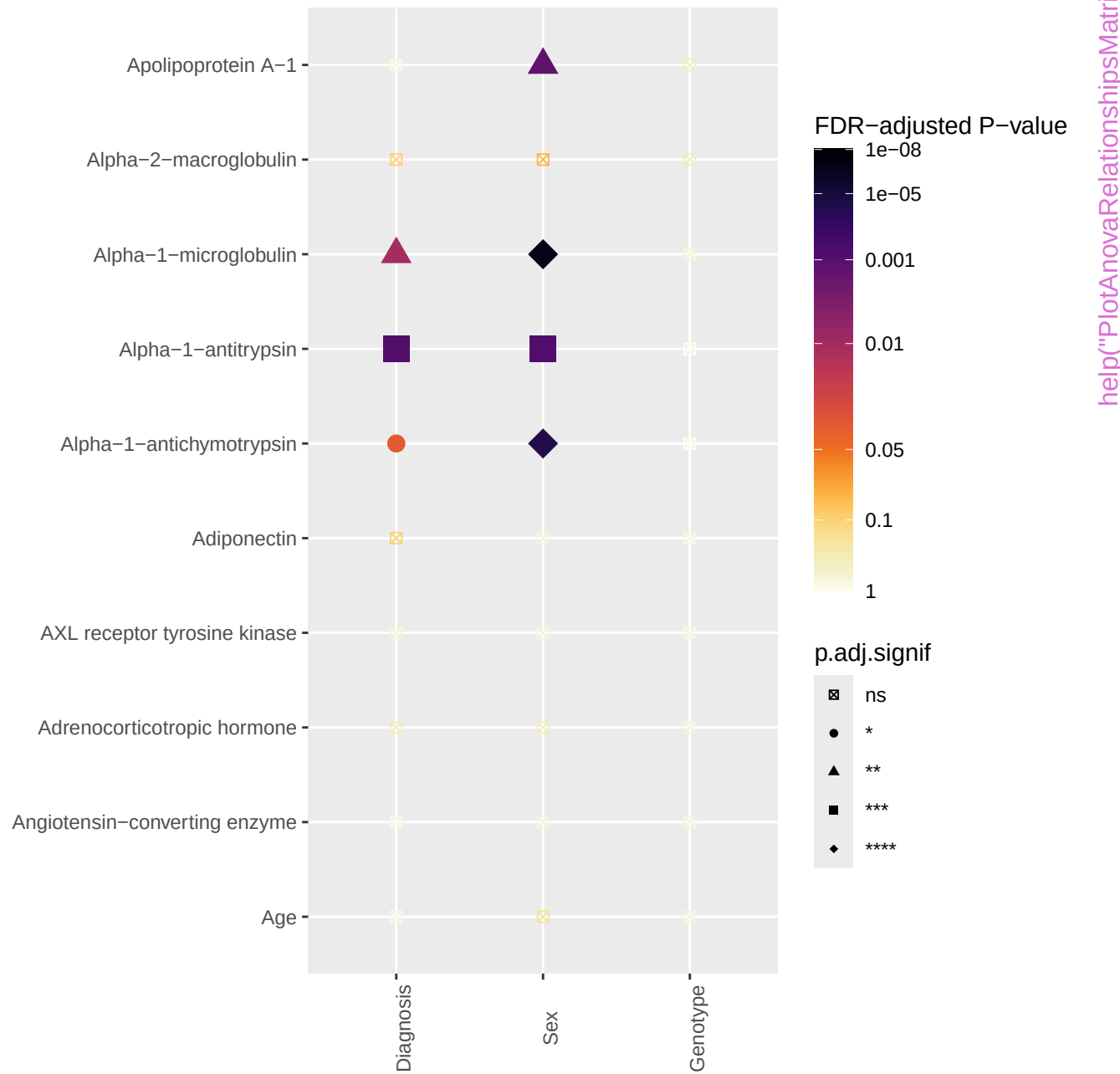
FDR Correction



No Multiple Comparison Correction



FDR Correction



FDR Correction

Alpha-1-antitrypsin

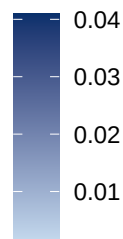
Adiponectin

AXL receptor tyrosine kinase

Diagnosis

Sex

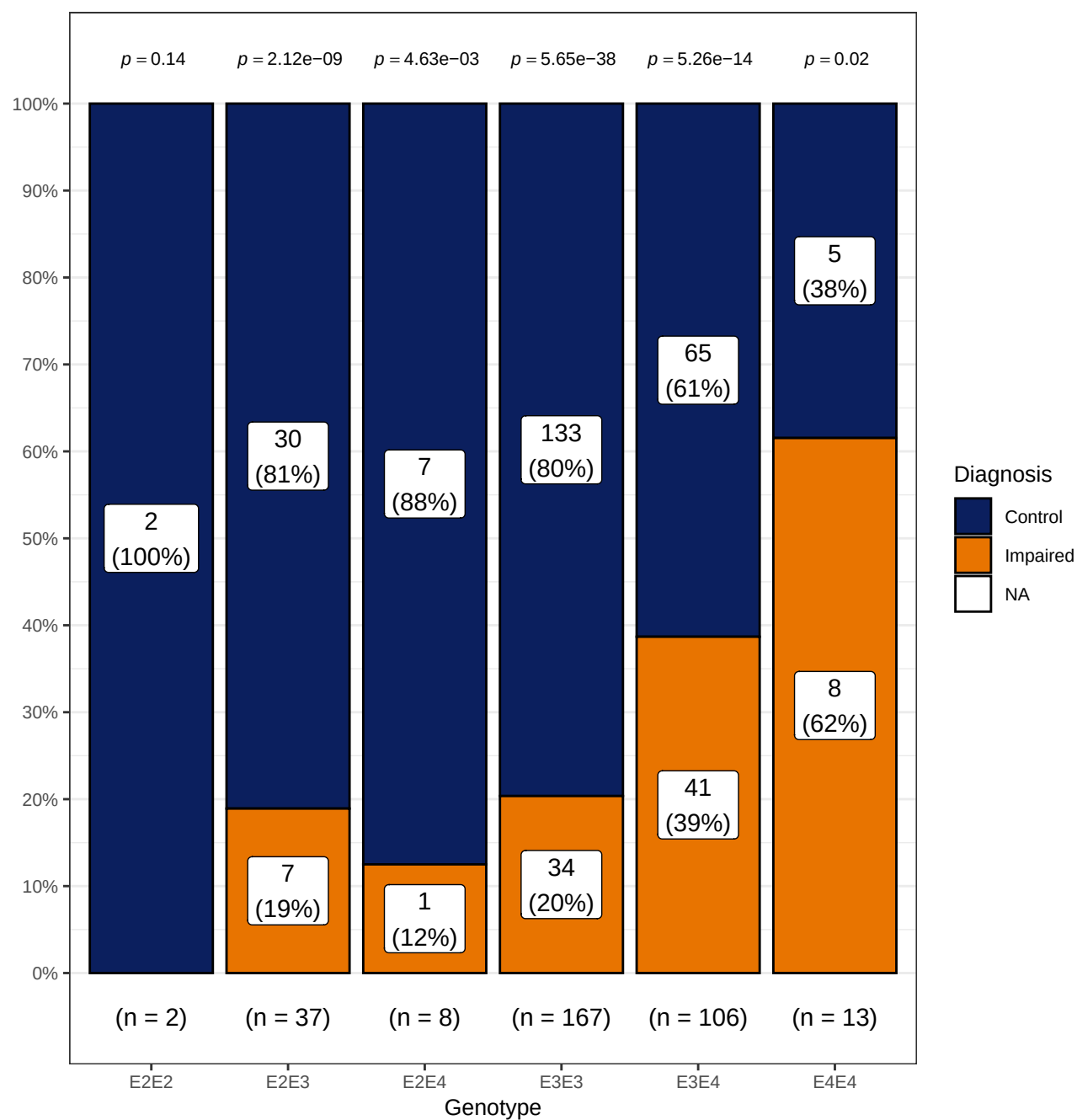
Effect Size



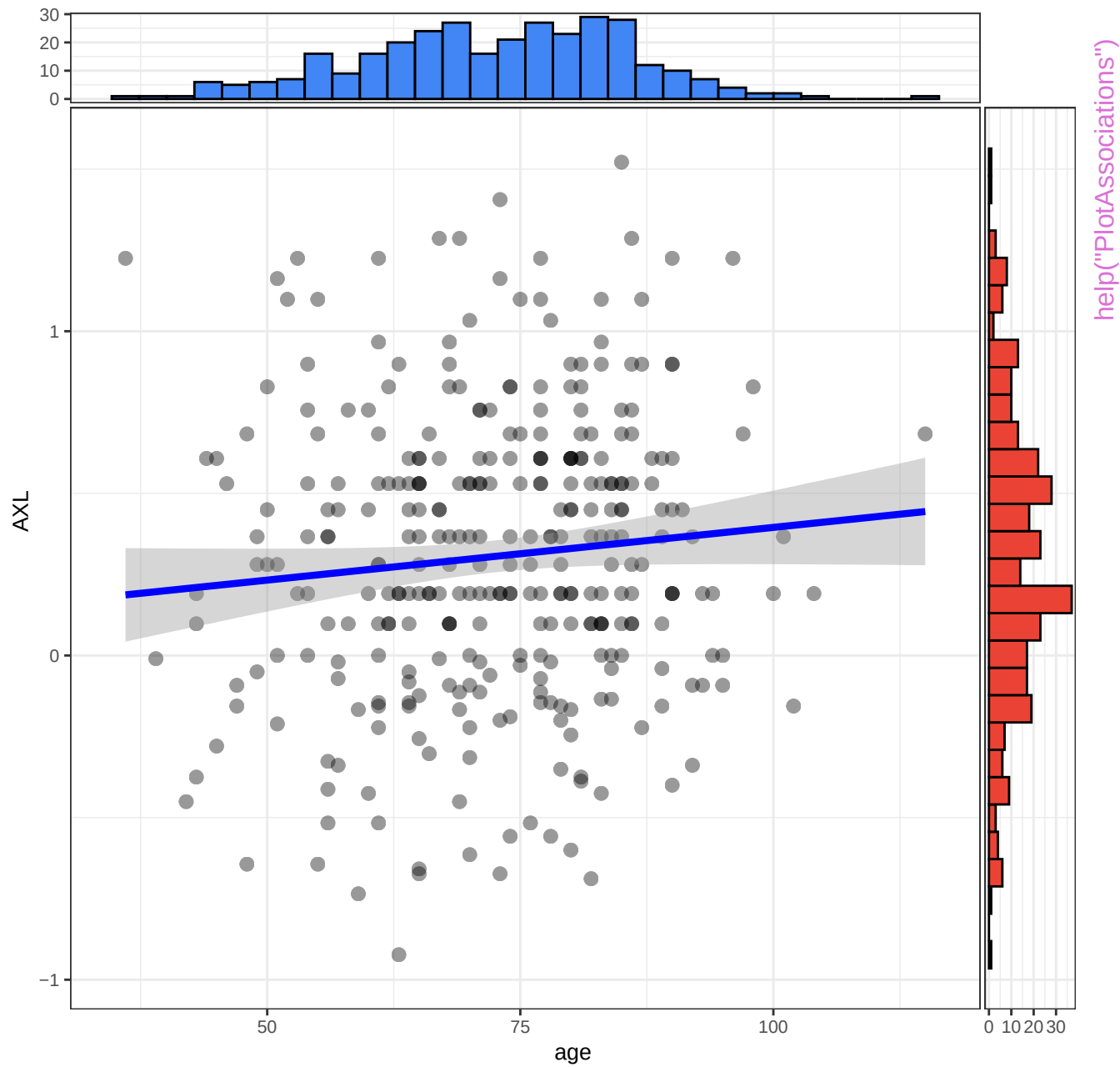
p.adj.signif



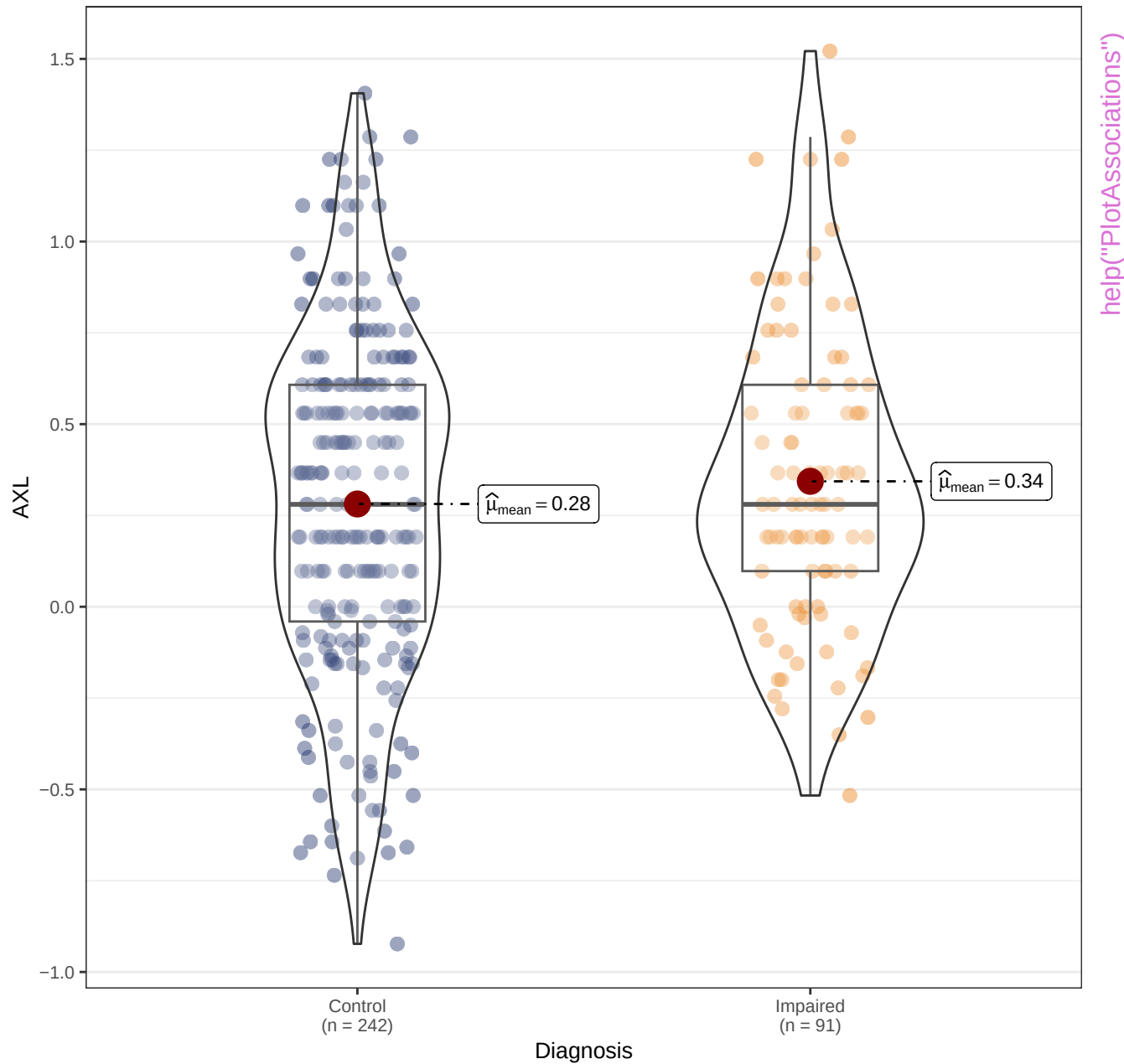
help("PlotAnovaRelationshipsMatrix")



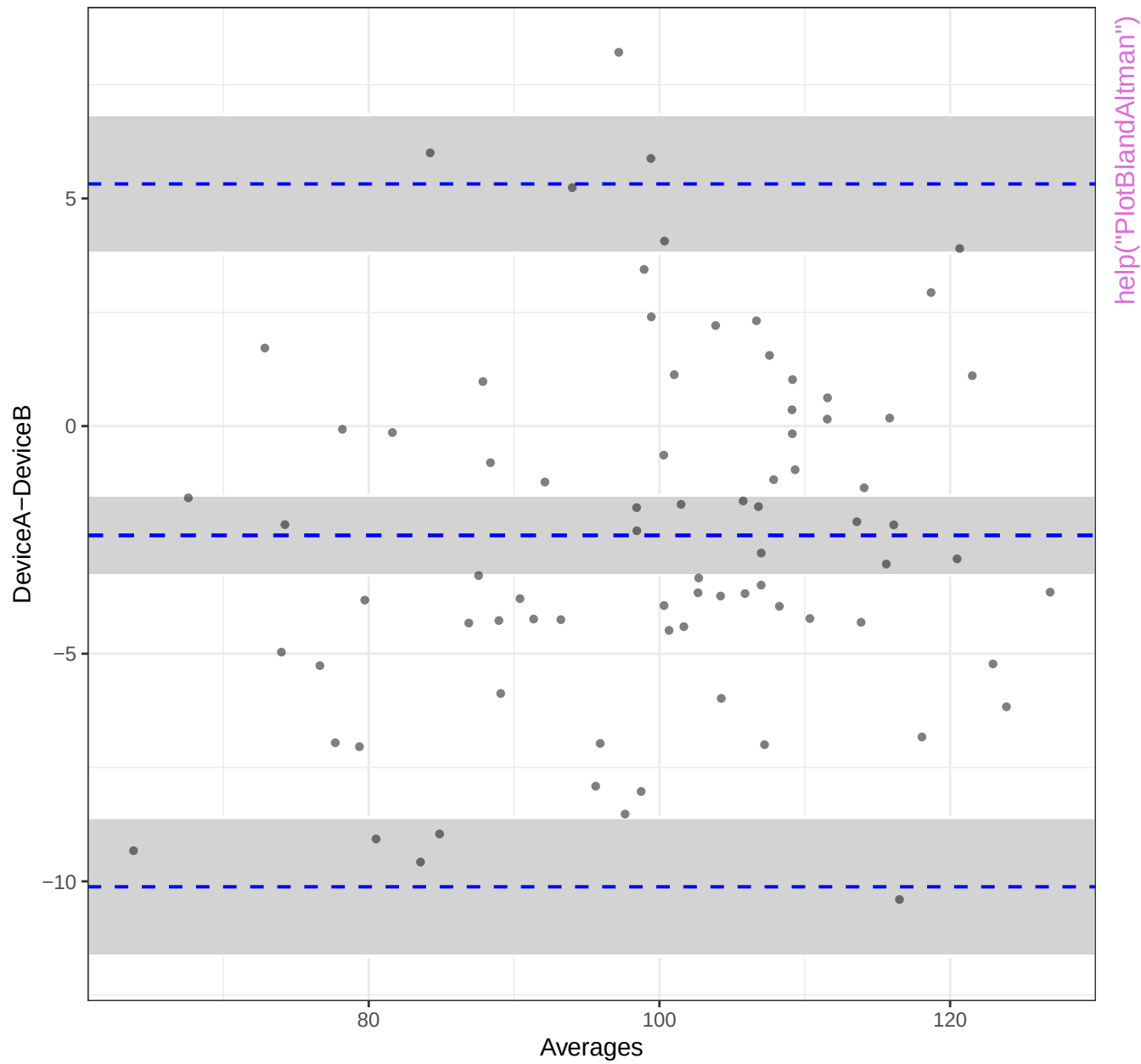
$t_{\text{Student}}(320) = 1.72$, $p = 0.09$, $\hat{r}_{\text{Pearson}} = 0.10$, $\text{CI}_{95\%} [-0.01, 0.20]$, $n_{\text{pairs}} = 322$



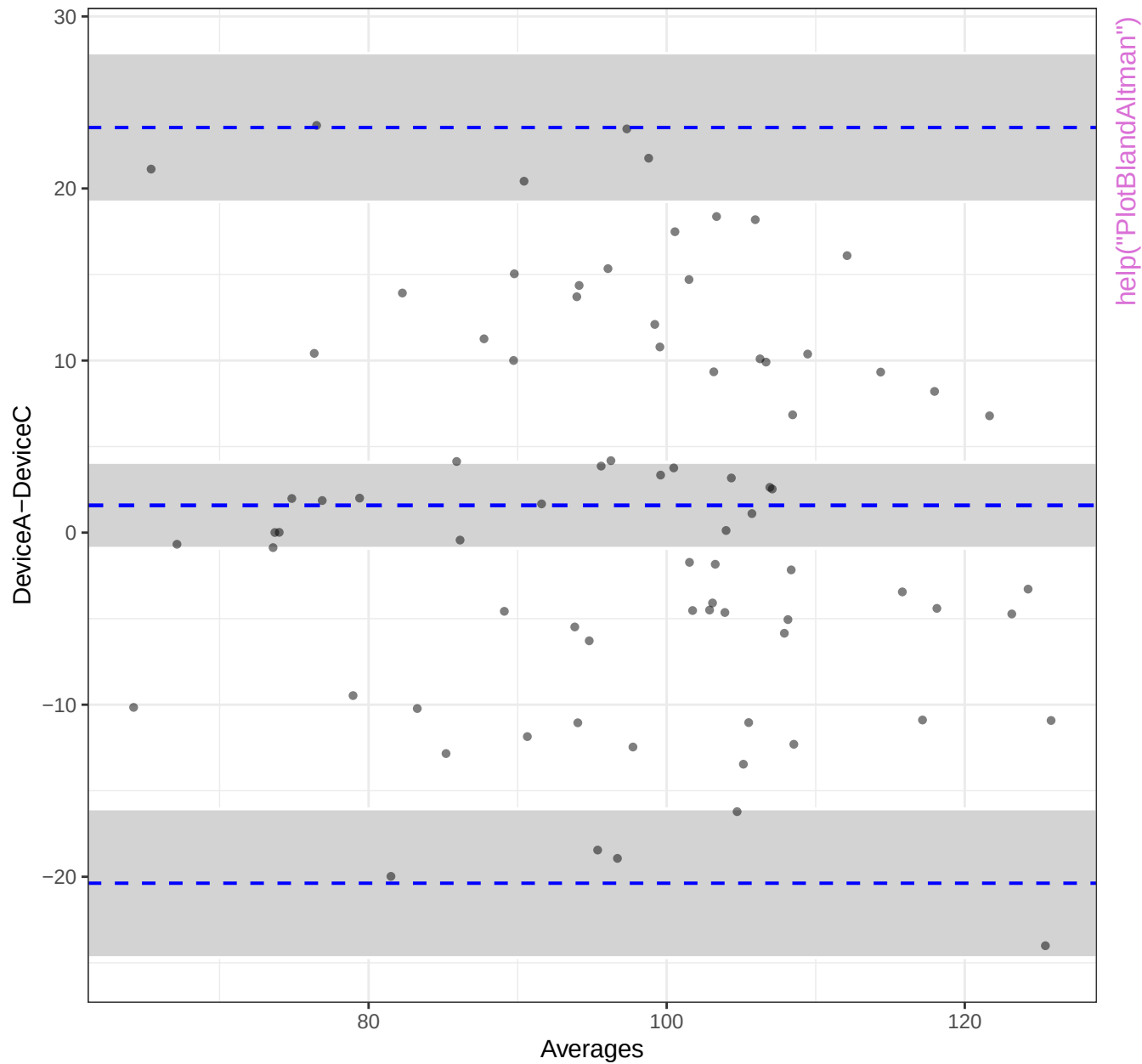
$t_{\text{Welch}}(179.77) = -1.18, p = 0.24, \hat{g}_{\text{Hedges}} = -0.14, \text{CI}_{95\%} [-0.38, 0.09], n_{\text{obs}} = 333$



Mean Difference = -2.4
Limits of Agreement: [-10.12 , 5.32]

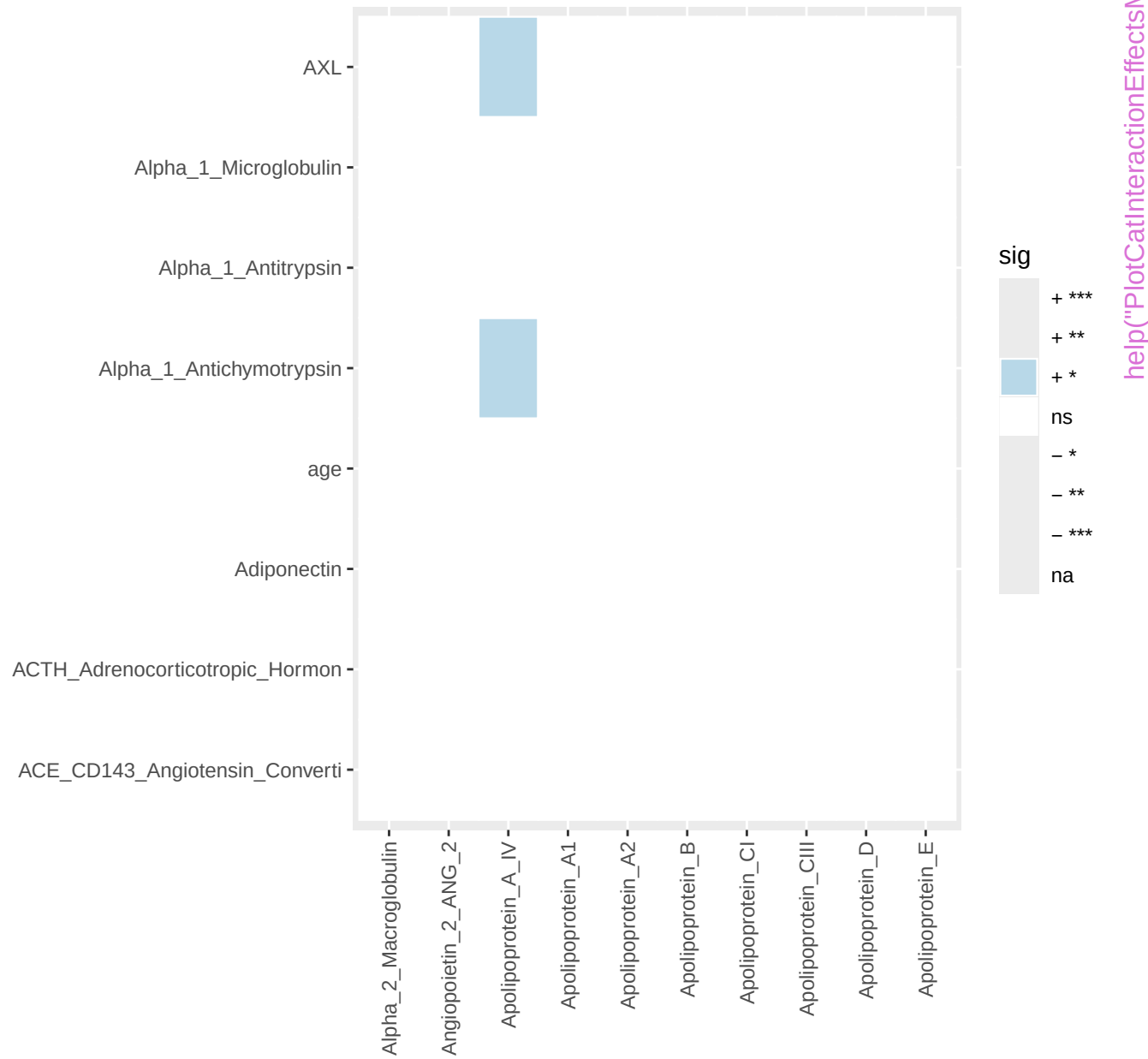


Mean Difference = 1.58
Limits of Agreement: [-20.37 , 23.54]



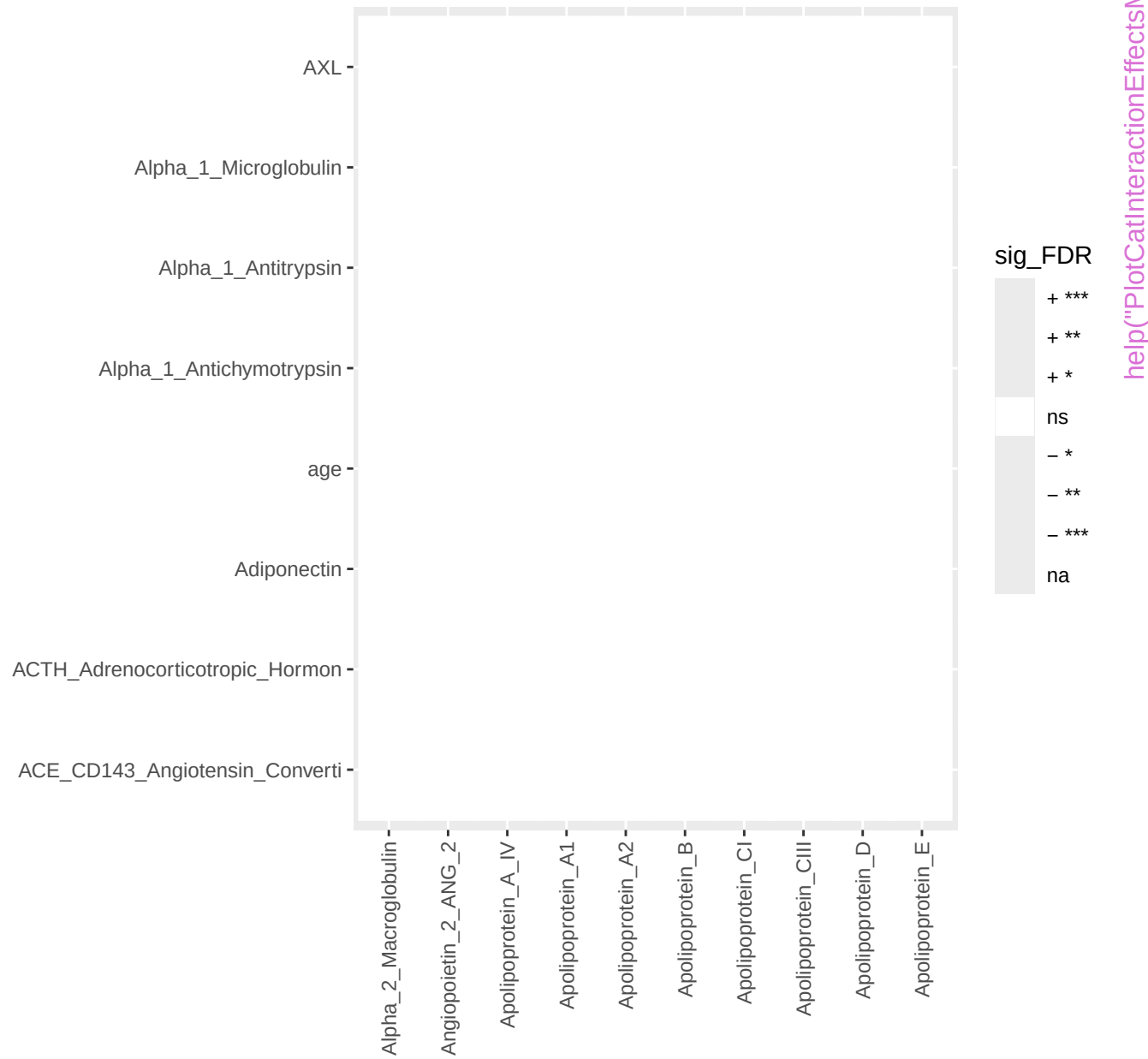
Diagnosis

No Multiple Comparison Correction



Diagnosis

FDR Correction



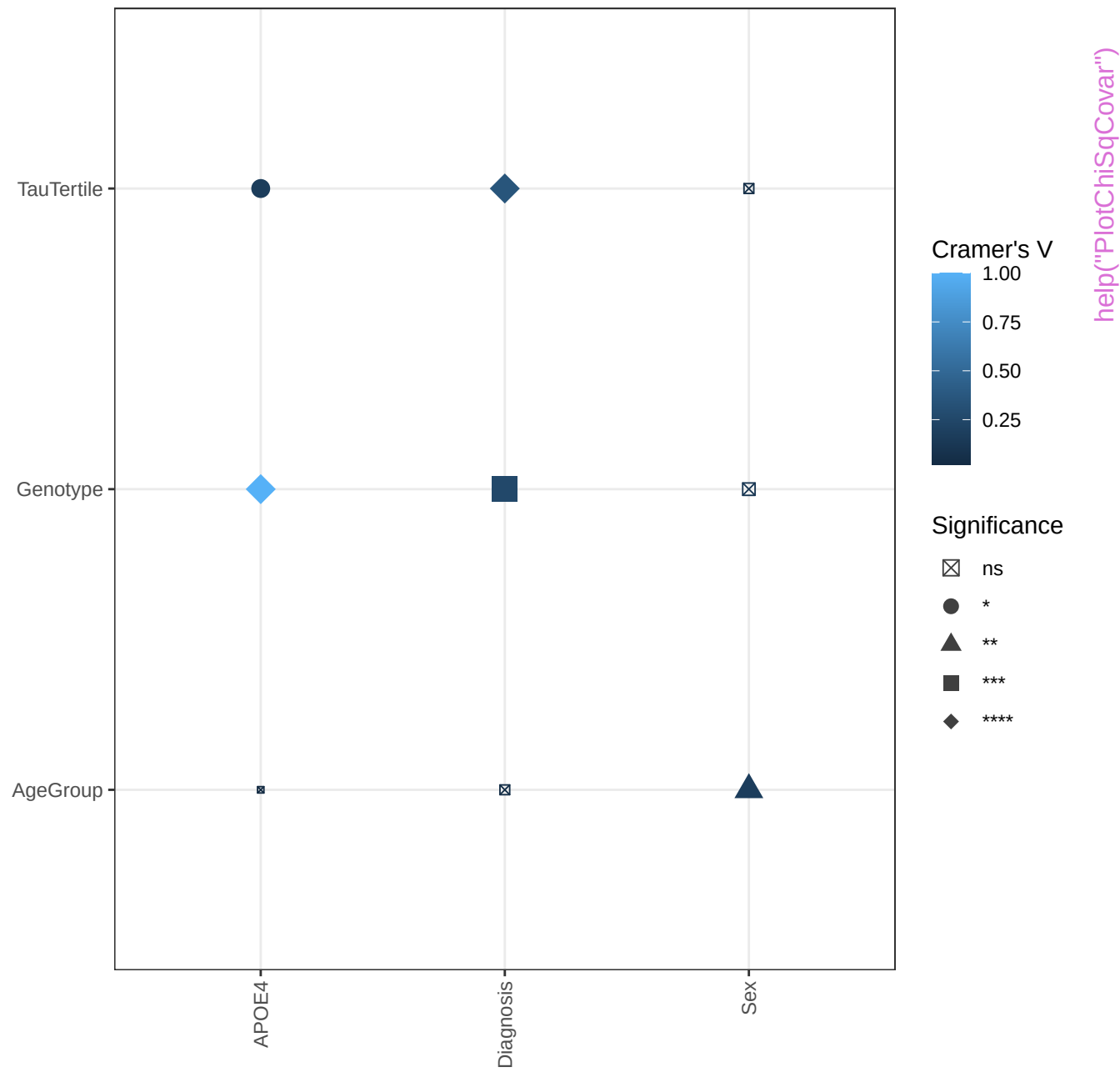
Genotype

Diagnosis



help("PlotCategoricalDistributions")

No multiple comparison correction



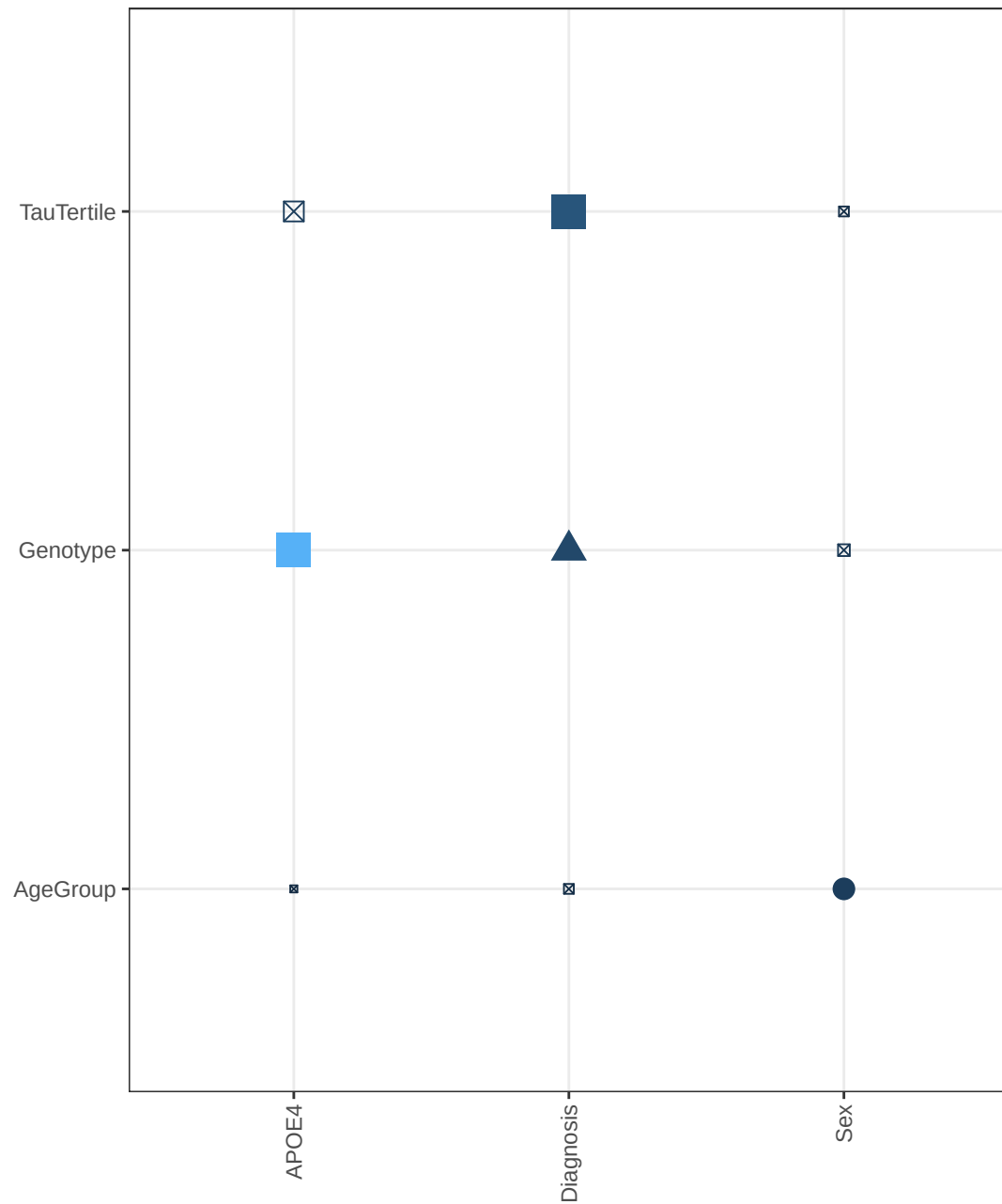
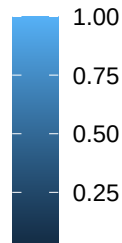
FDR correction

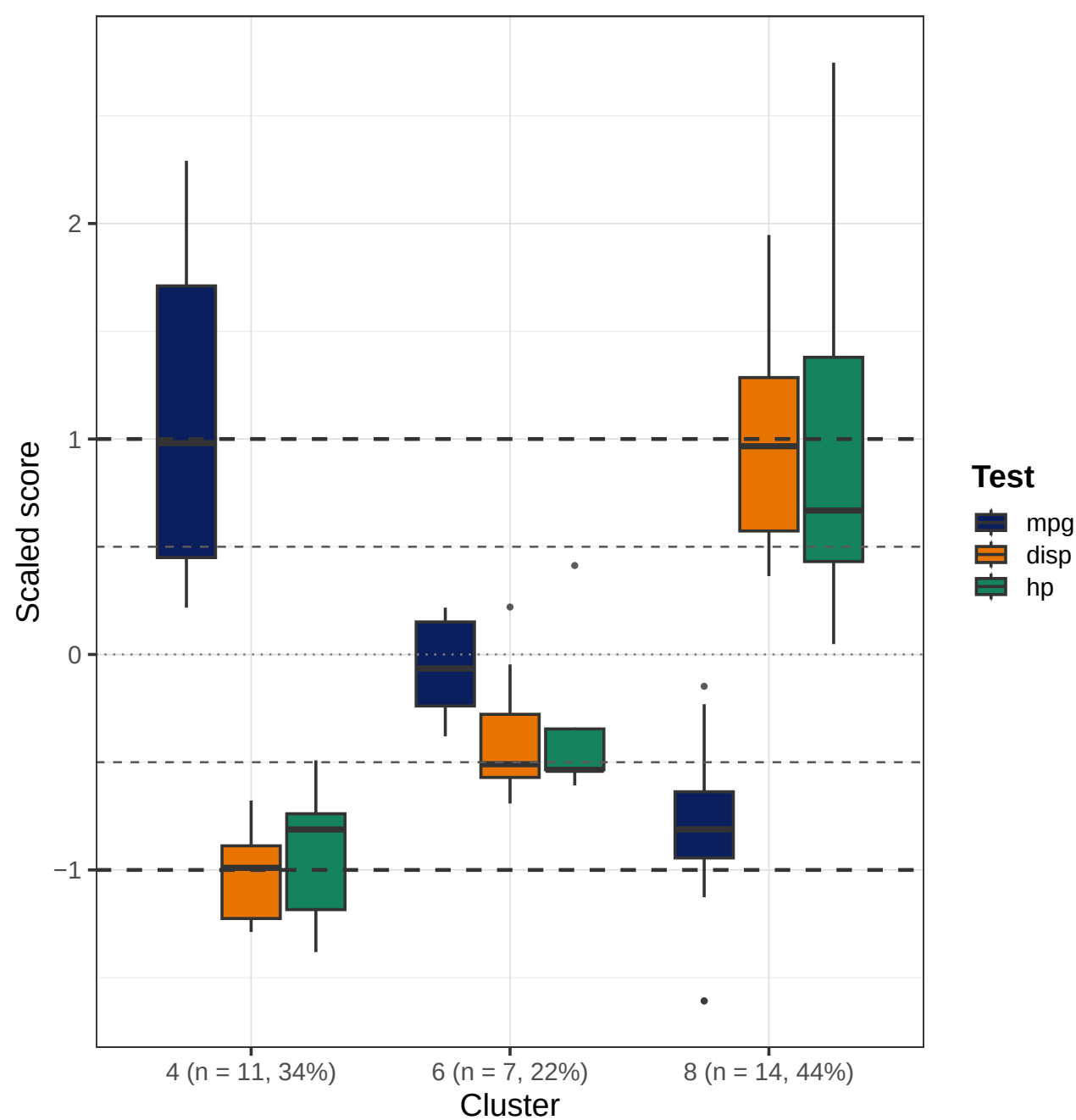
help("PlotChiSqCovar")

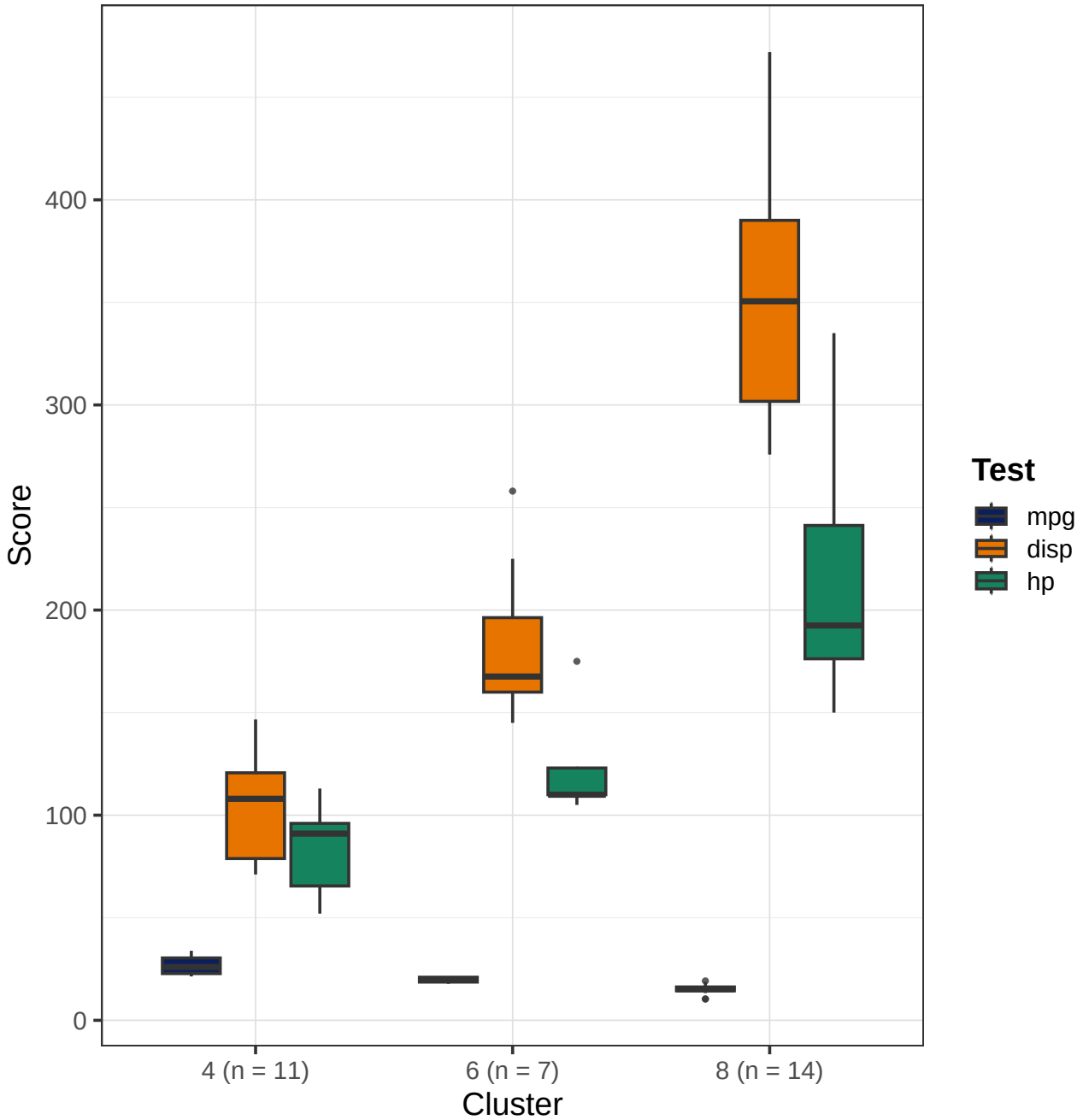
Significance

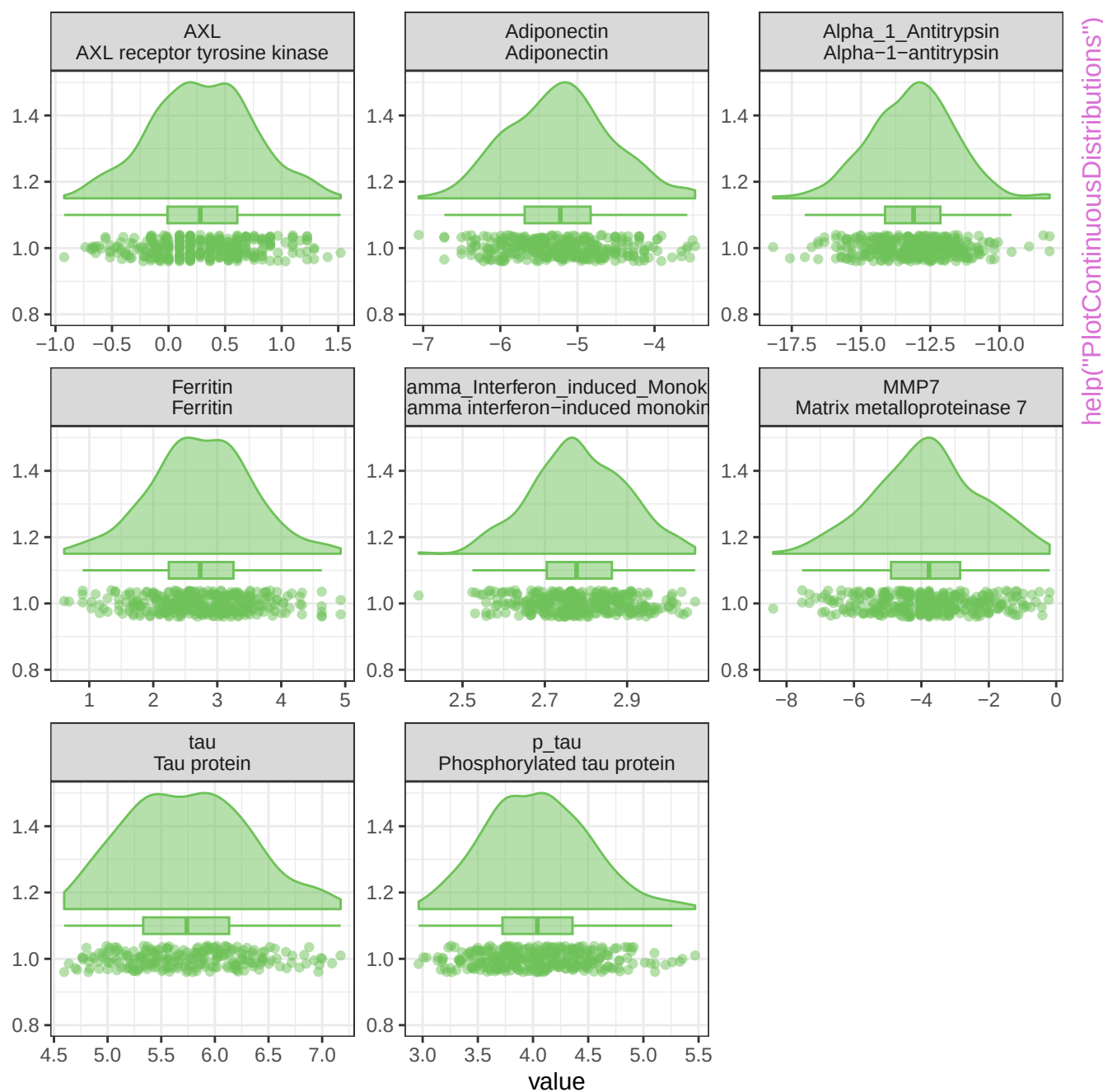
- ⊠ ns
- *
- ▲ **
- ****

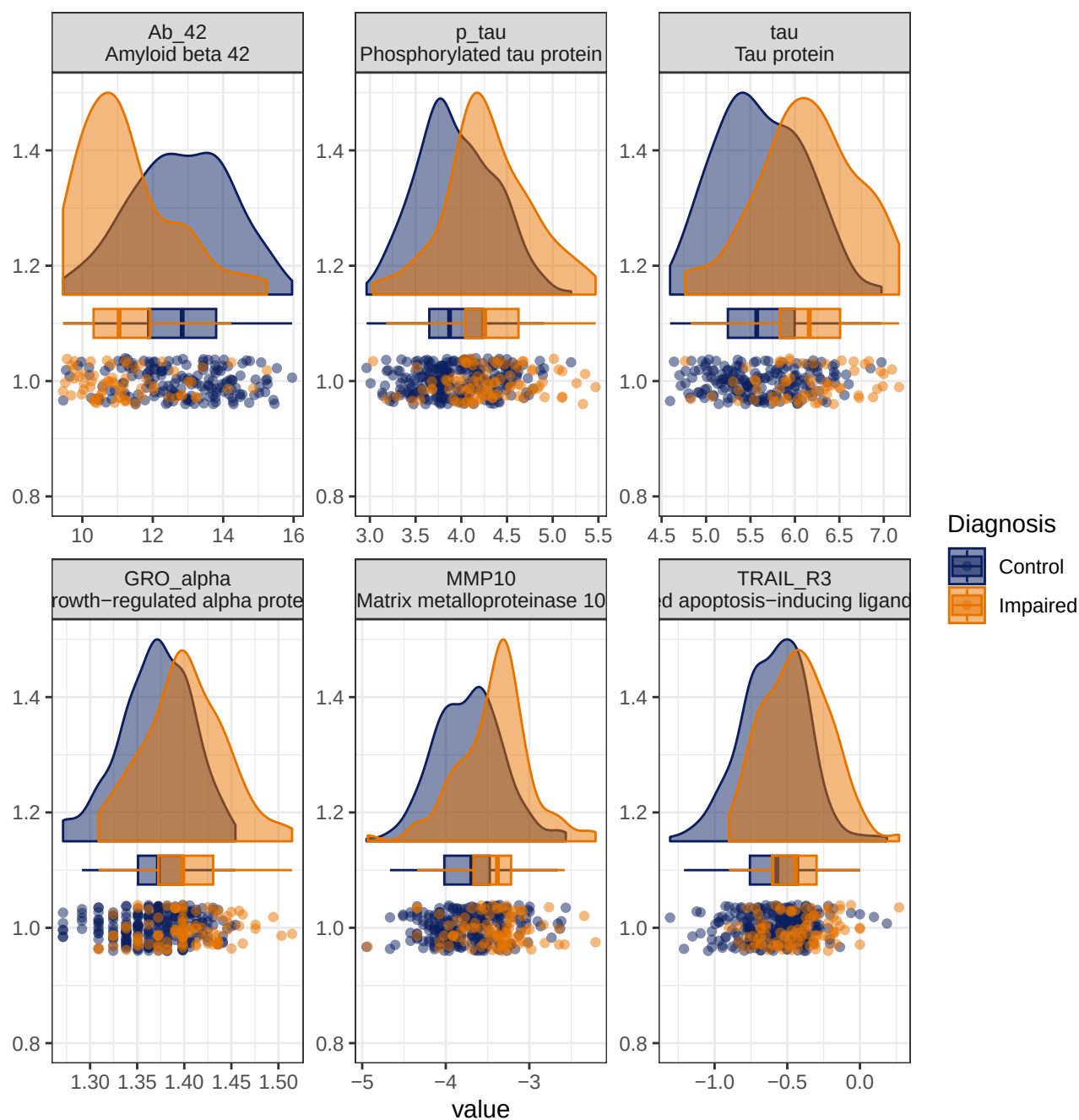
Cramer's V

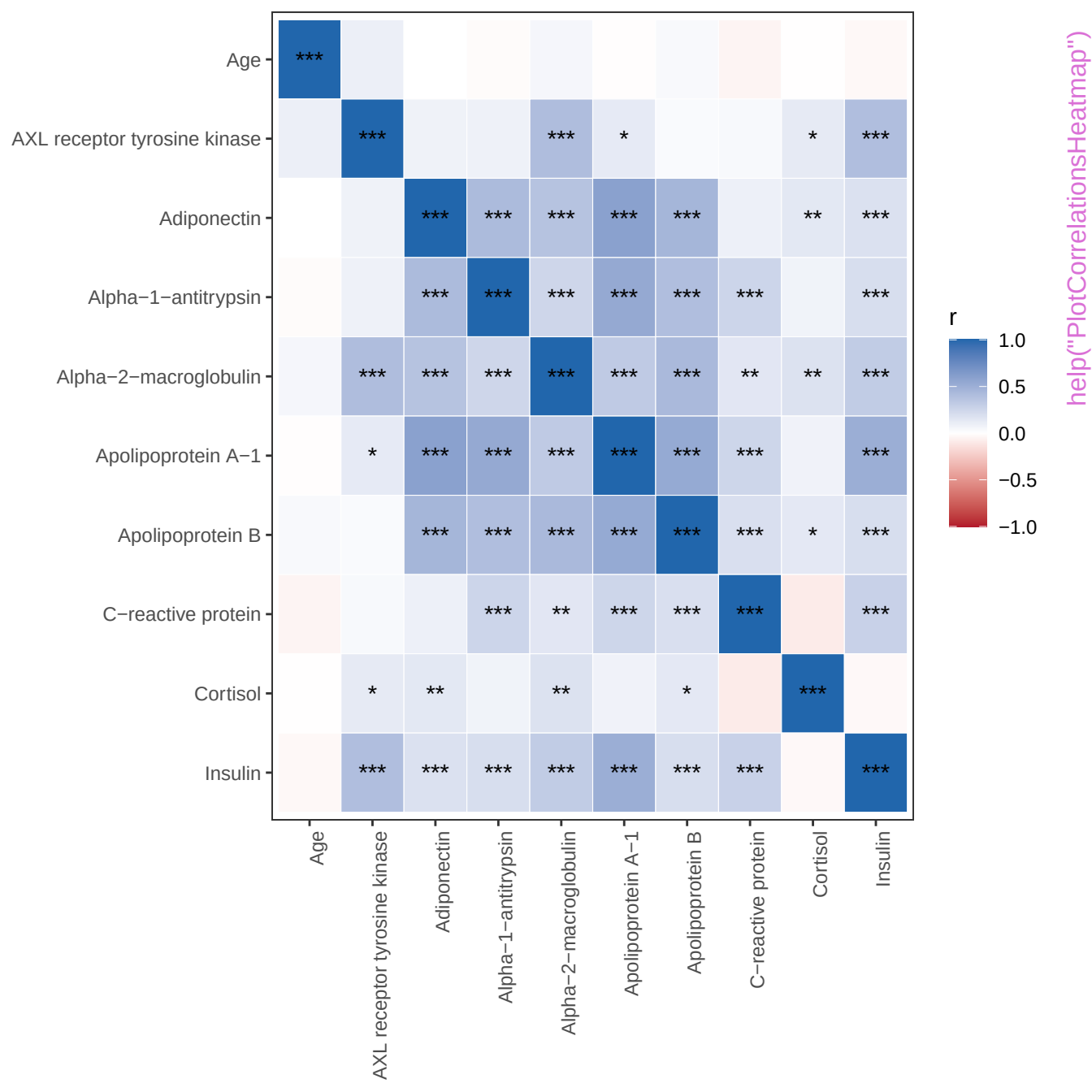


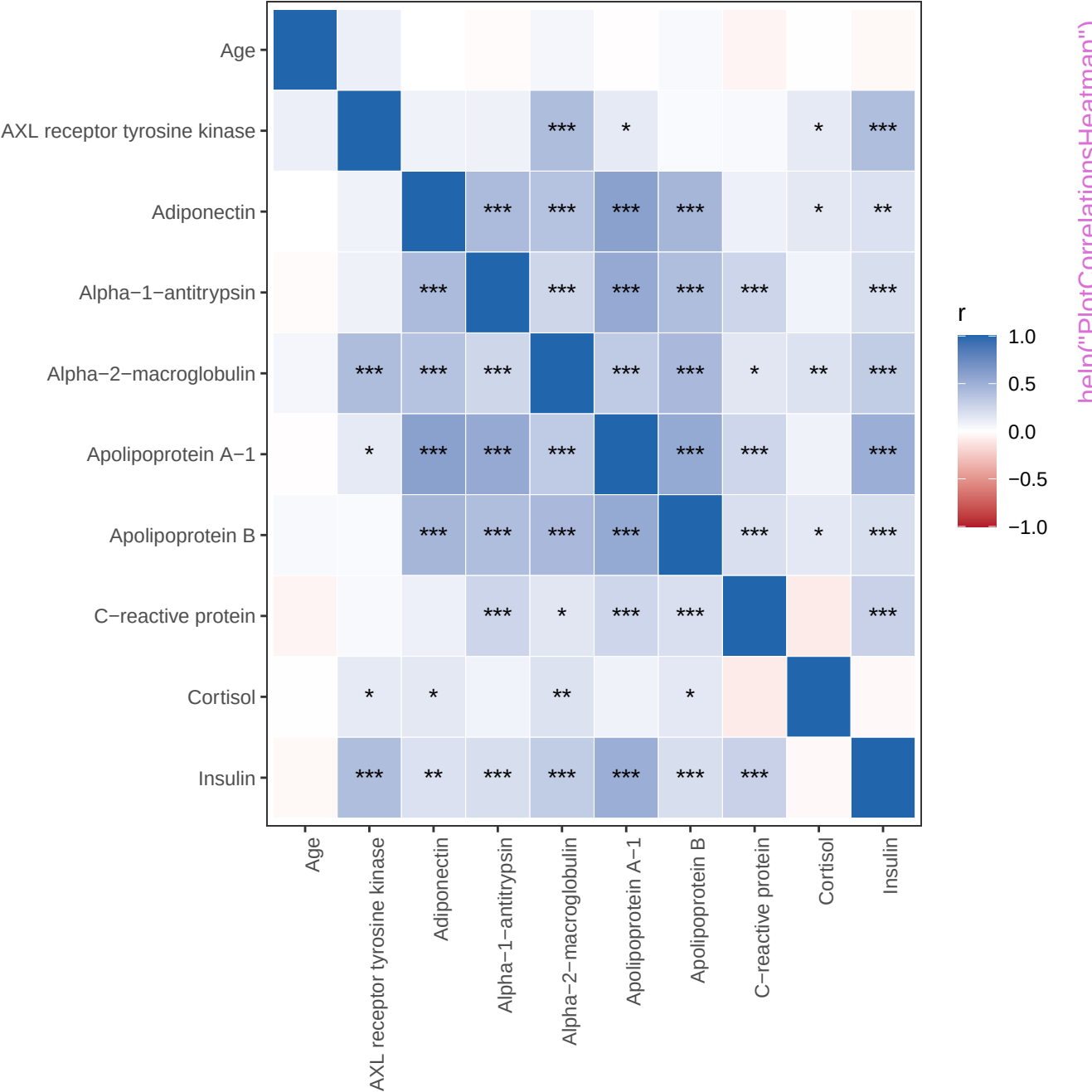




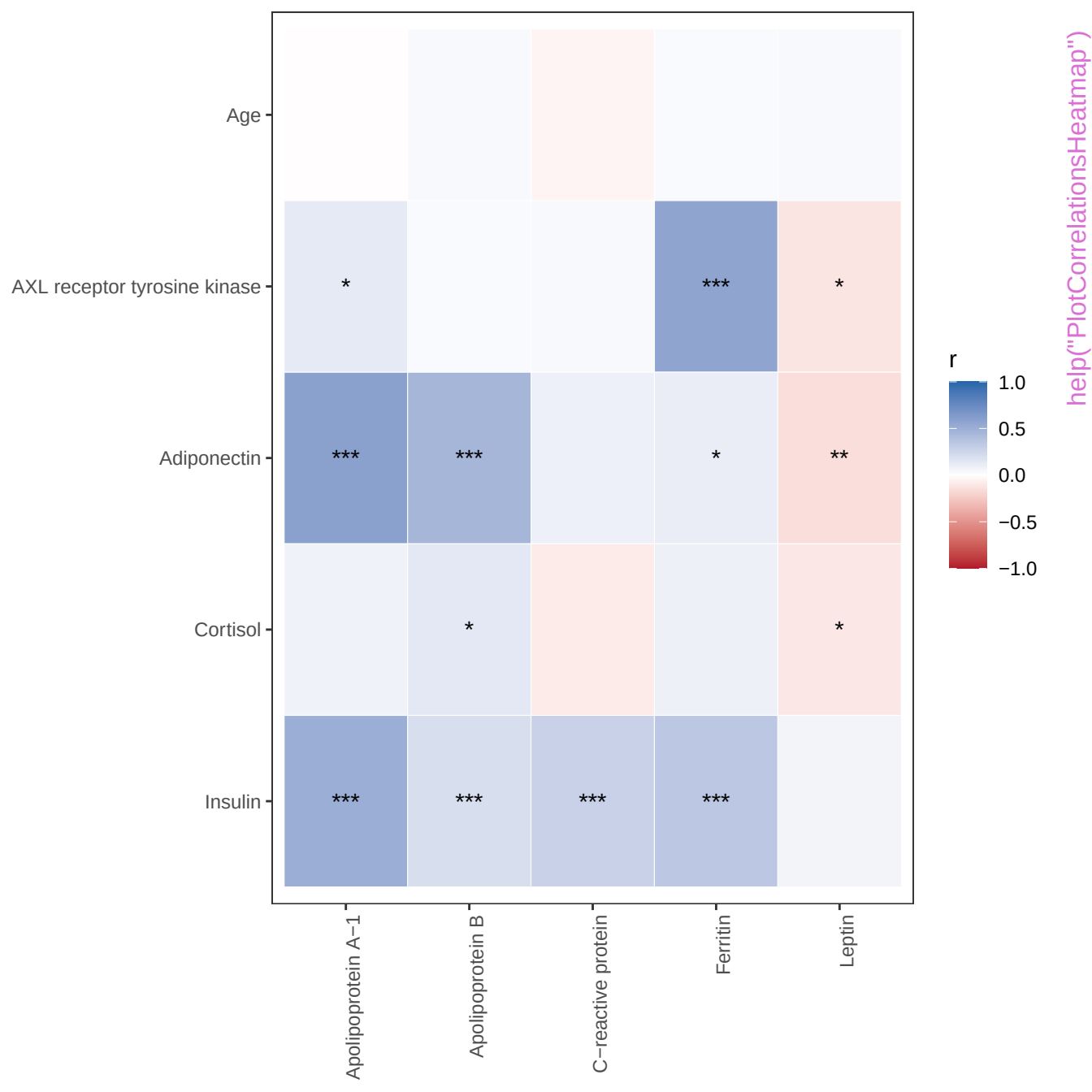






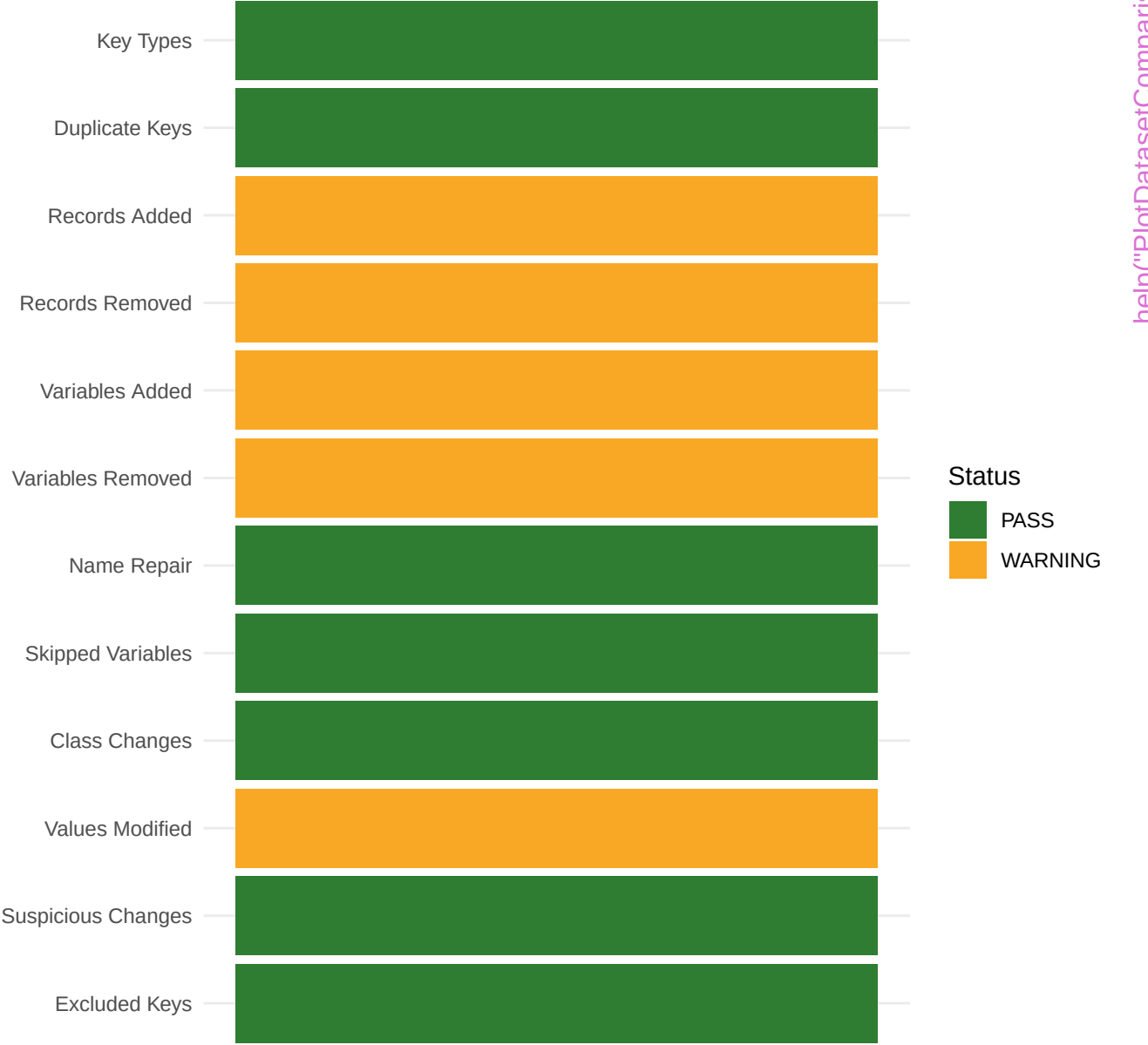


help("PlotCorrelationsHeatmap")



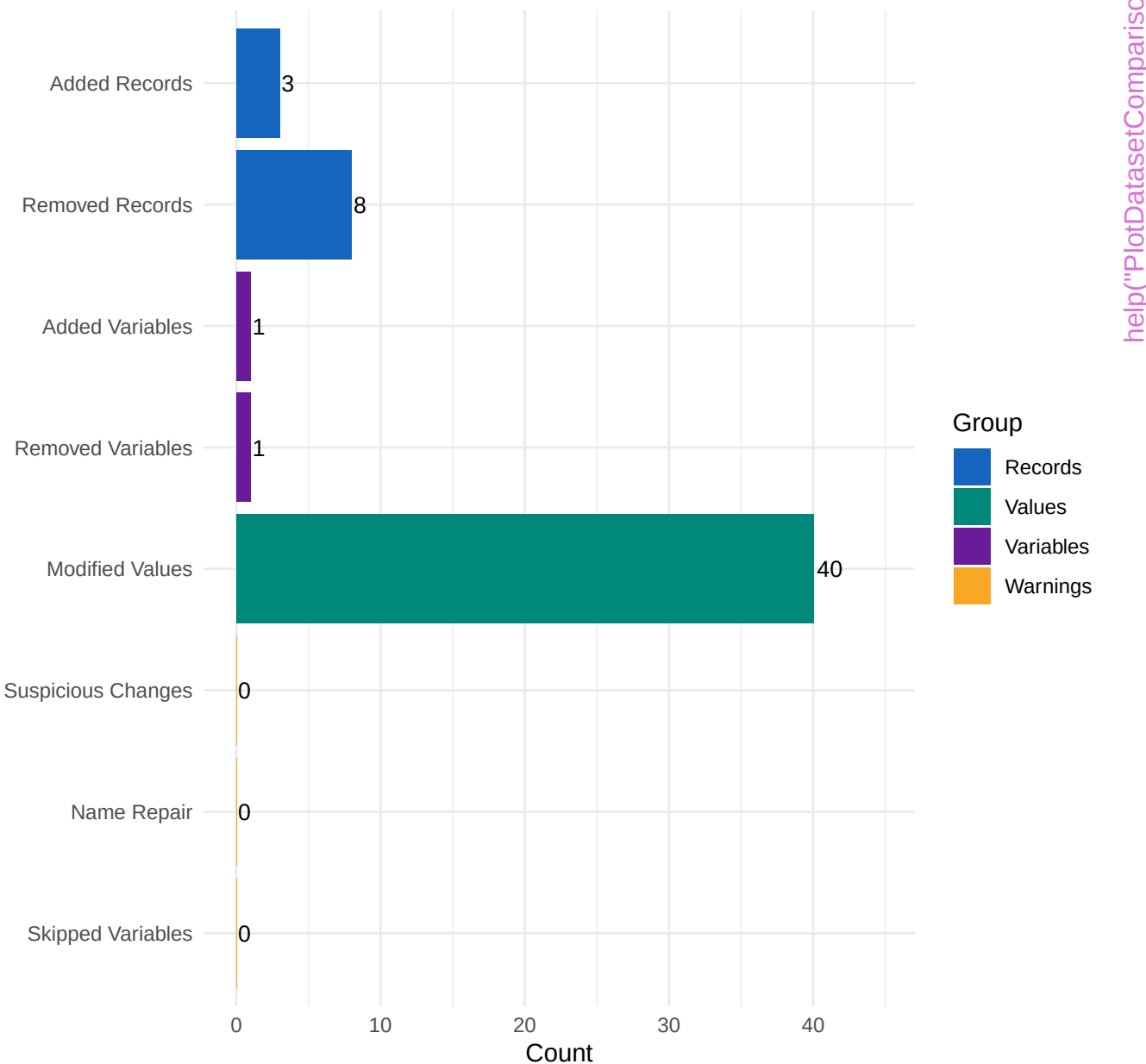
Dataset comparison checks

PASS, WARNING, and FAIL status by comparison check



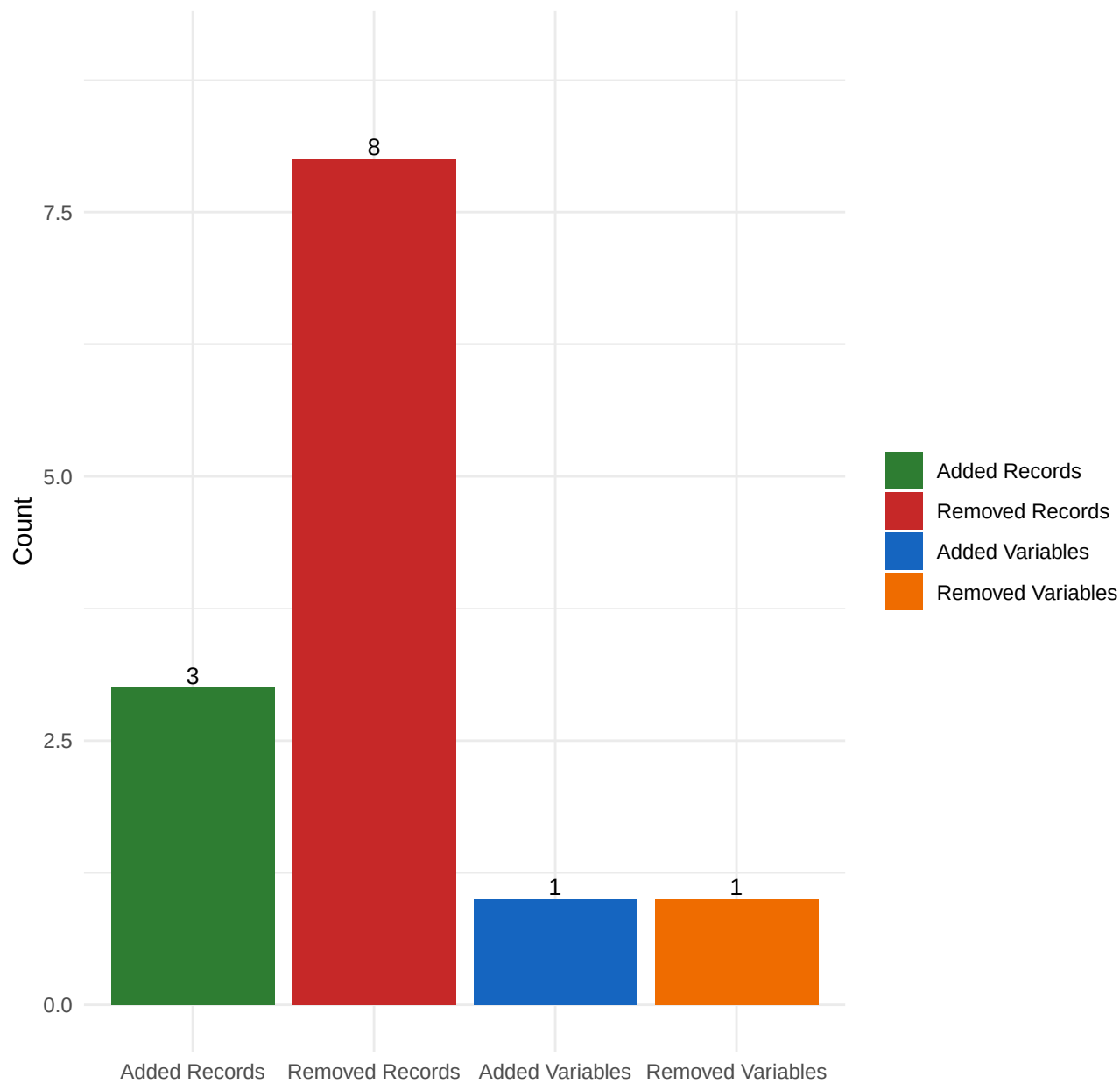
Dataset comparison summary

High-level record, variable, value, and warning counts



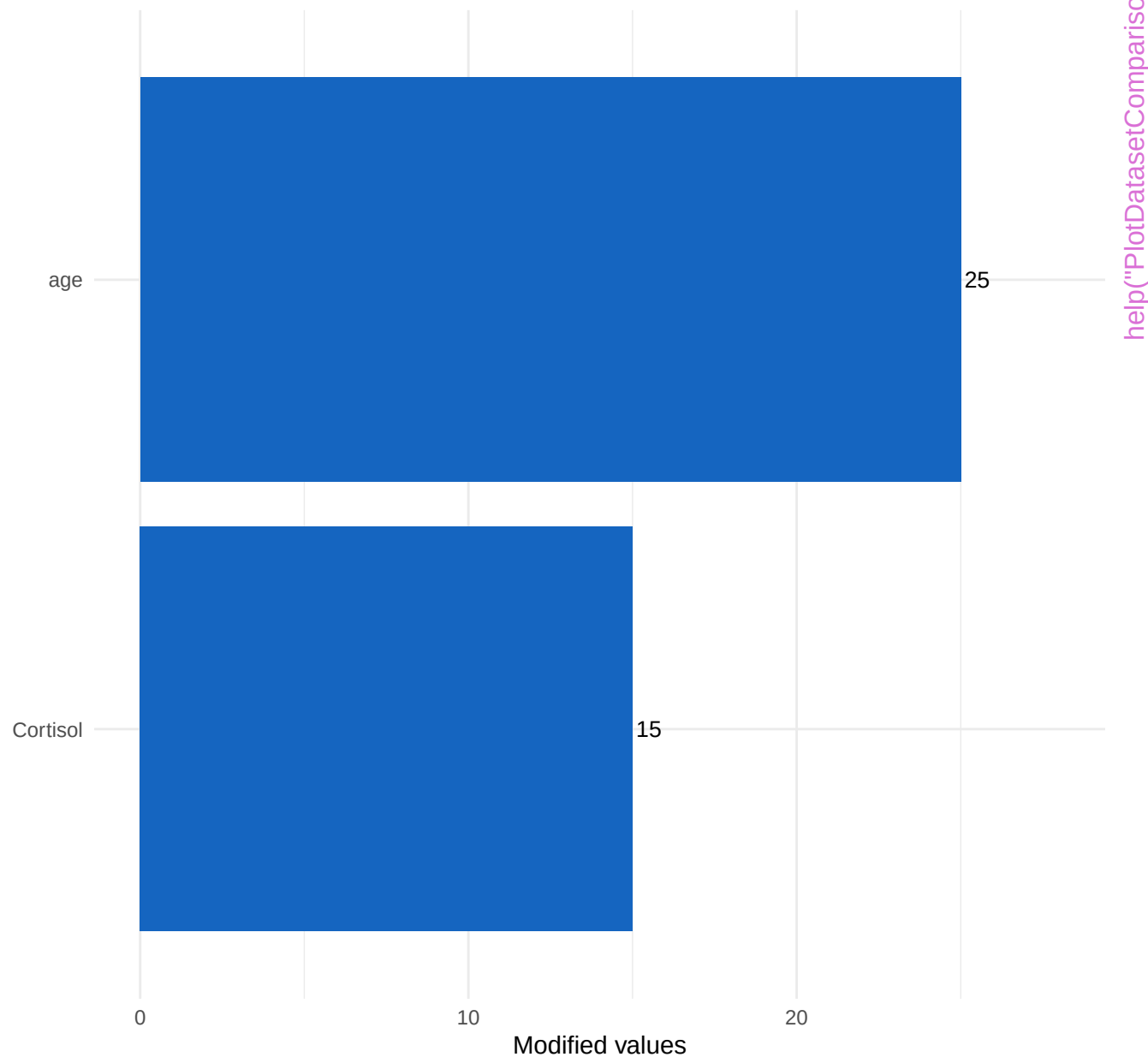
Record and structure changes

Records and variables added or removed



Variable-level changes

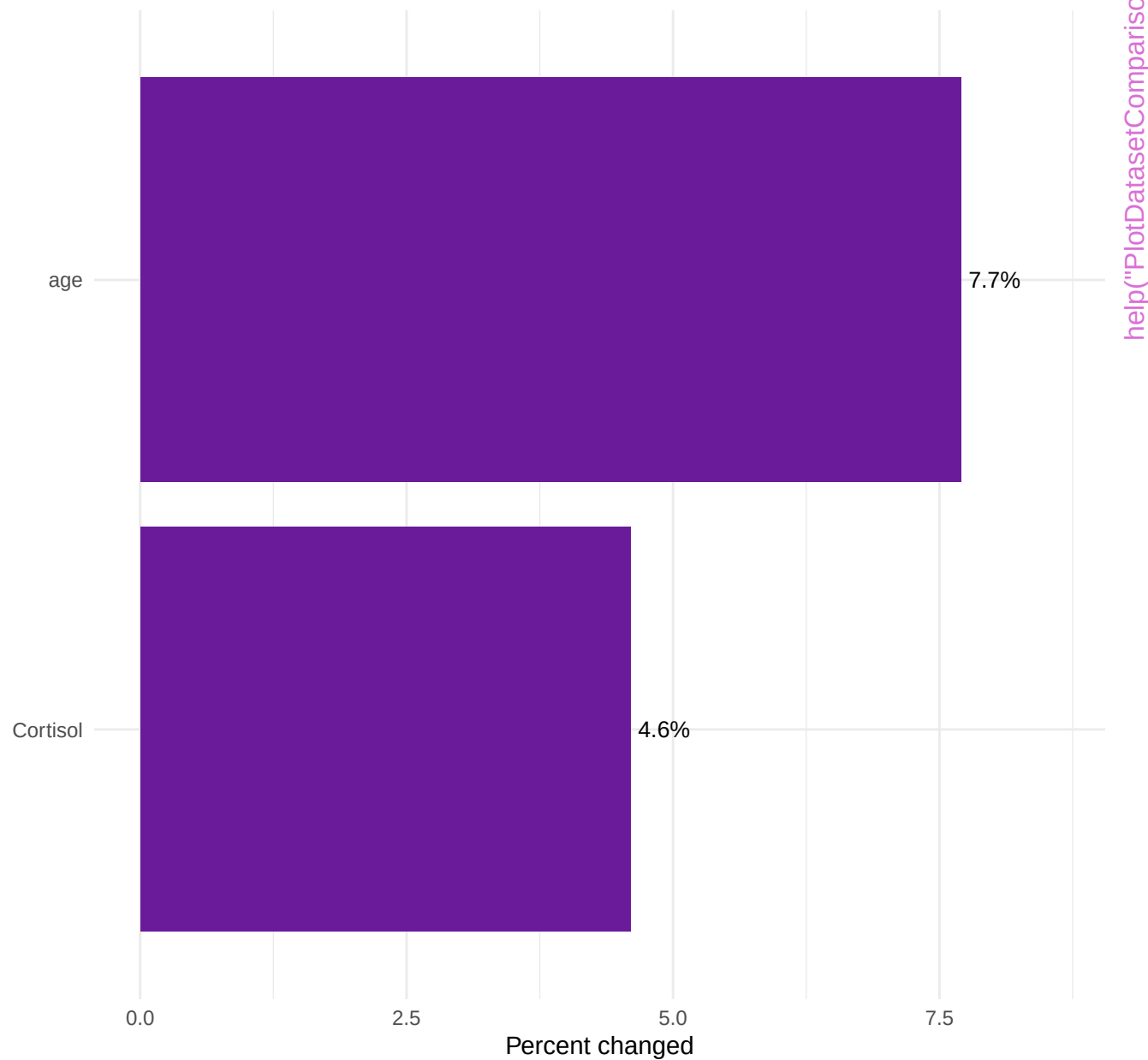
Top 2 variables by modified values

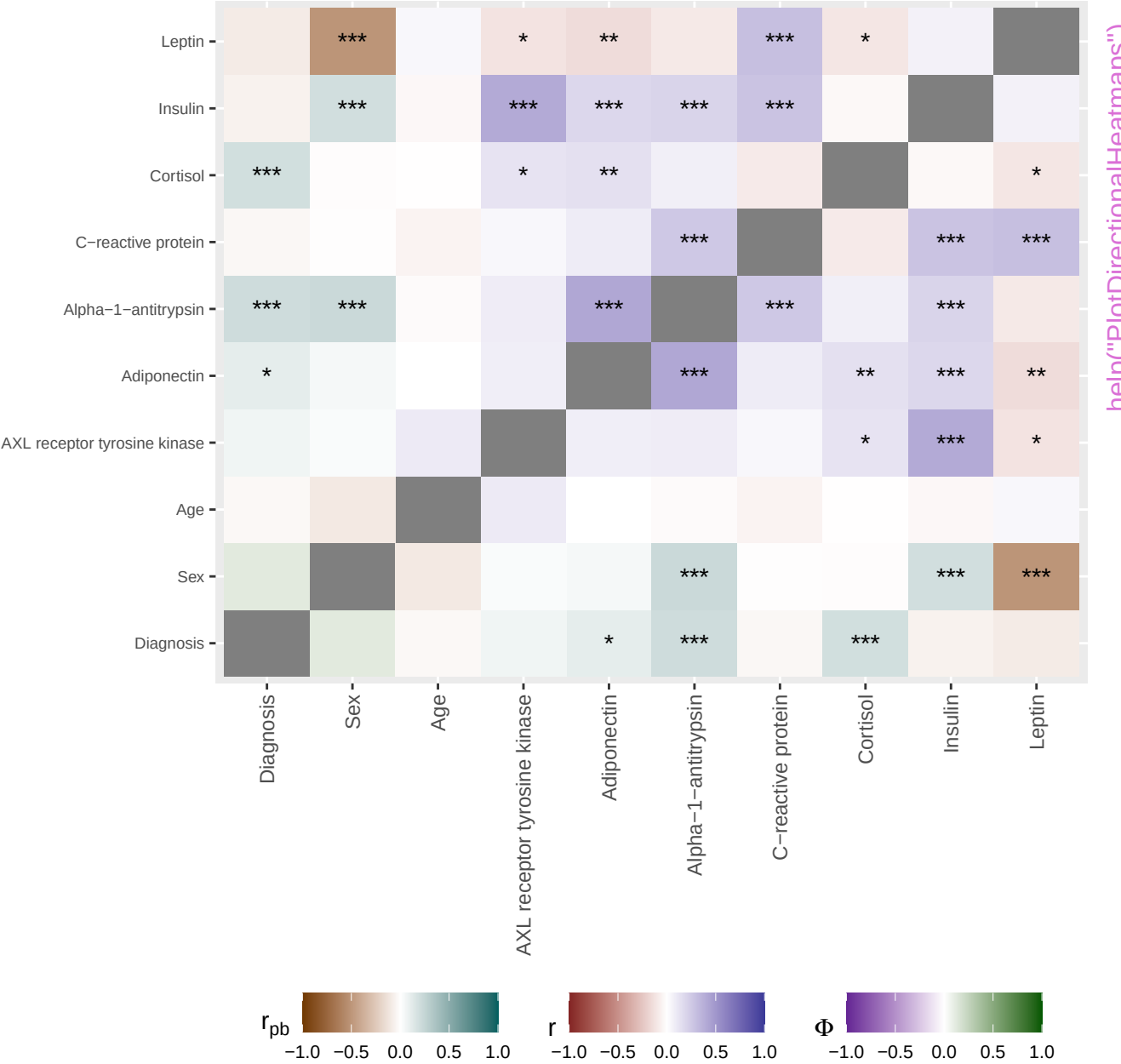


help("PlotDatasetComparison")

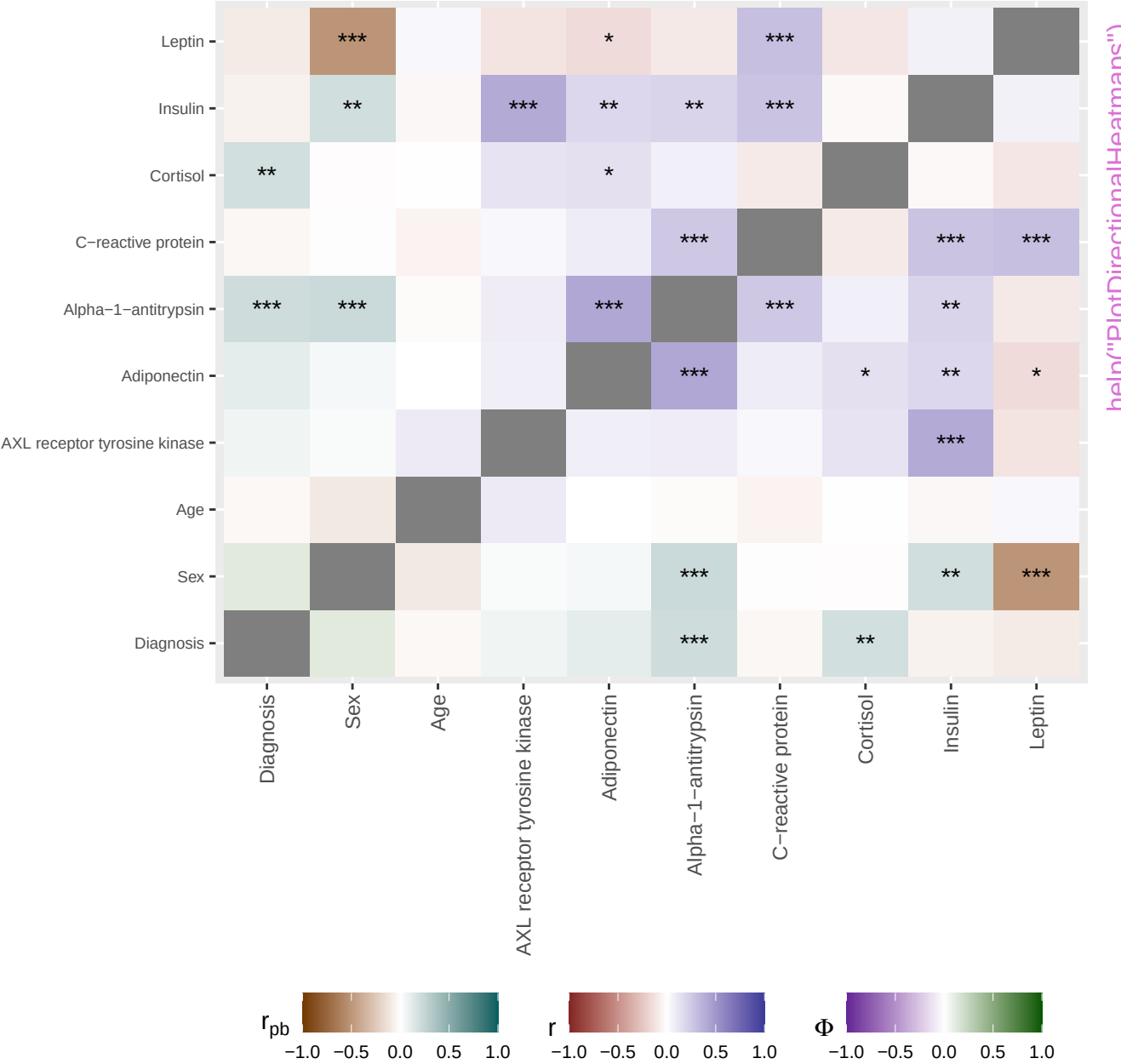
Top changed variables

Top 2 variables by percent changed





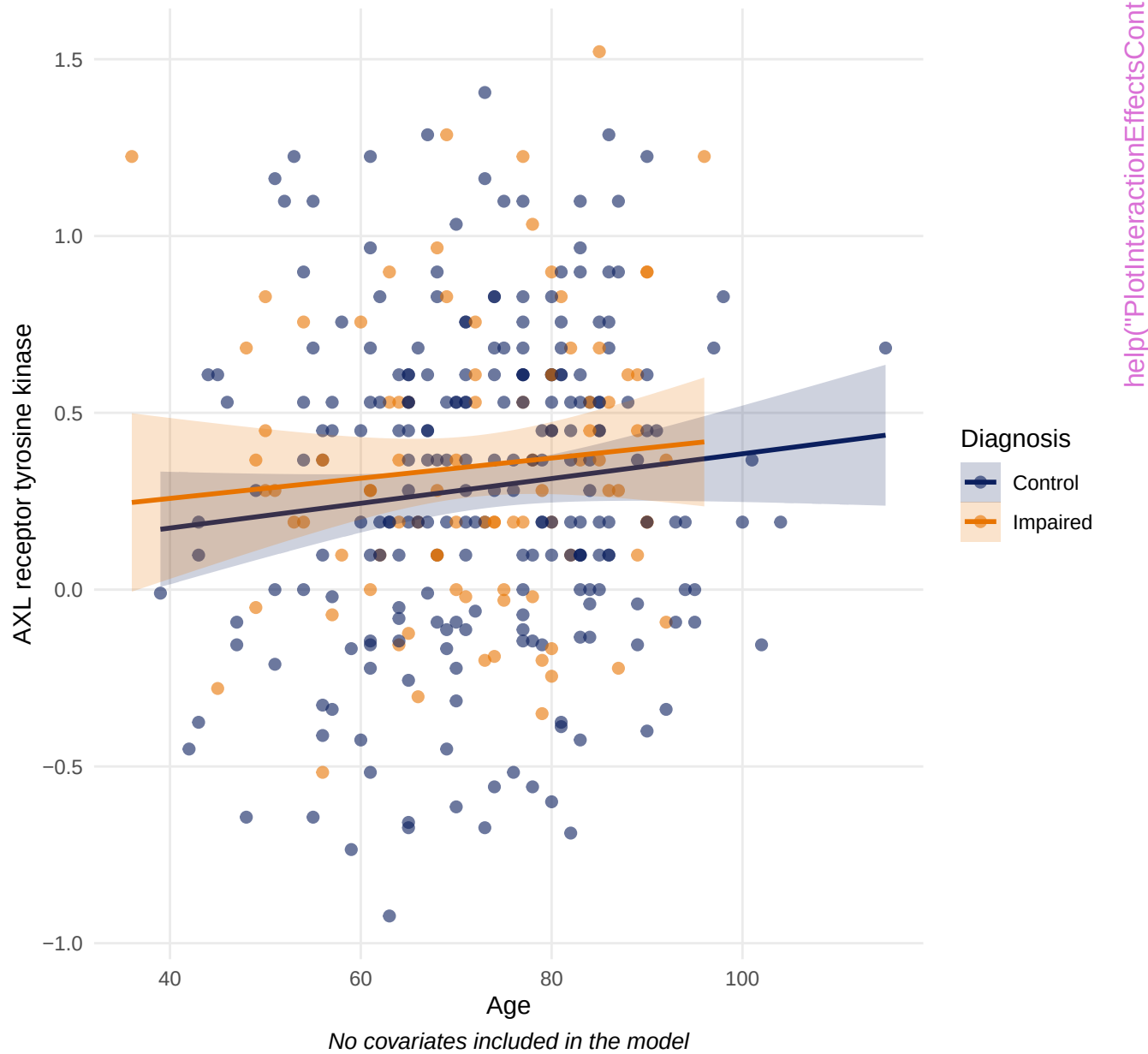
help("PlotDirectionalHeatmaps")



help("PlotDirectionalHeatmaps")

Interaction with Diagnosis

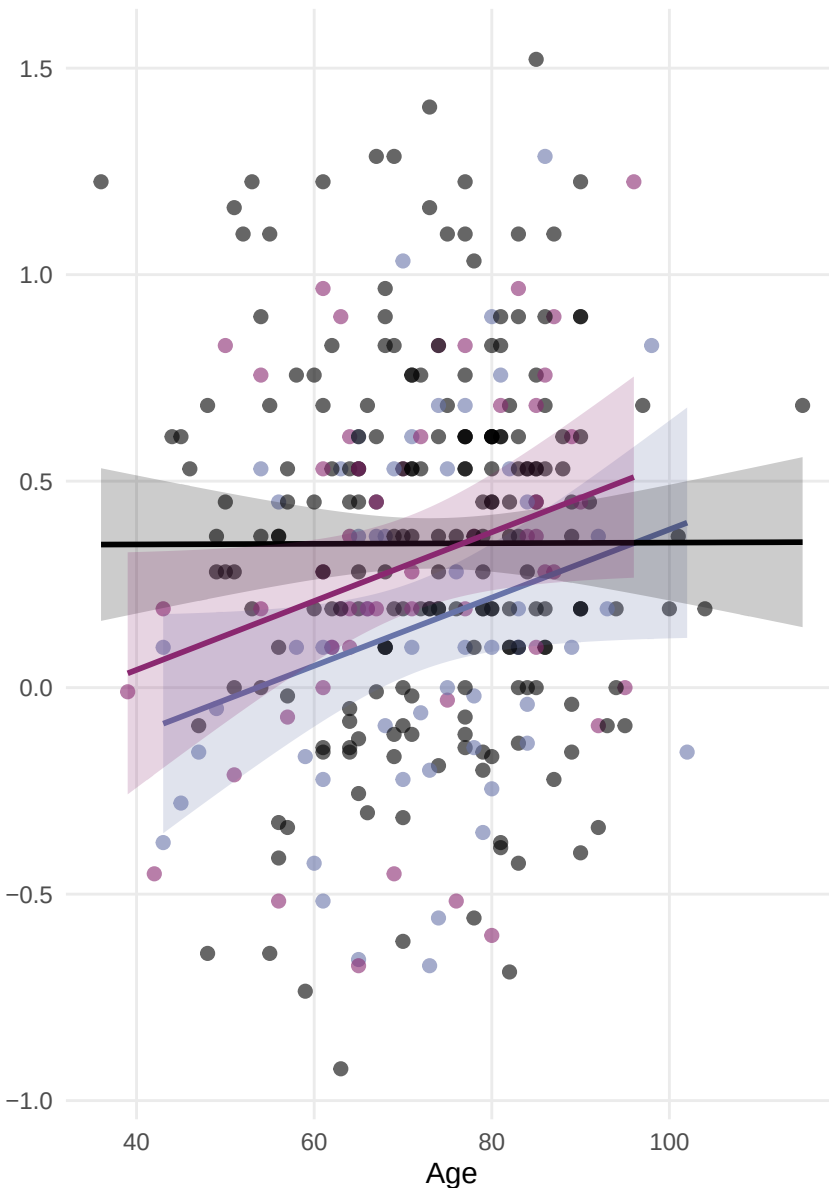
Interaction p-value: 0.8789



Interaction with Adiponectin

Interaction p-value: 0.7488

AXL receptor tyrosine kinase



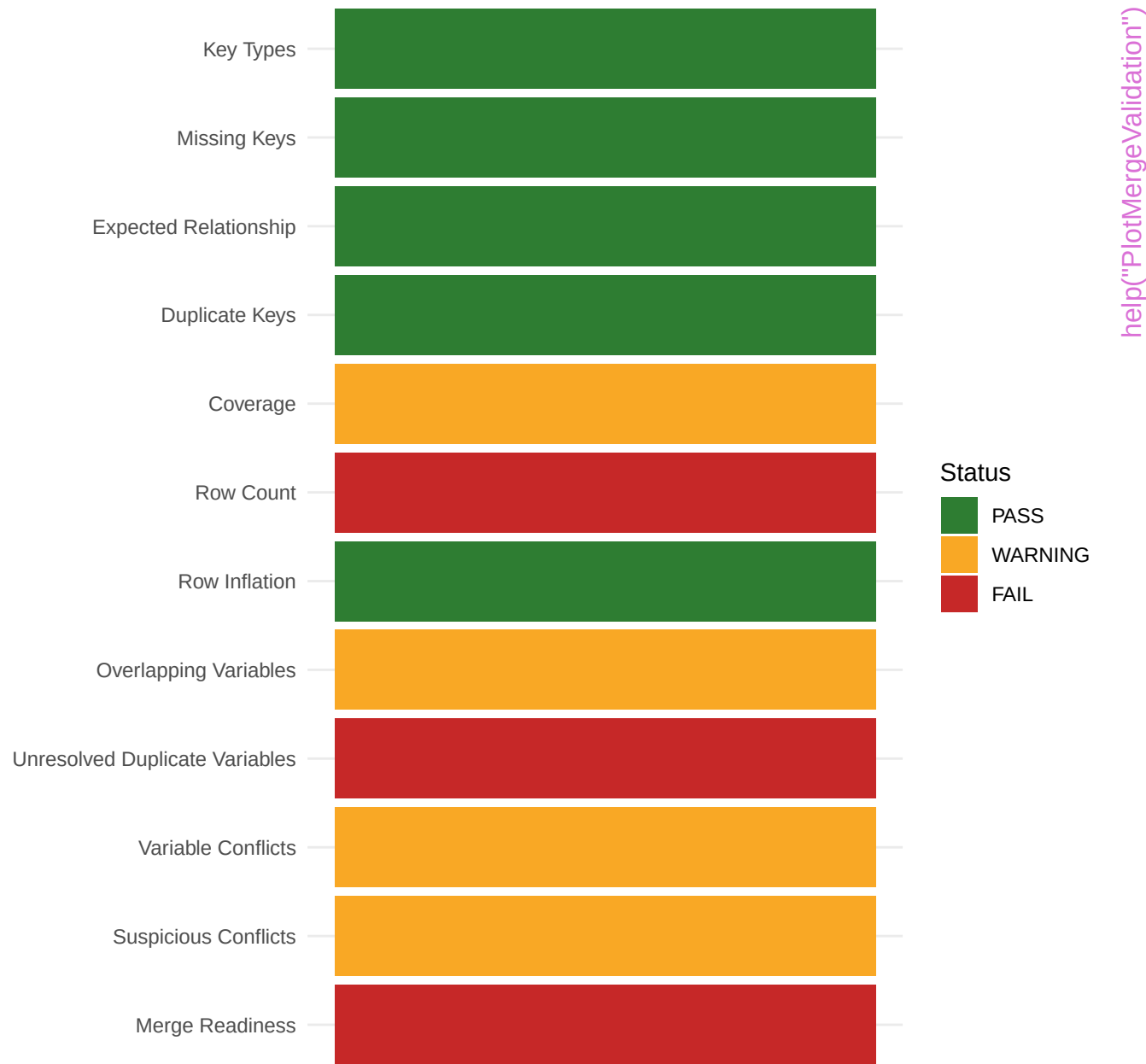
Adiponectin

- Adiponectin = -5.9 (-1 SD)
- Adiponectin = -5.2 (Mean)
- Adiponectin = -4.6 (+1 SD)

No covariates included in the model

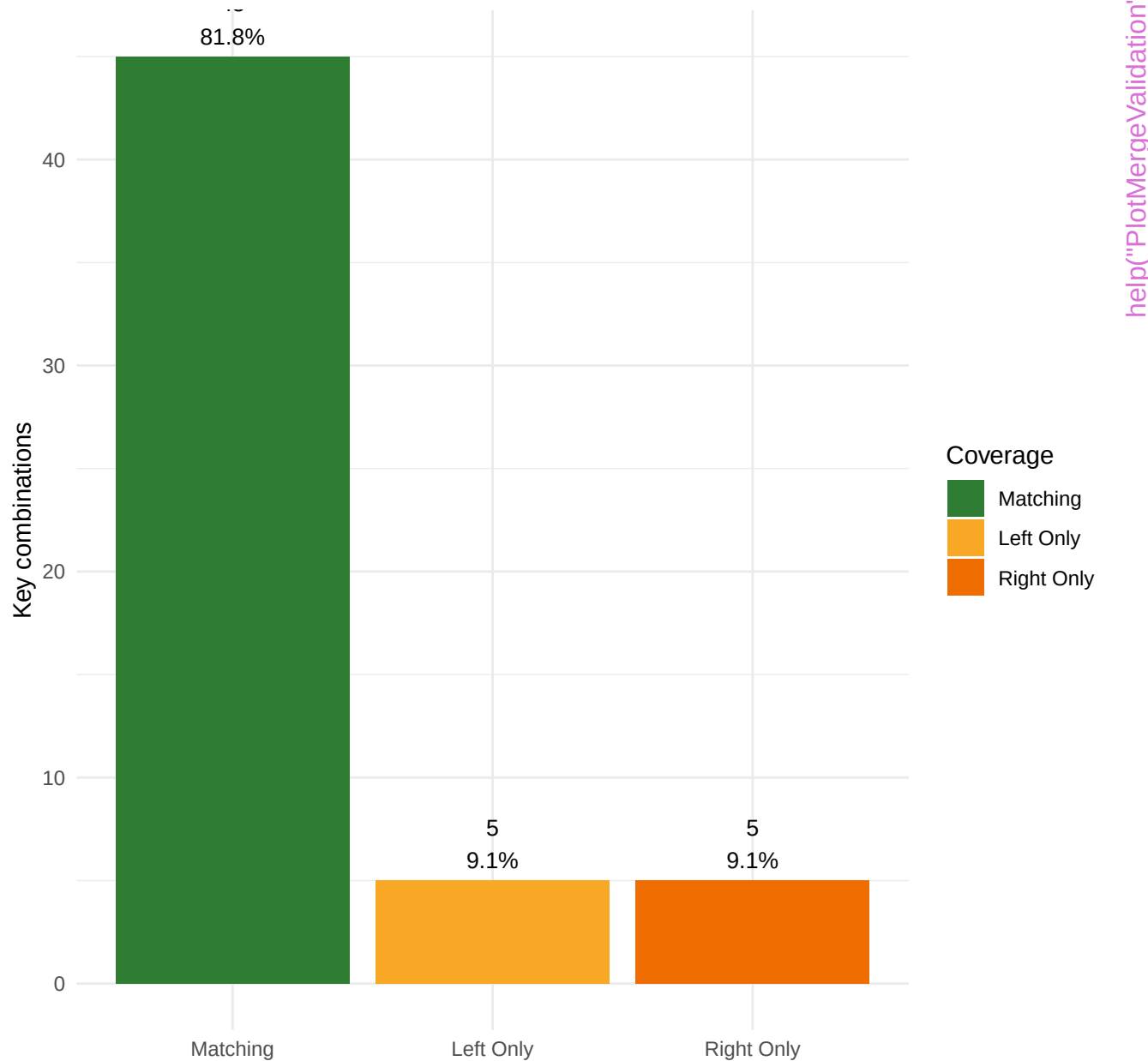
help("PlotInteractionEffectsContinuous")

Merge validation checks



Merge coverage

Key combinations by source overlap



Join audit

id



Join role



- Specified Key
- Overlap Not In Keys

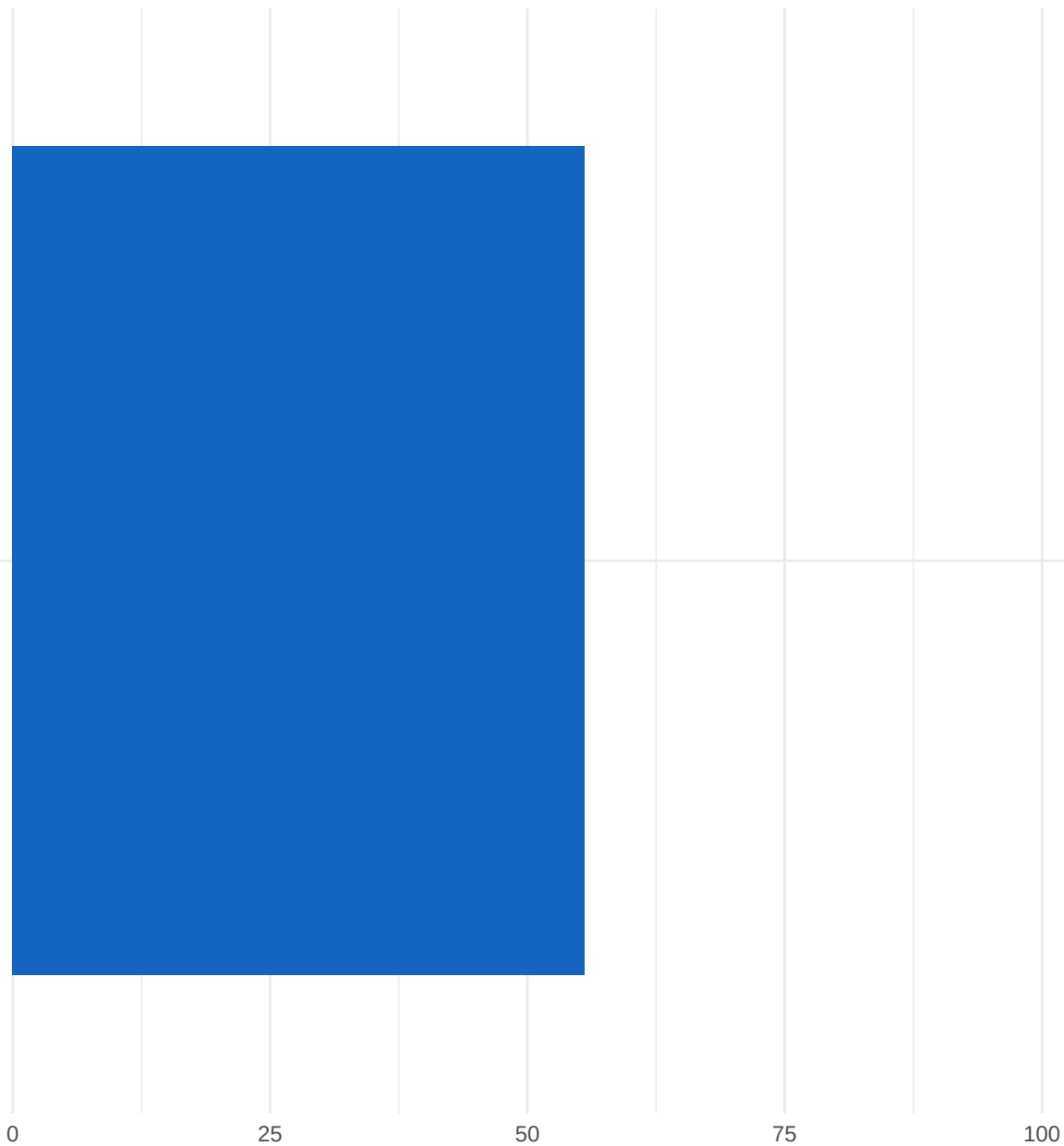
site



help("PlotMergeValidation")

Duplicate-variable agreement

site



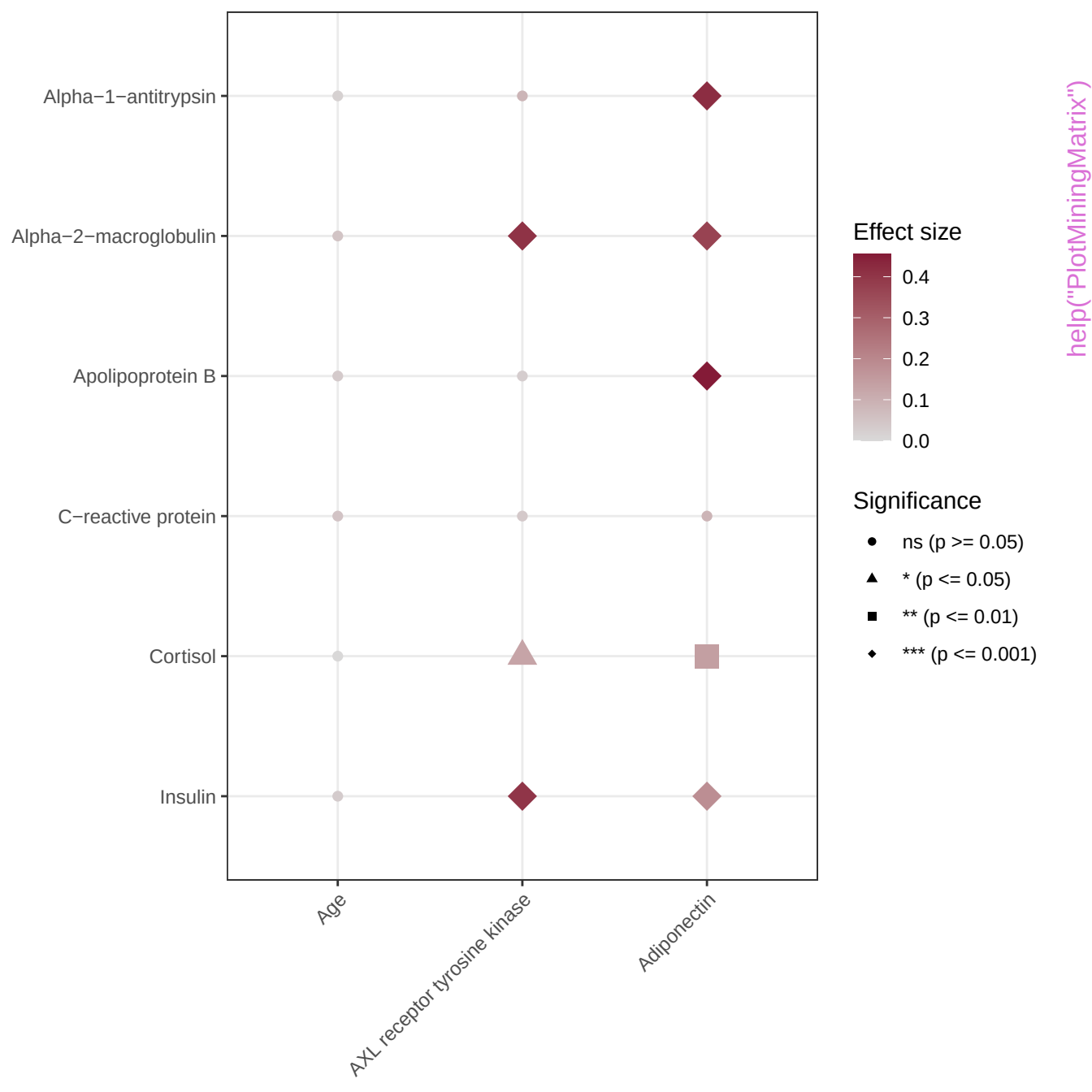
help("PlotMergeValidation")

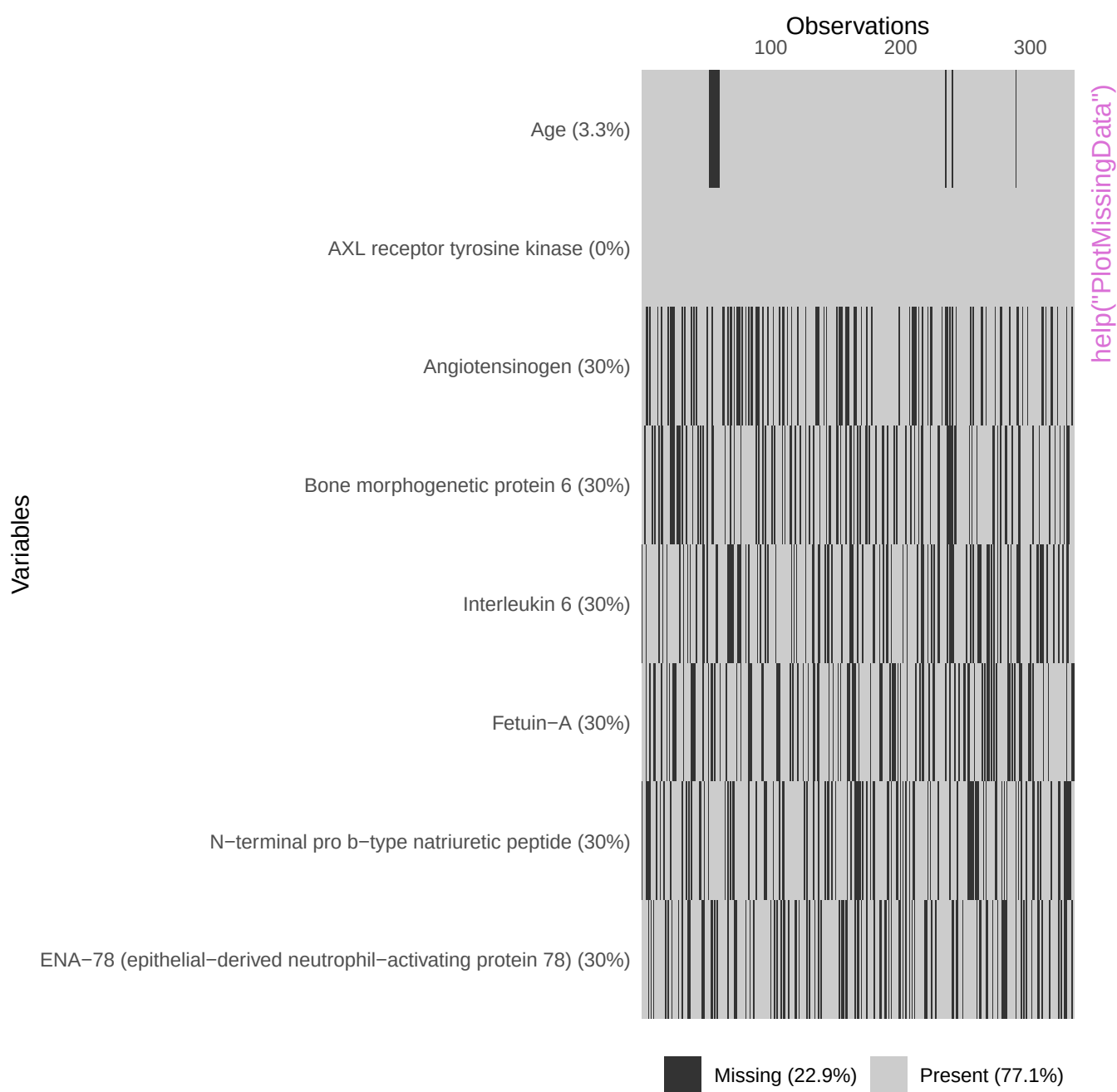
Duplicate-variable conflicts

site

Conflicts

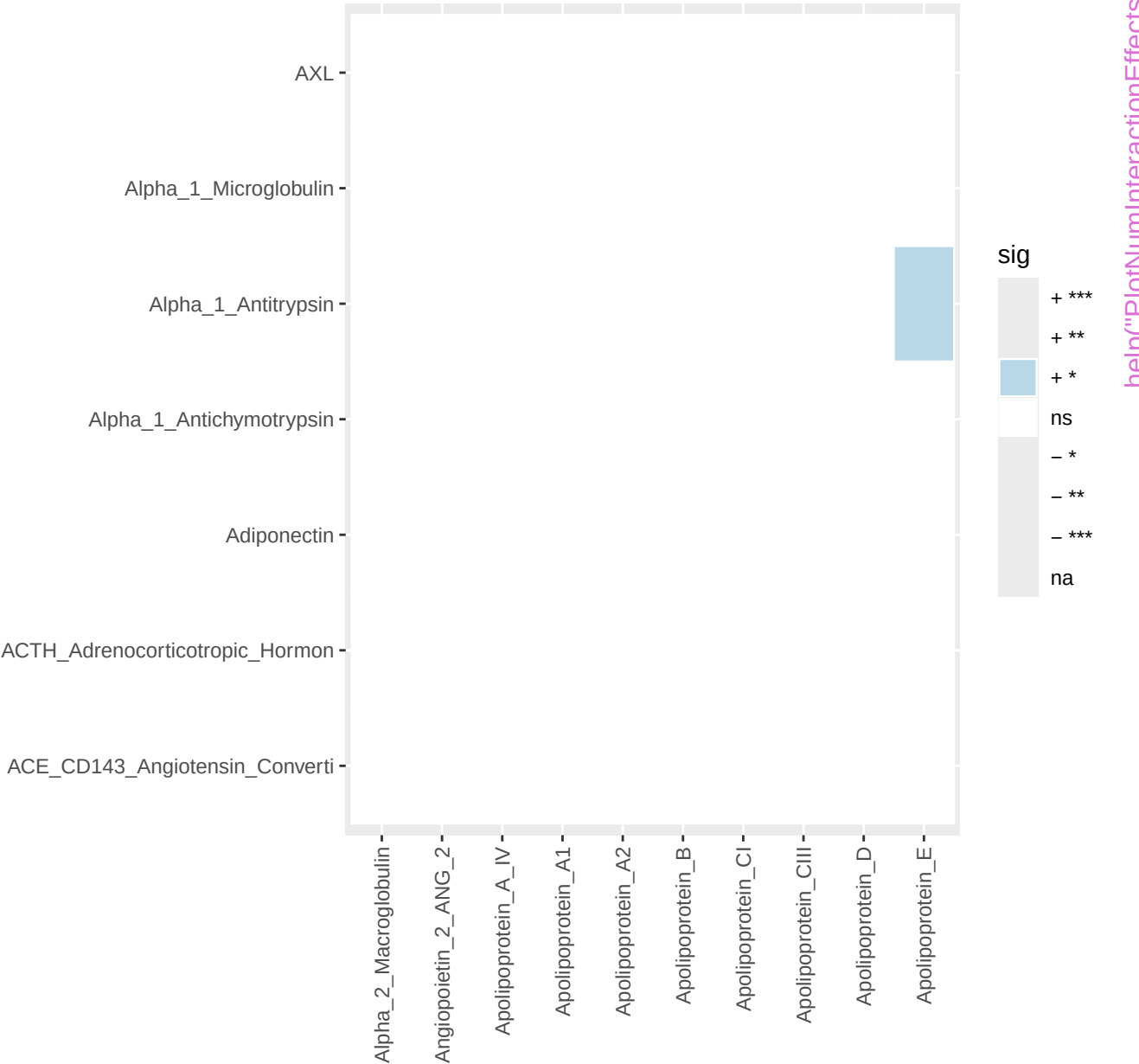
help("PlotMergeValidation")





age

No Multiple Comparison Correction



sig

+ ***

+ **

+ *

ns

- *

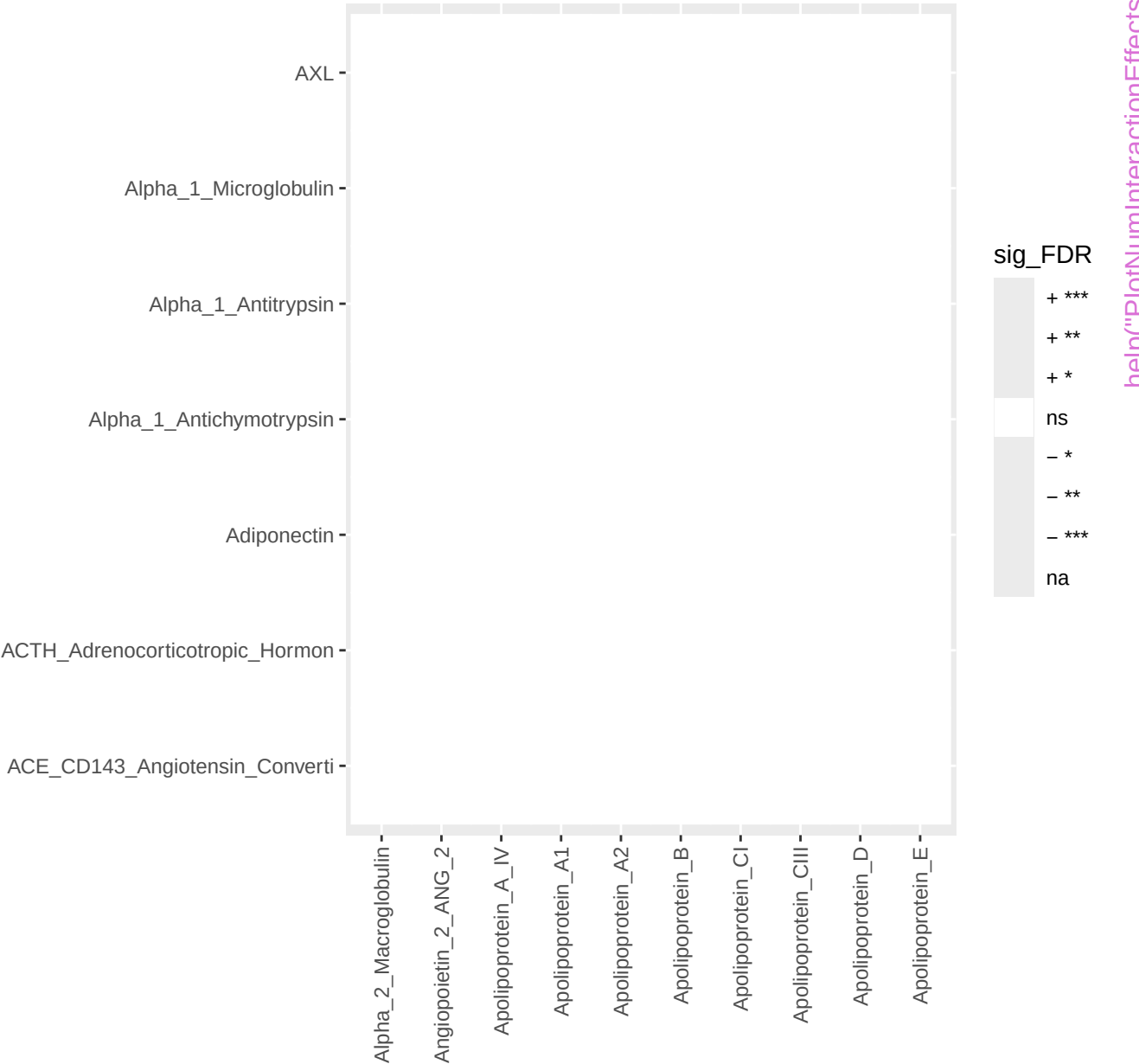
- **

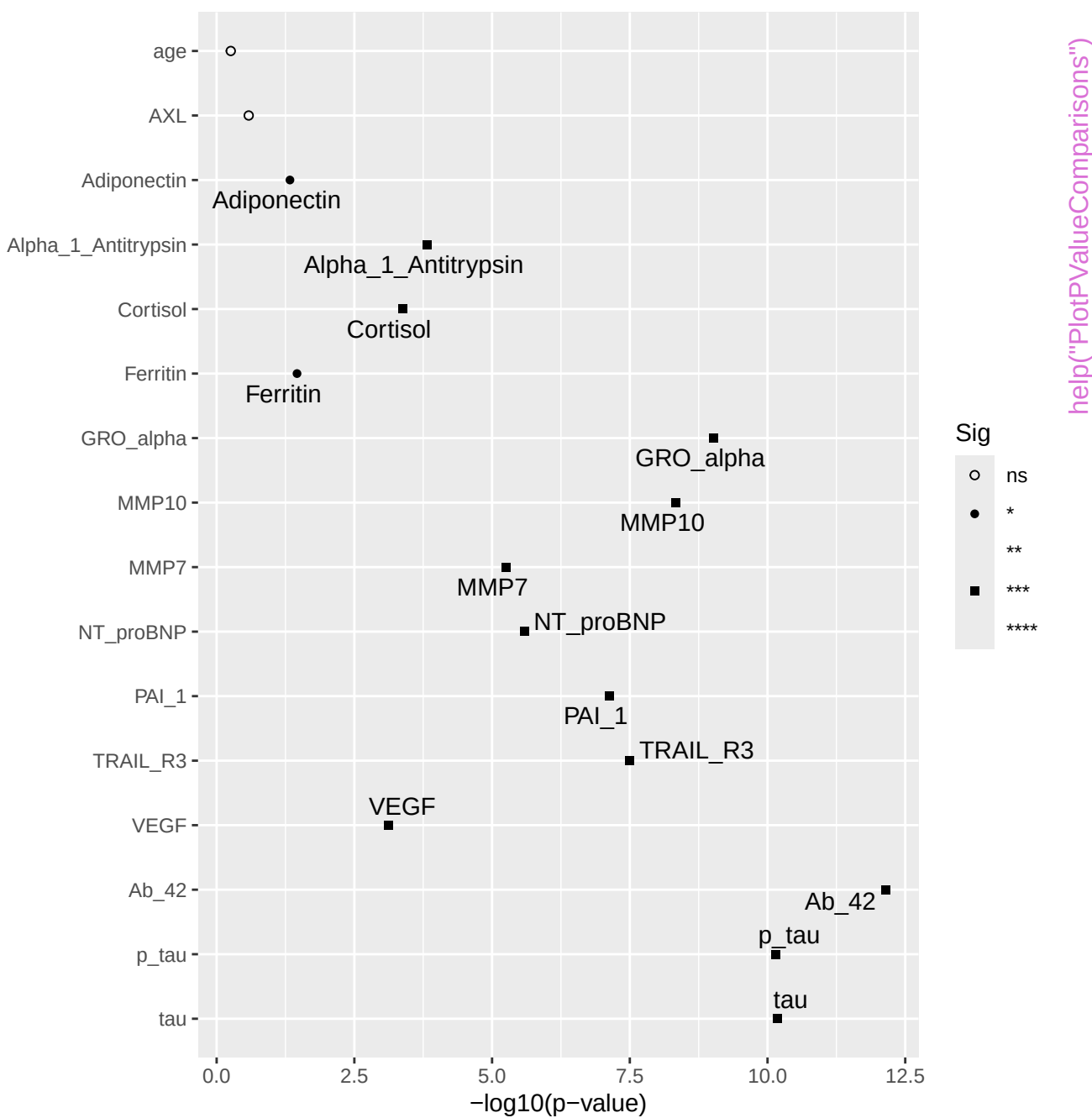
- ***

na

help("PlotNumInteractionEffectsMatrix")

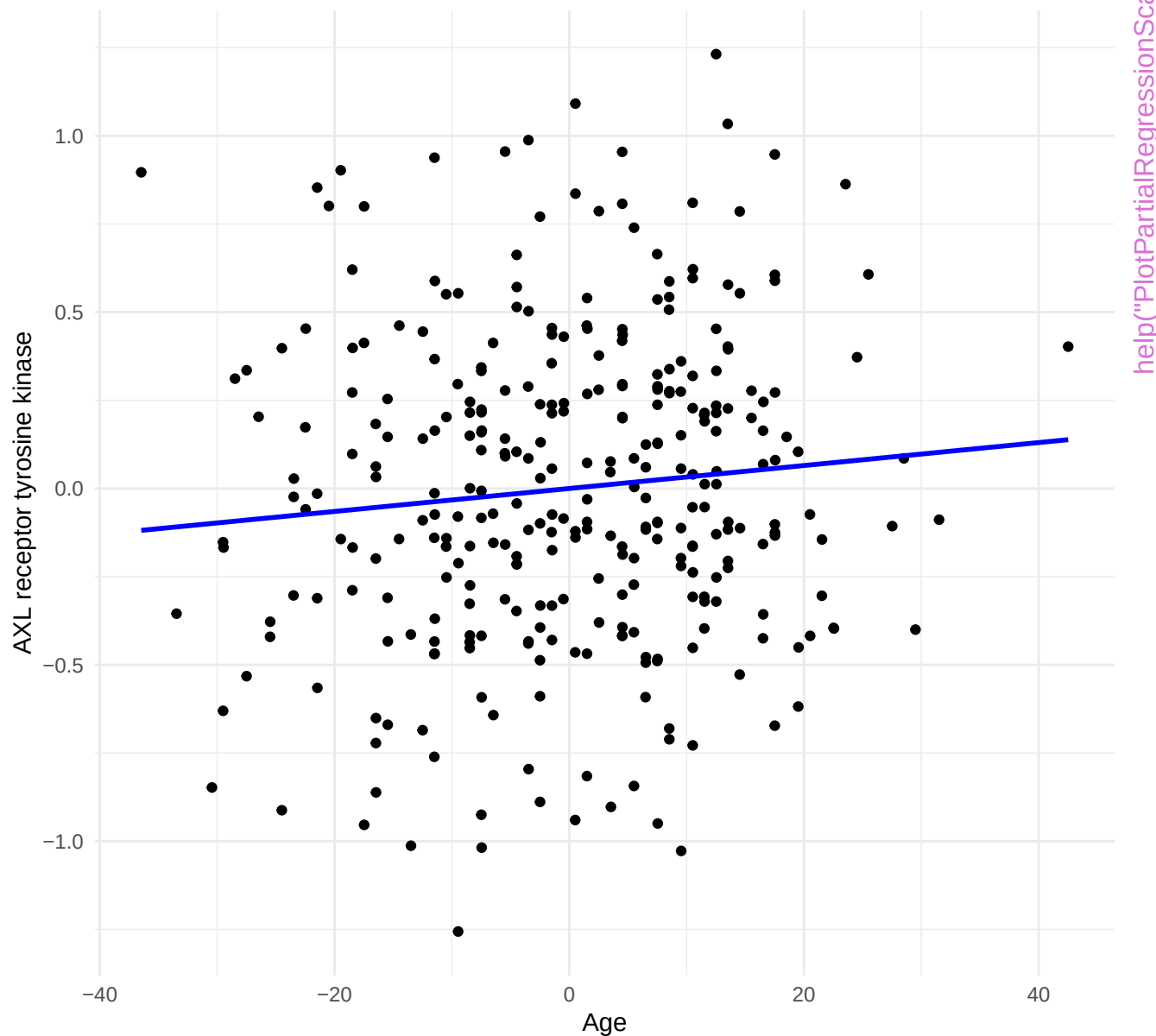
age
FDR Correction



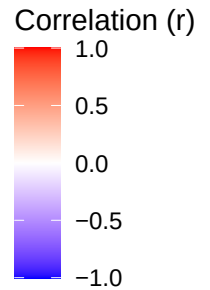
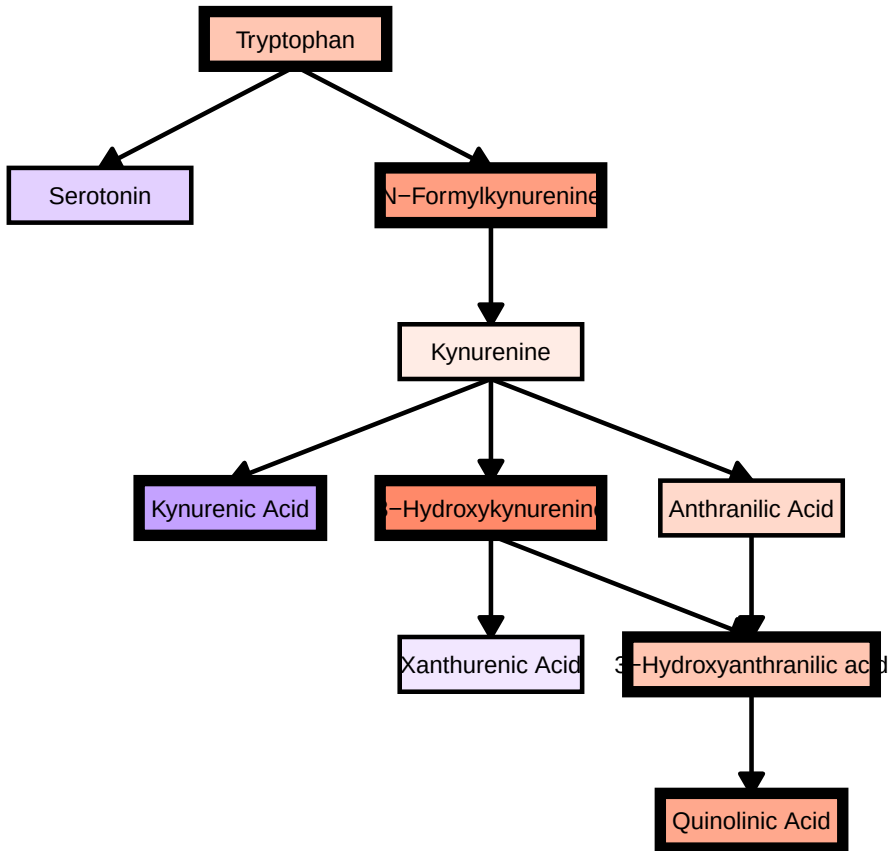


Partial Regression Plot

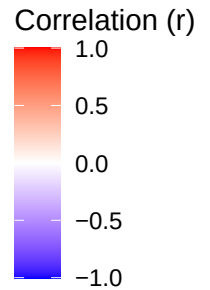
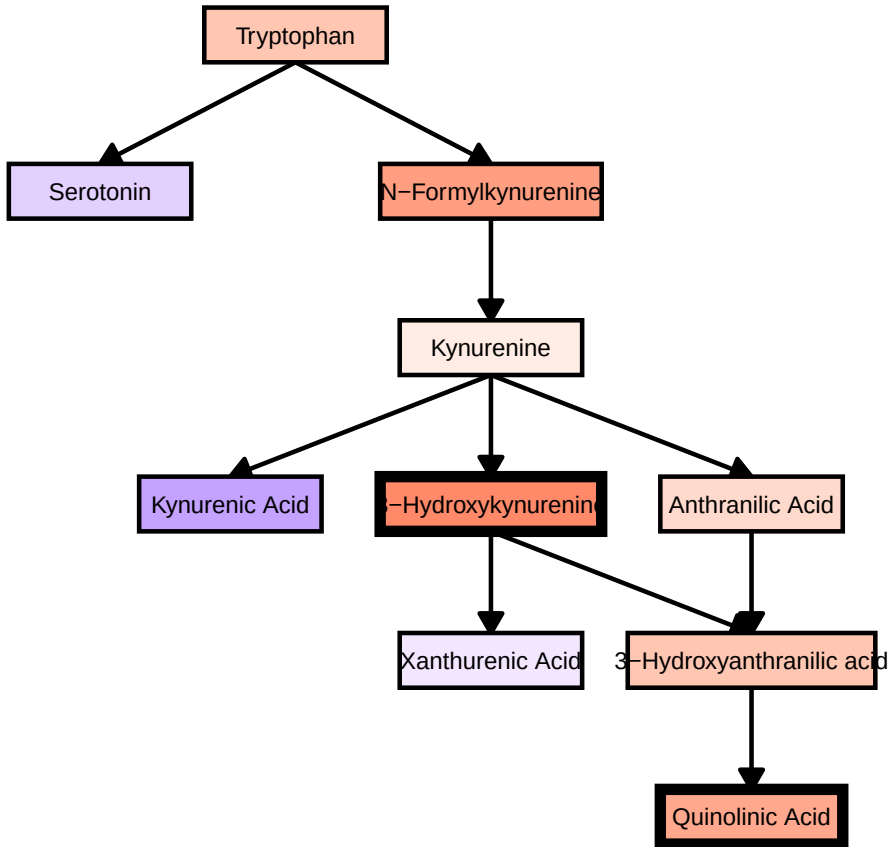
Residual AXL = 0 + 0 * age_resid | R-squared = 0.01, n = 322, p = 0.085



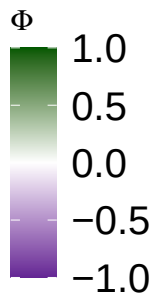
Kynurenine pathway



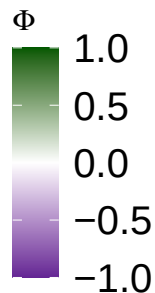
Kynurenine pathway

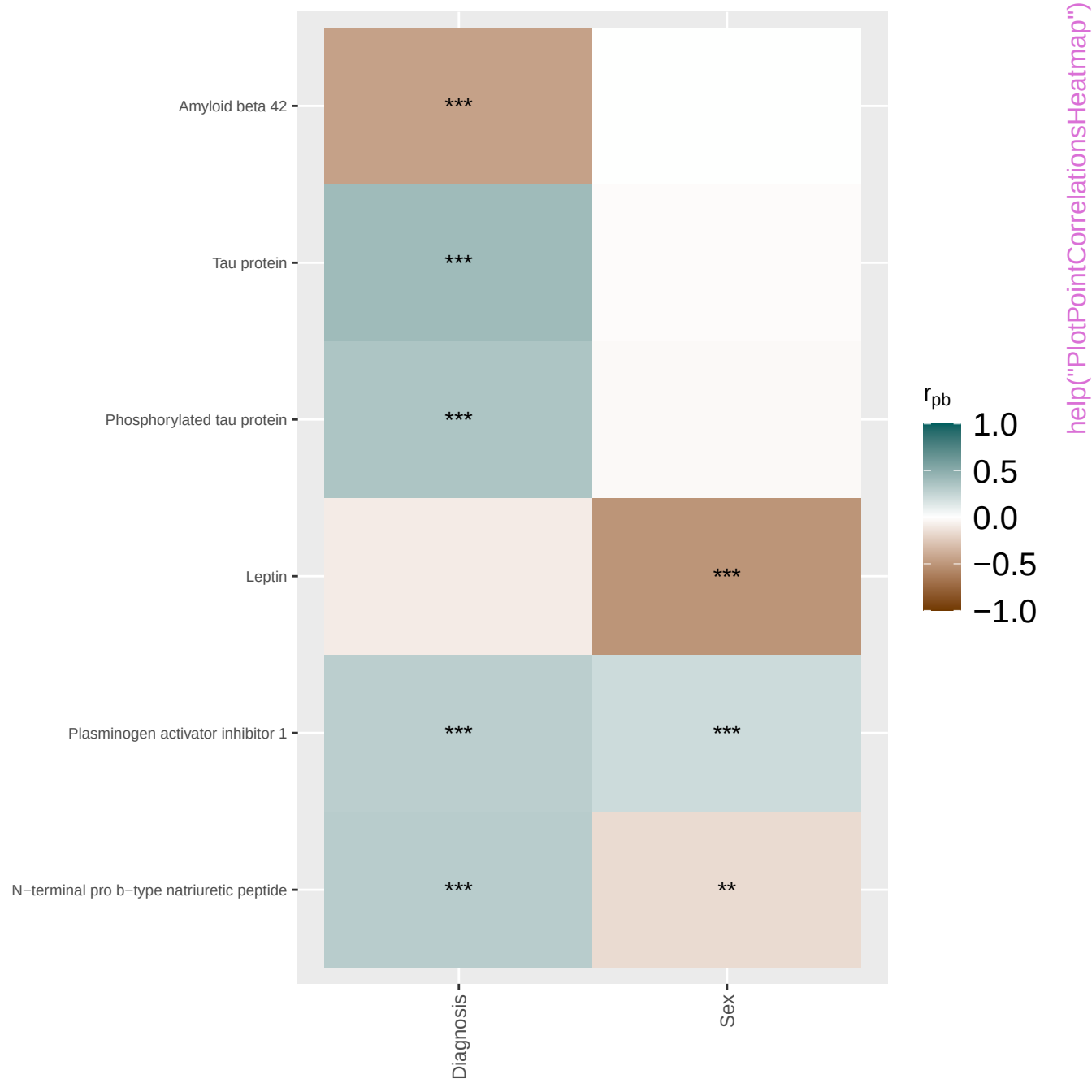


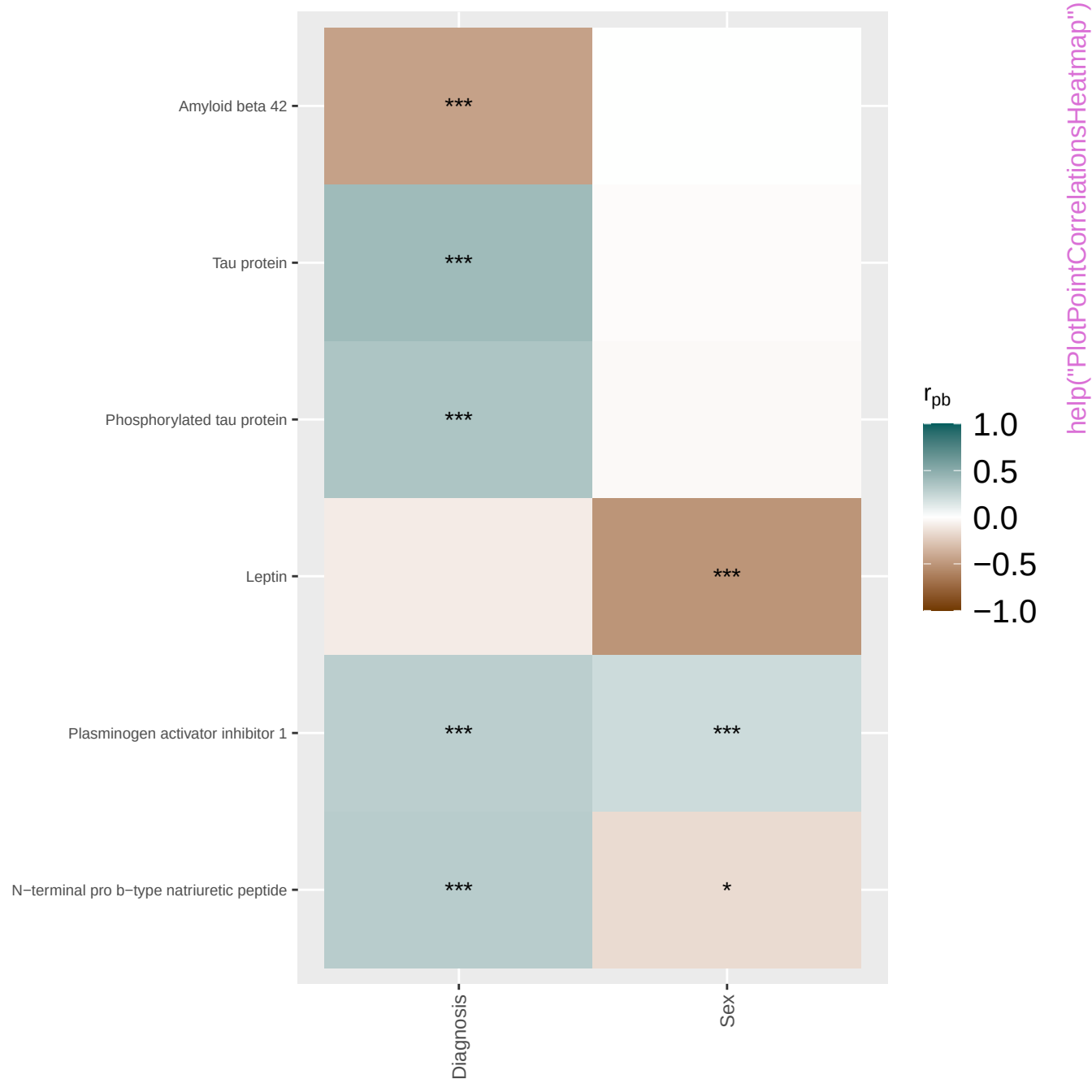
help("PlotPhiHeatmap")

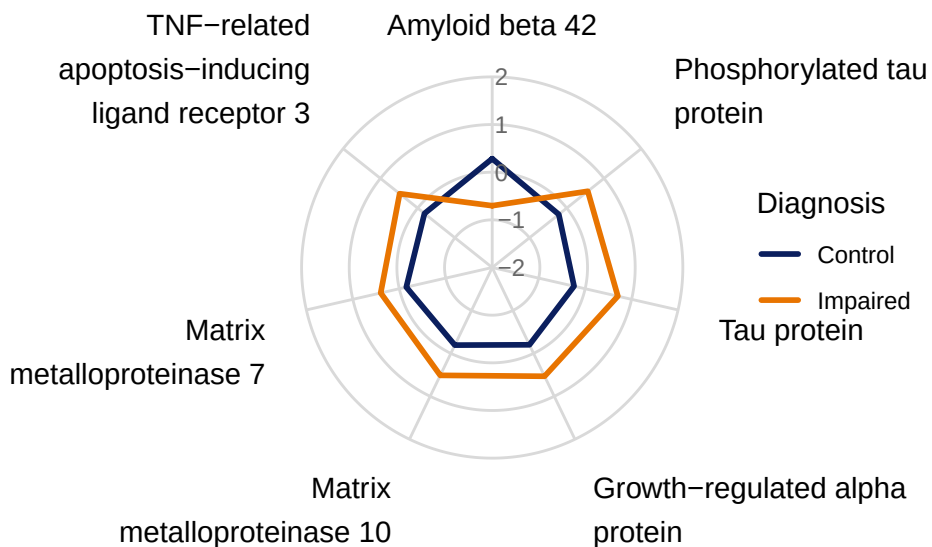


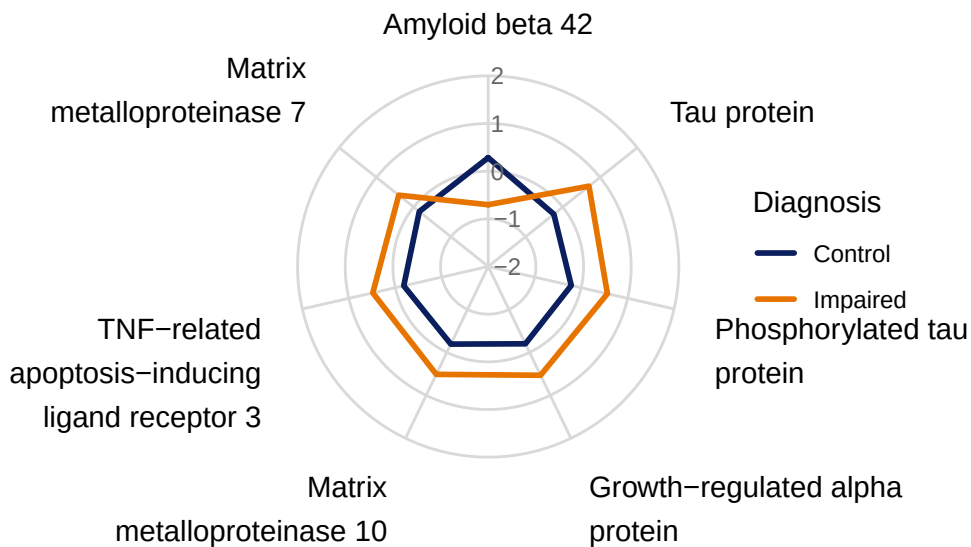
help("PlotPhiHeatmap")

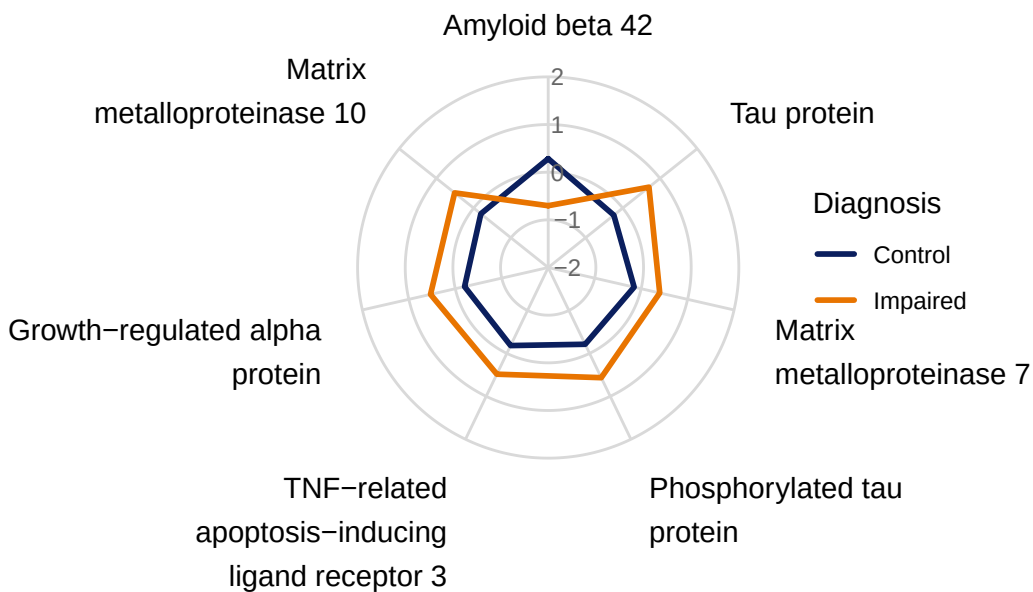


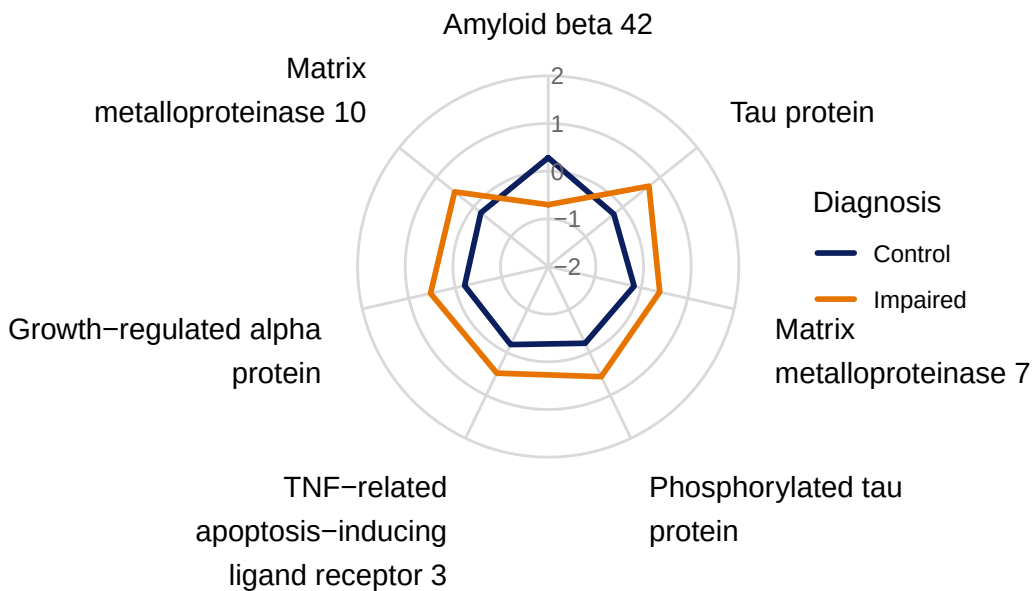


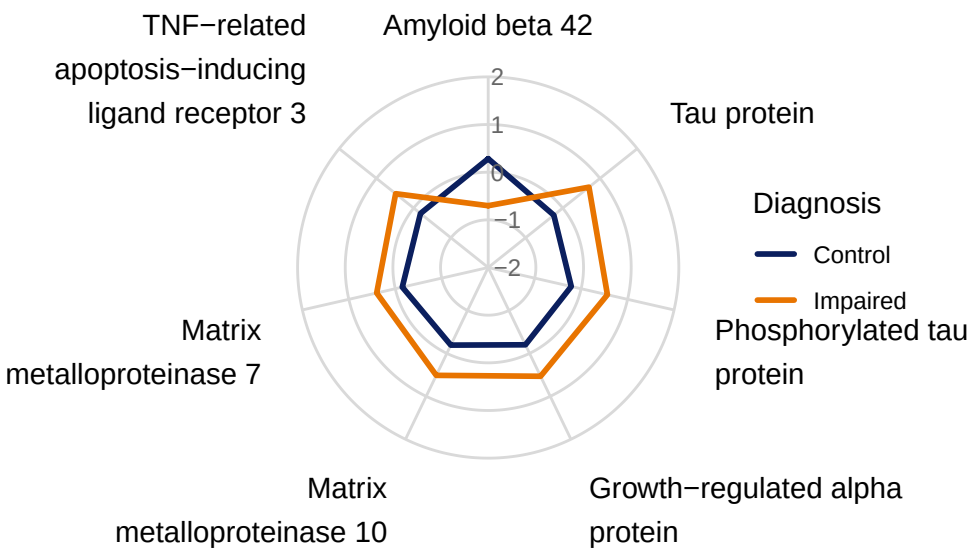














Control (n=166)



Impaired (n=67)

Ab_42

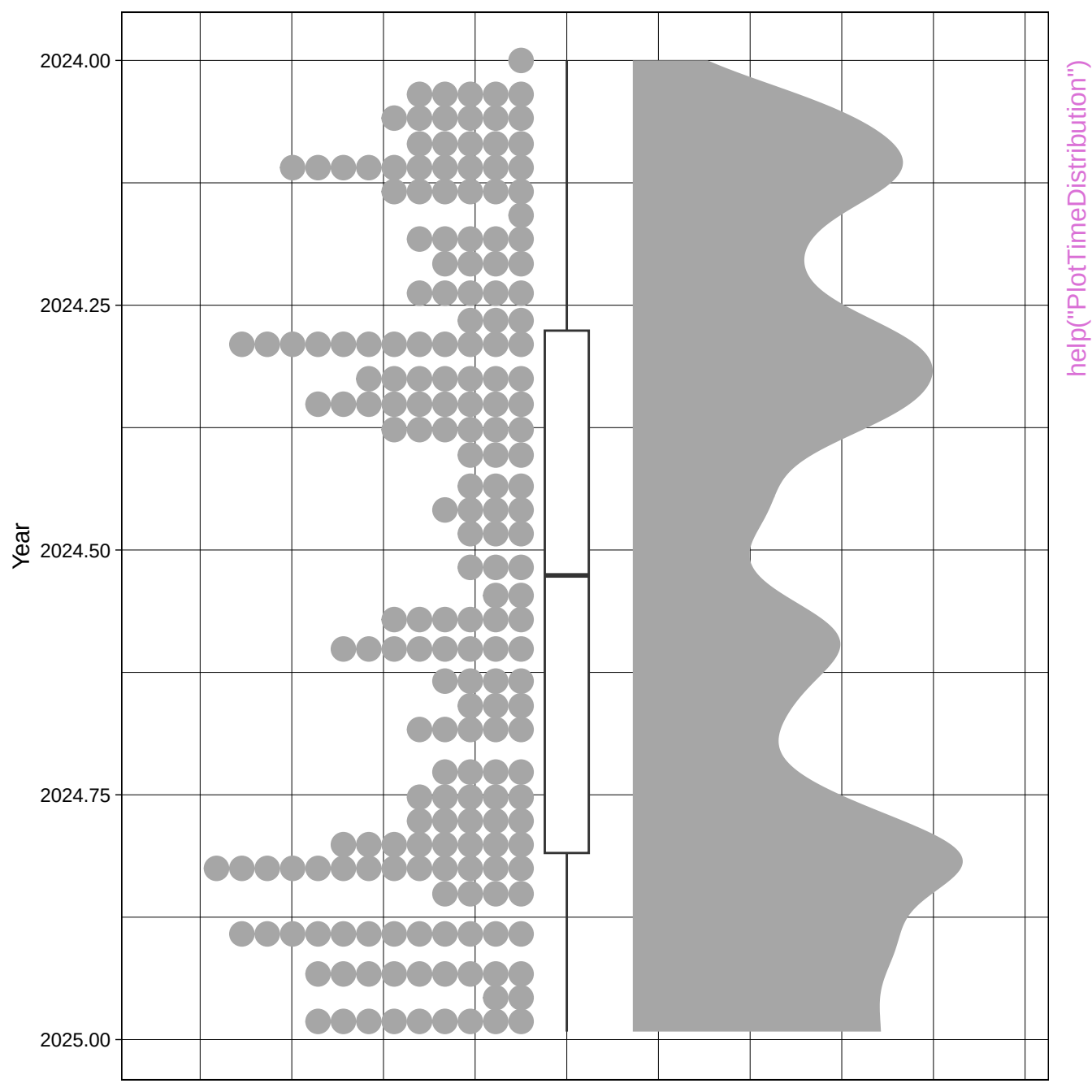
16

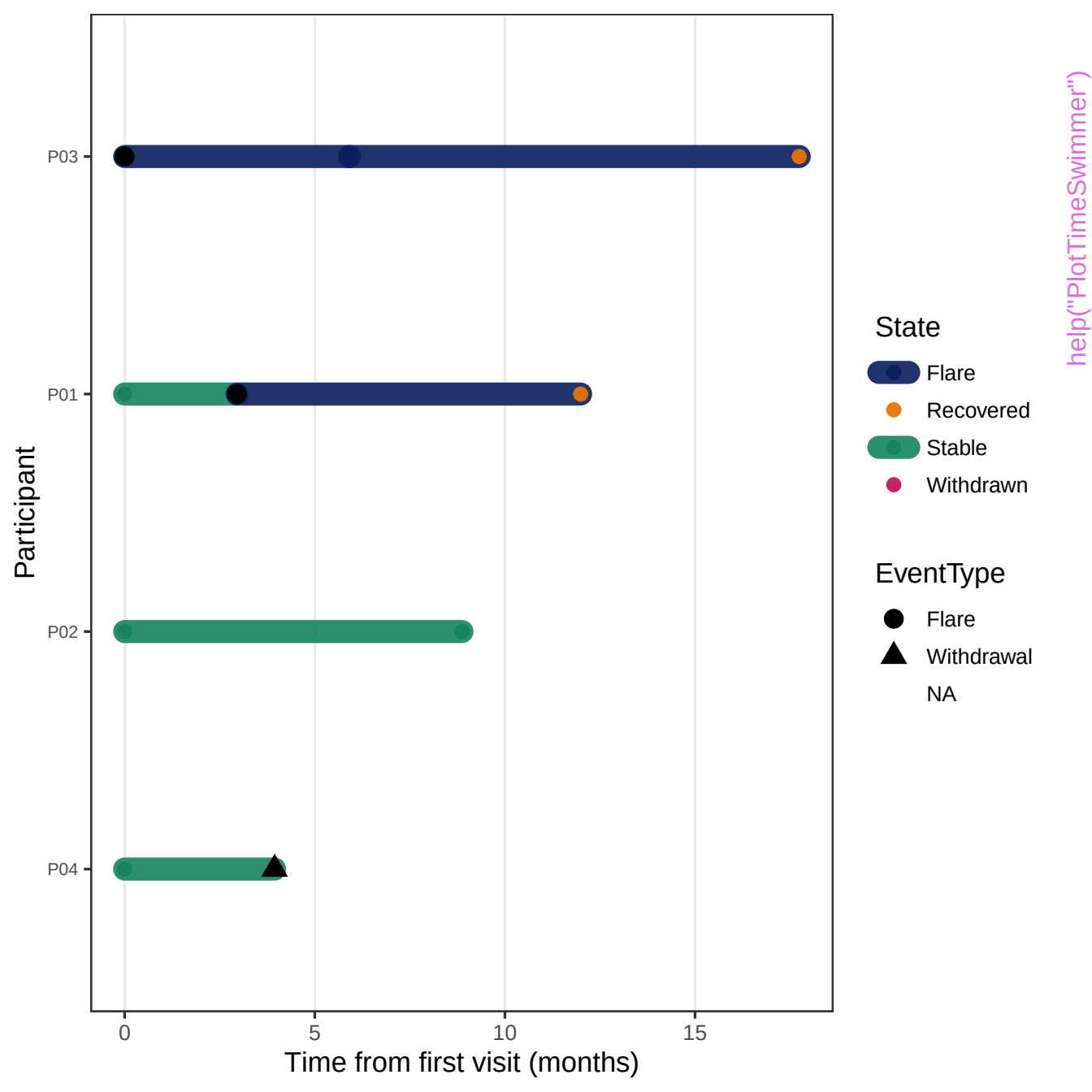
14

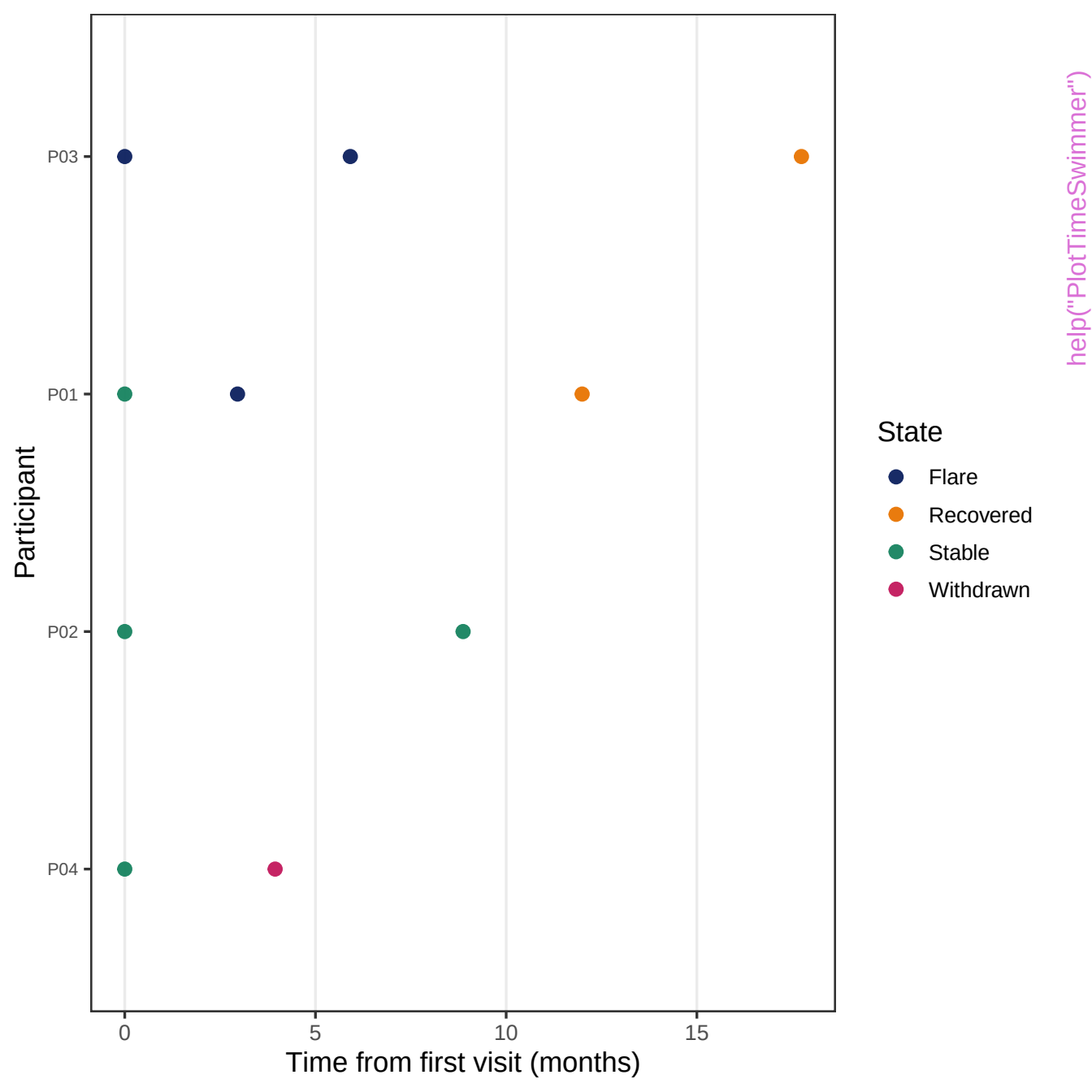
12

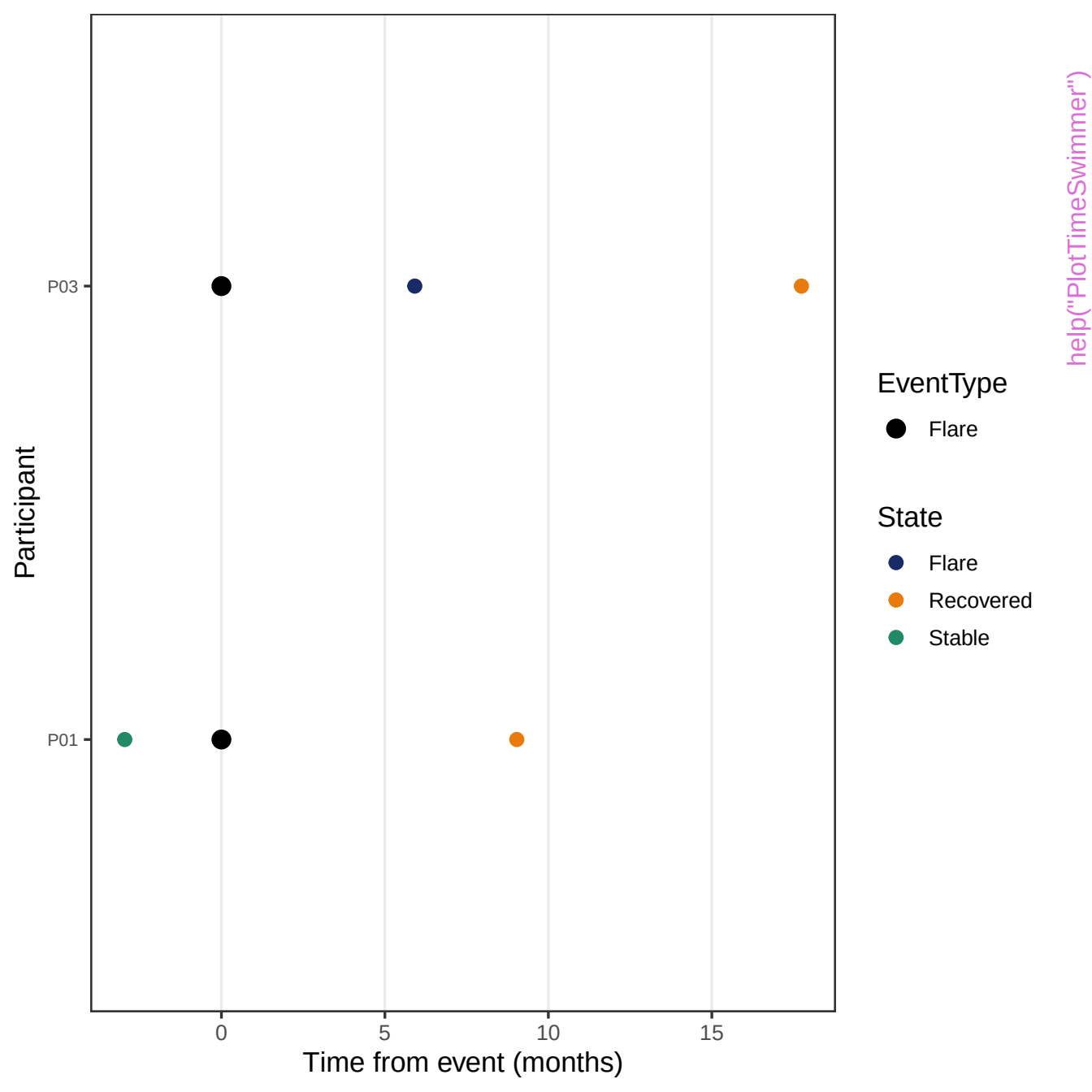
10

help("PlotSplitViolin")

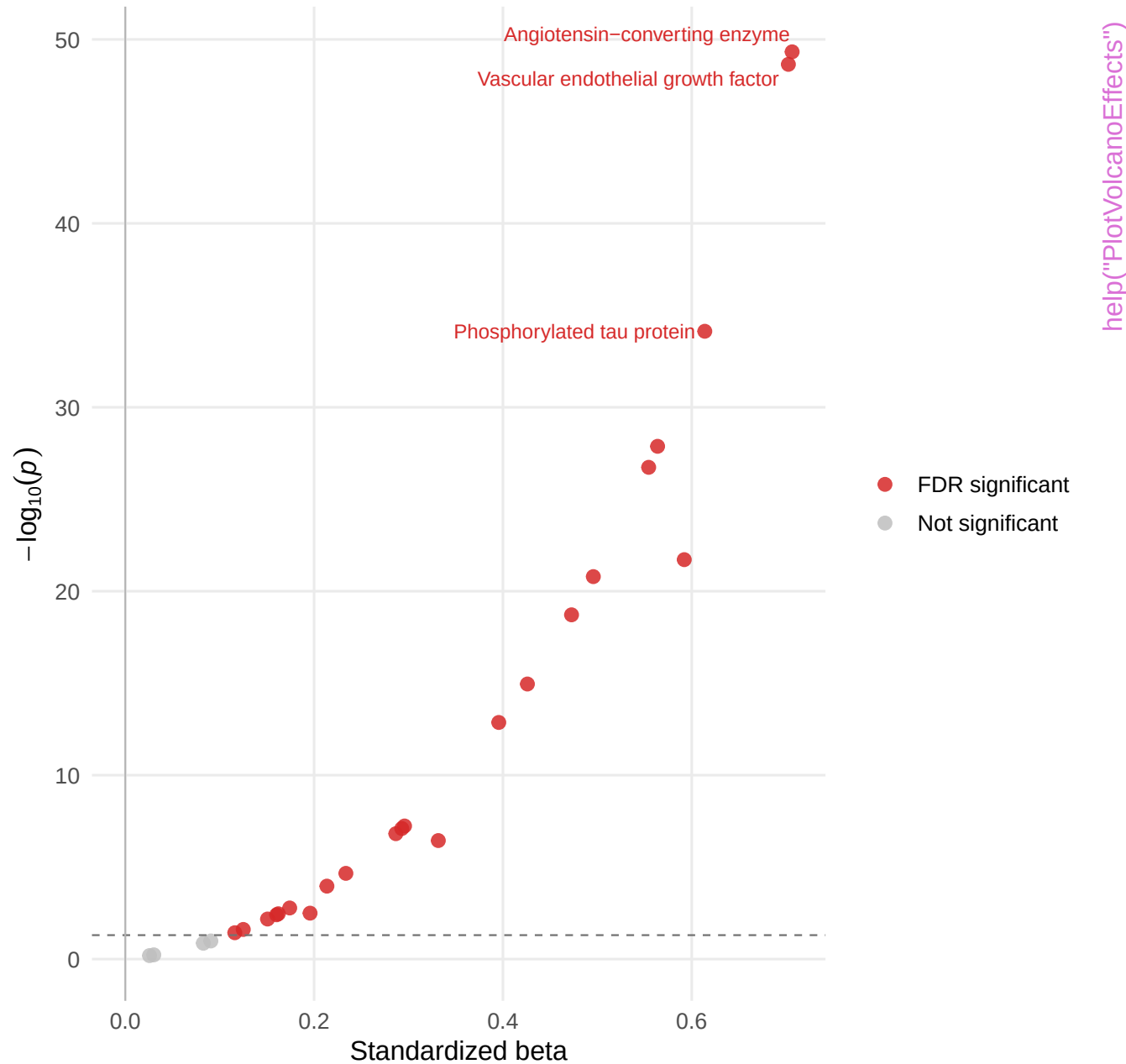




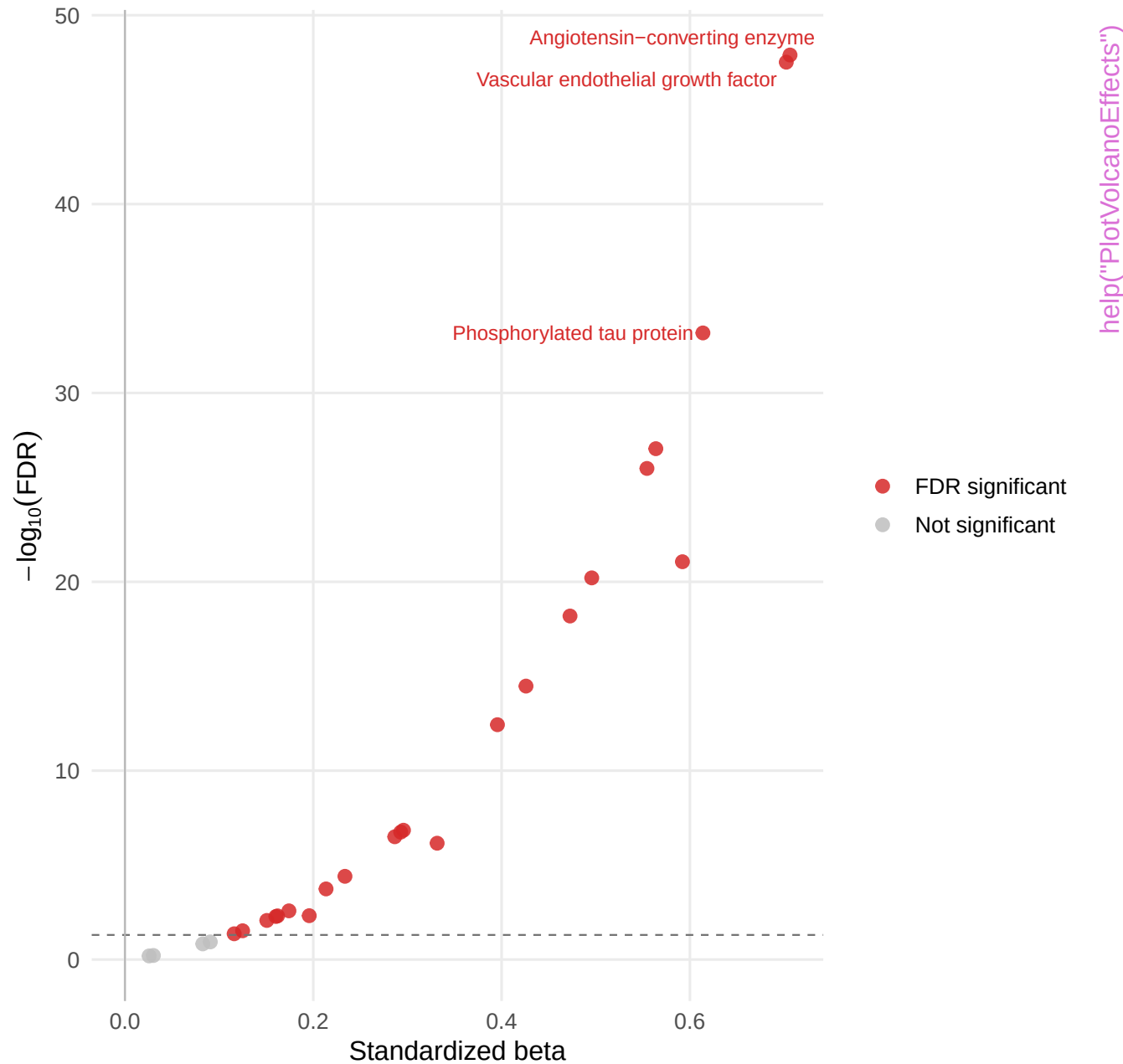




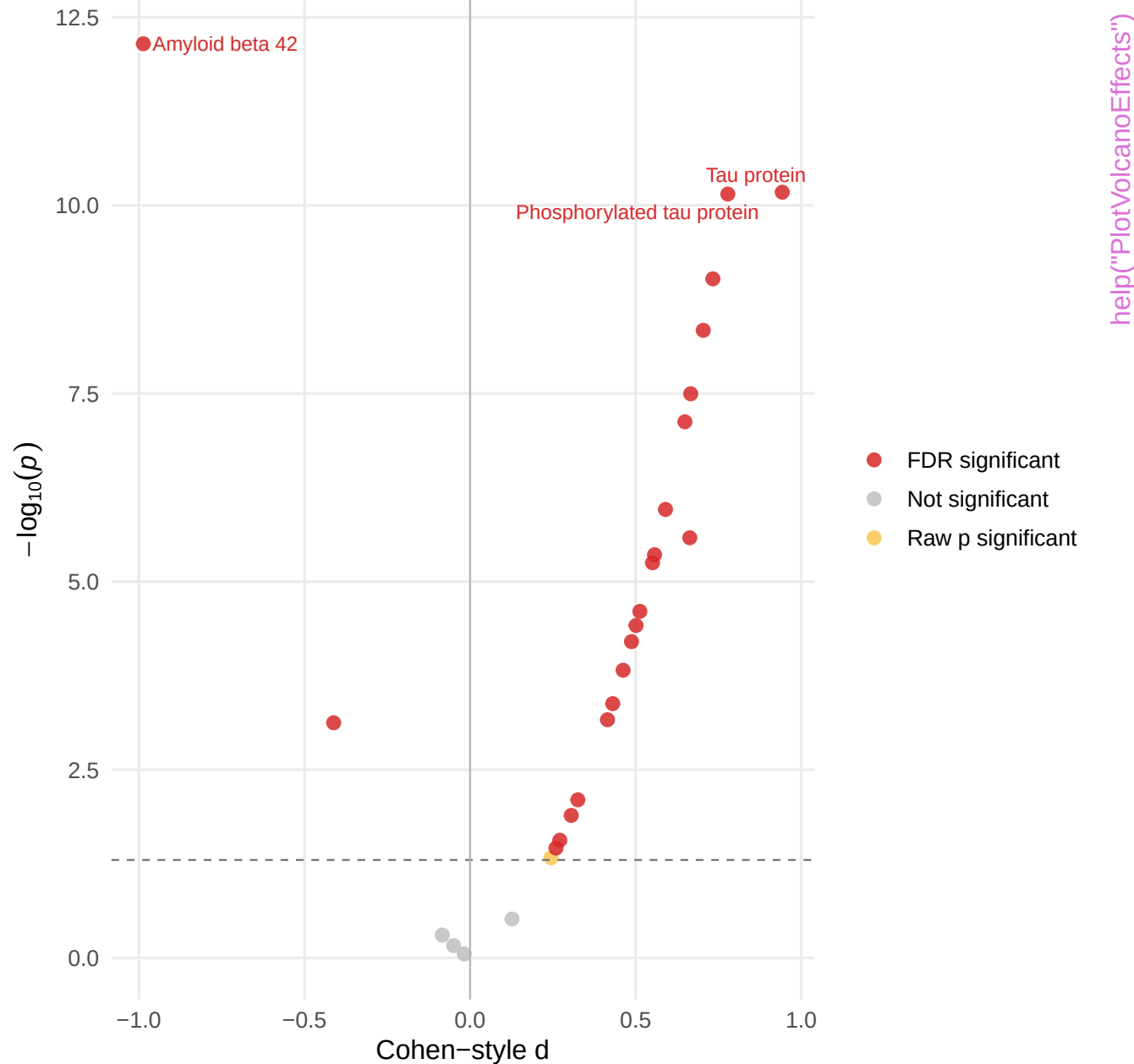
Volcano plot using raw p-values: AXL



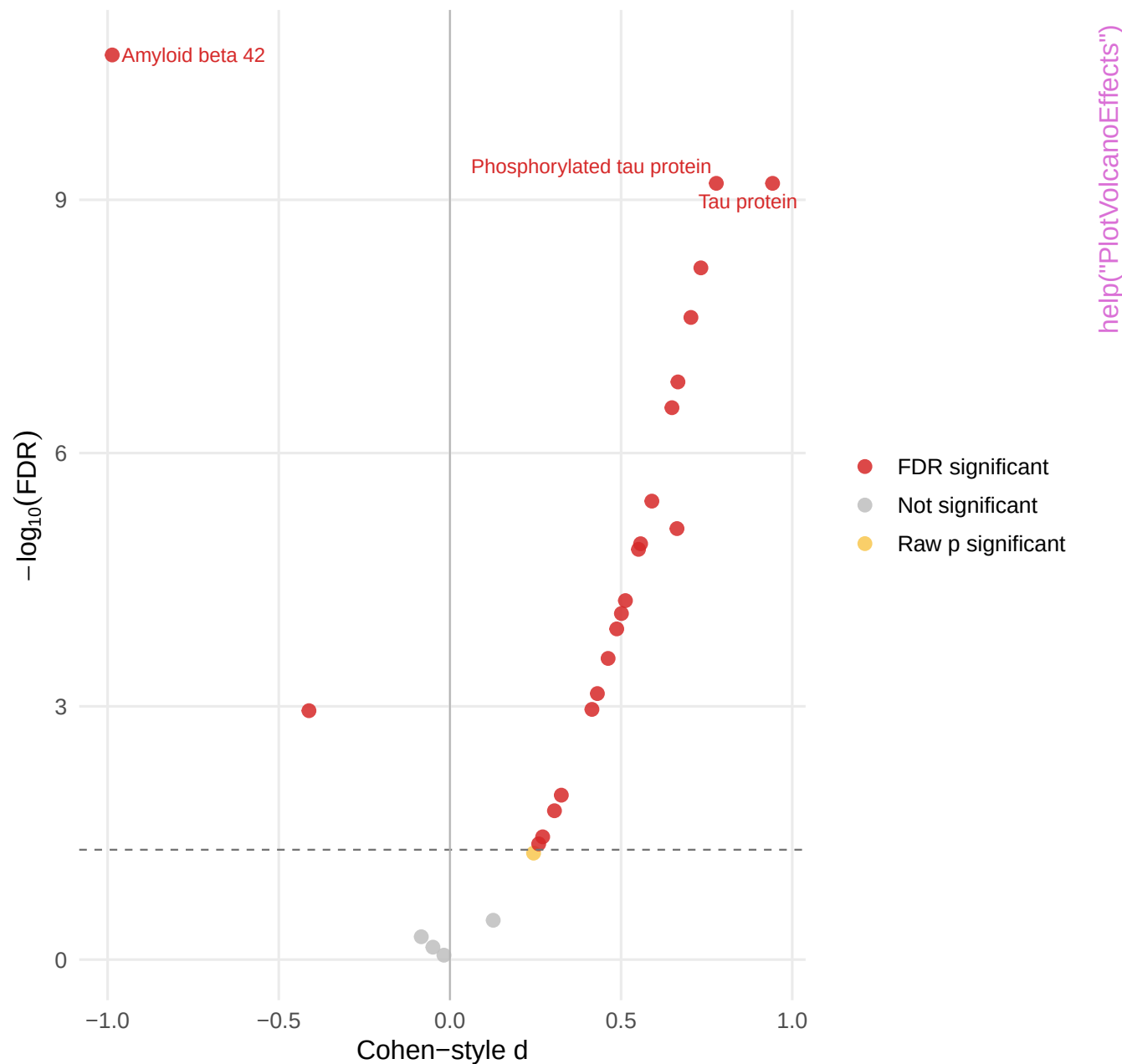
Volcano plot using FDR-adjusted p-values: AXL

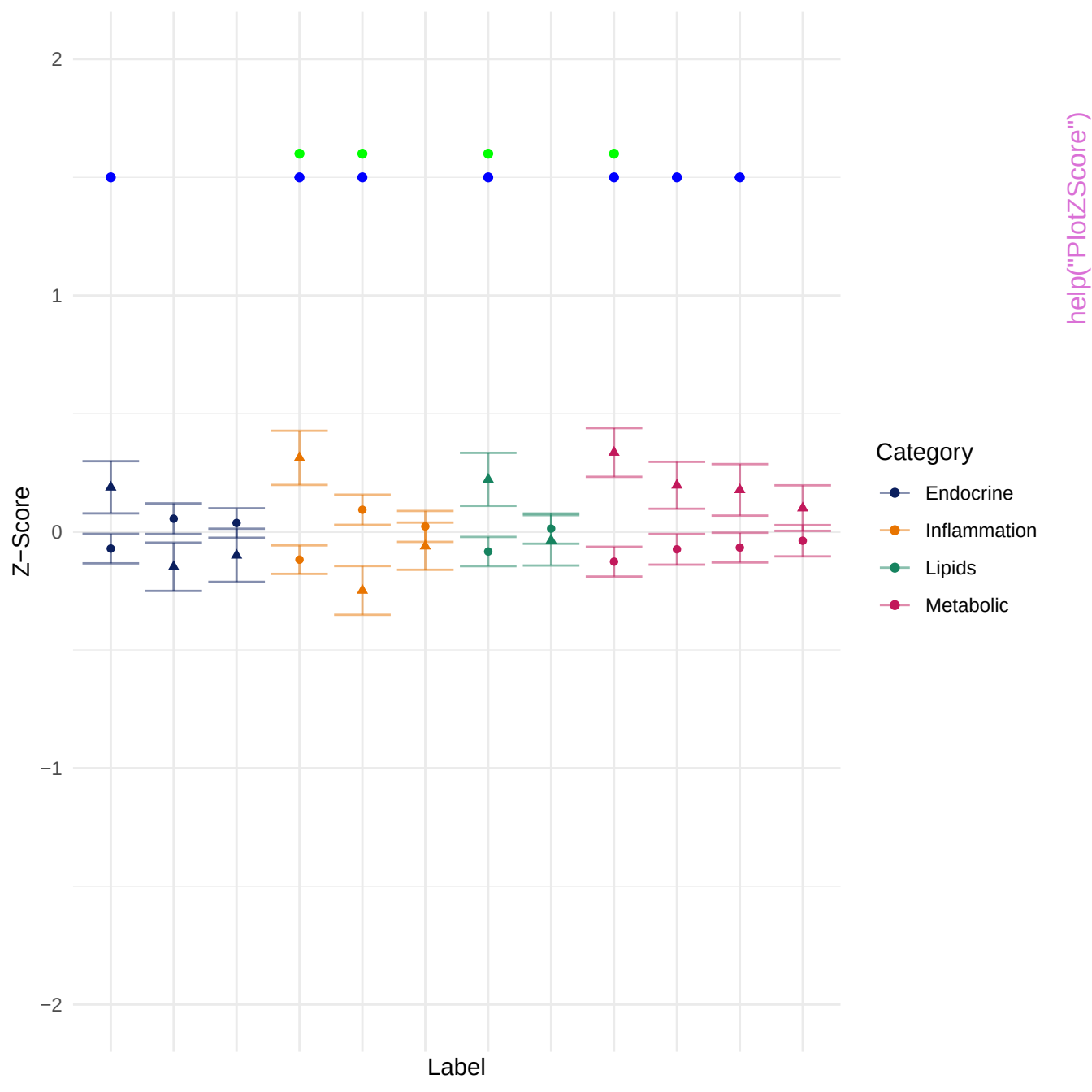


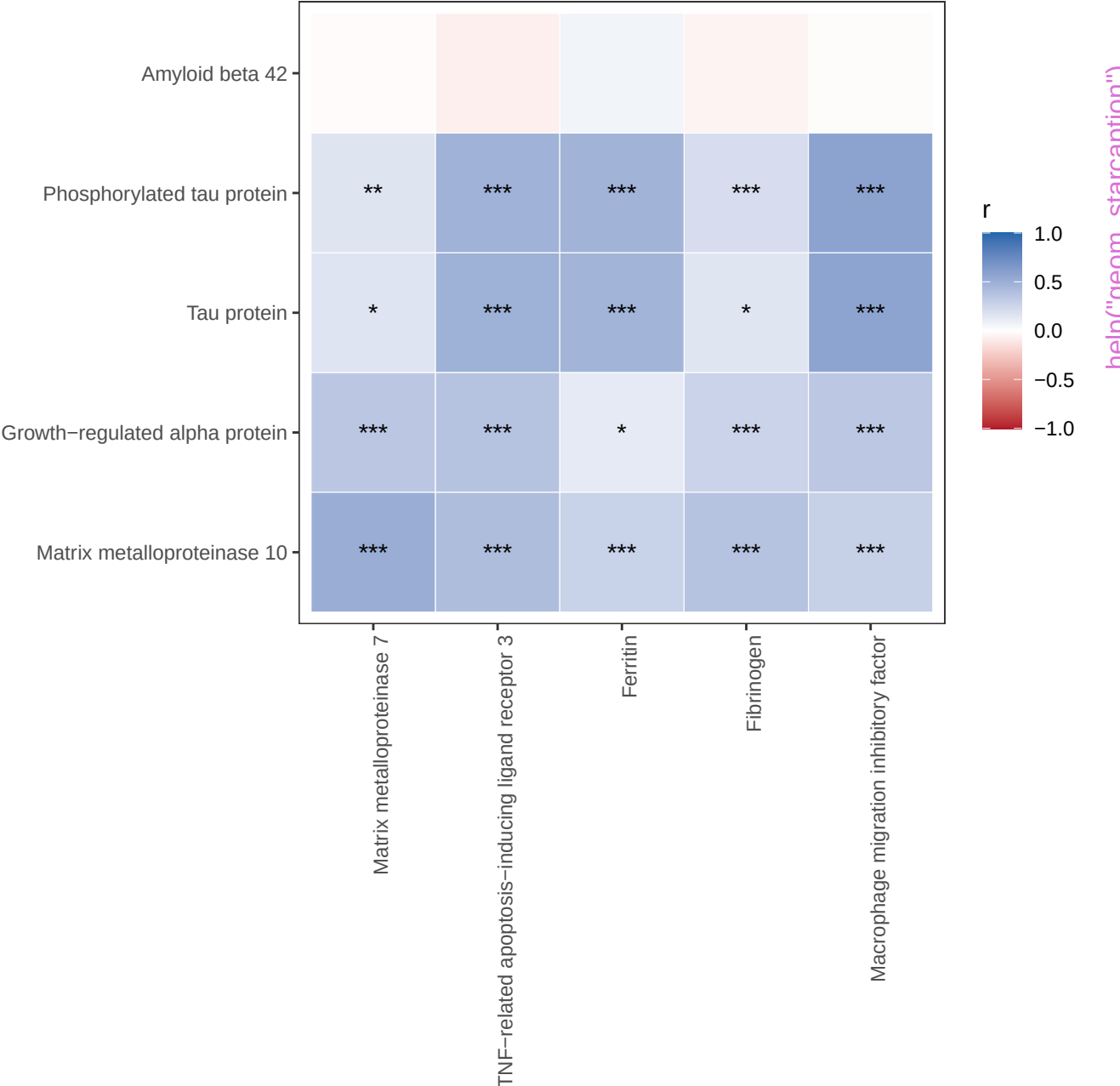
Volcano plot using raw p-values: Diagnosis



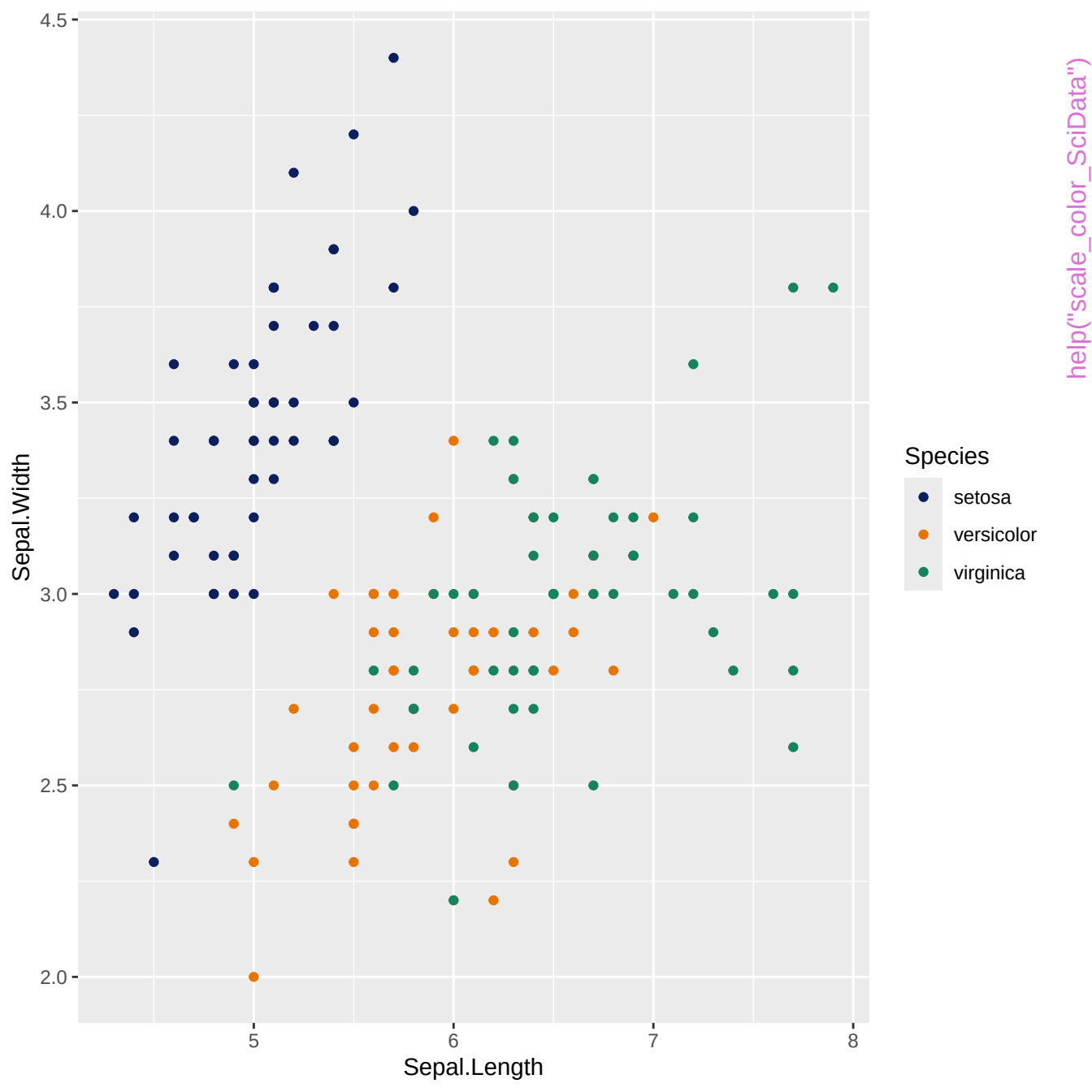
Volcano plot using FDR-adjusted p-values: Diagnosis

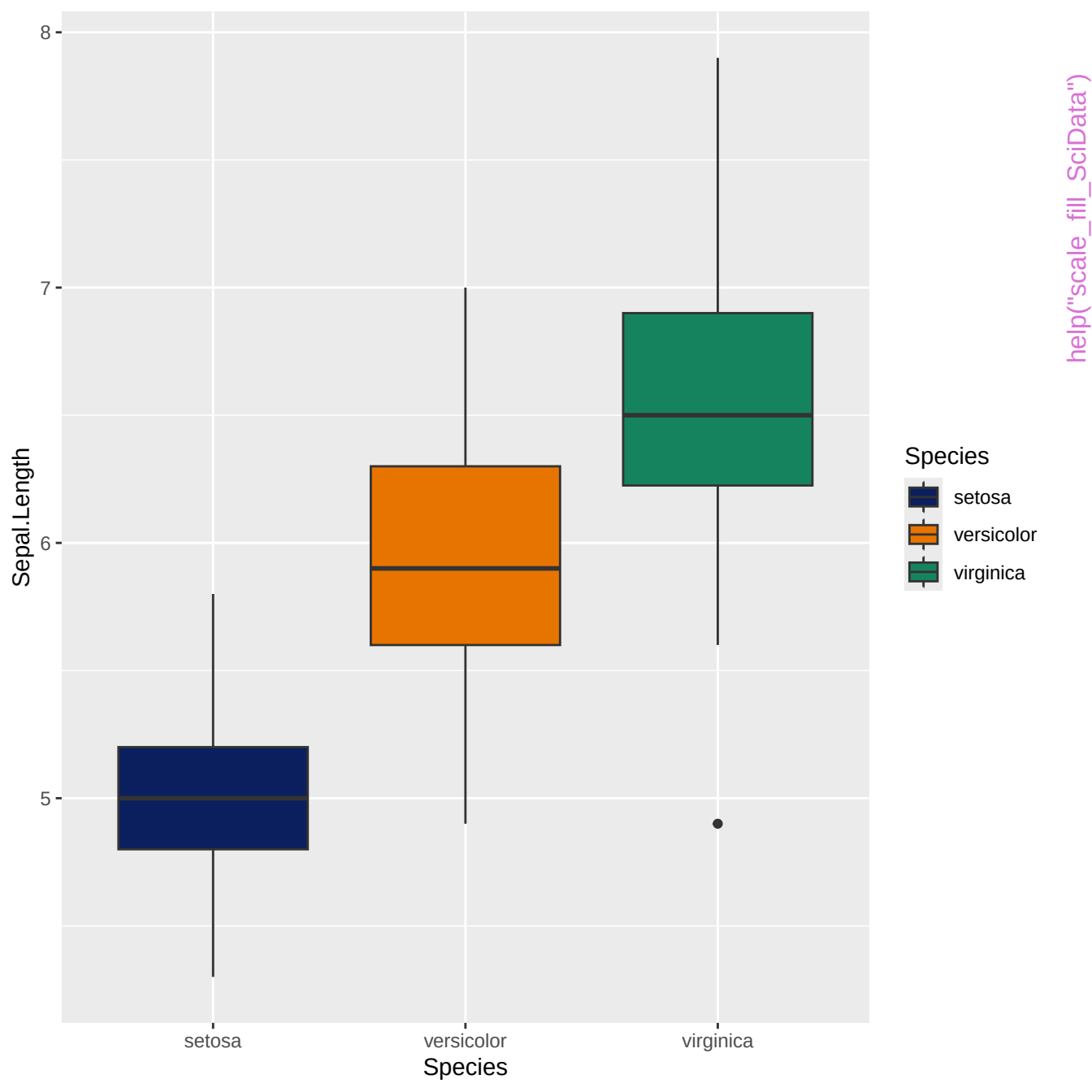






* = p <= 0.05, ** = p <= 0.01, *** = p <= 0.001





Winsorization pulls outliers to the limits

Both panels share the raw data's x range

